ESE 4115 Complete Study Pack

Power Plant Engineering · KUET · past papers 2020, 2021, 2022, 2023, 2024 · answers from the course slides

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 6 h for Part A and 6 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2024 ×5, 2023 ×4, 2022 ×3, 2021 ×2, 2020 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Next if you have time: the 2024 exergy numericals (B7, 25 marks each) appeared only once, so they sit low. They are long; learn them only after the pass line. The Brayton calculator (B5) and the hydro calculator (A14) let you redo every year's numbers in minutes, so practise those first: A3, A8 and A14 are numericals that come almost every year.

Section A: Fundamentals, load curves, economics, forecasting, Bangladesh, hydro

Block 1: very likely and easy · 2.5 h

Topic	Chance	Level	Covered
A10 Load forecasting: need, factors, methods (similar-day, additive, peak vs energy)	Very likely 100%	Easy	11%
A4 Effects of variable load on power plant design and operation	Very likely 100%	Easy	18%
A2 Load curve, load duration curve and the plant factors (load, demand, diversity, capacity, use)	Very likely 93%	Easy	25%
A9 Tariff: objectives, requirements and types (flat, block, two-part, maximum demand, ToU)	Very likely 93%	Easy	31%
A7 Cost of generation, fixed and operating costs, depreciation	Very likely 87%	Easy	36%
A5 Site selection for a power plant and the interconnected grid	Very likely 80%	Easy	43%

Block 2: very likely, medium effort · 3.8 h

Topic	Chance	Level	Covered
A12 Hydro-electric plant: layout, essential features, dam, spillway, surge tank, forebay, hydrology terms, classification	Very likely 100%	Medium	54%

A14 Numericals: hydro power from catchment, rainfall and head	Very likely 100%	Medium	63%
A3 Numericals on load curve, load factor, capacity factor, use factor	Very likely 100%	Medium	70%
A13 Hydraulic turbines: selection, specific speed, governing of impulse (Pelton) turbine	Very likely 100%	Medium	77%
✔ Pass line: 77% of Section A marks covered, enough to choose 3 of the 4 questions (about 5.5 h)			
A11 Bangladesh power sector: fuel mix, generation cost, renewables, PSMP 2015, nuclear and coal plans	Very likely 93%	Medium	86%

Block 4: likely · 0.4 h

Topic	Chance	Level	Covered
A1 Power plant basics: classification, energy sources, base vs peak plant, choosing capacity	Likely 67%	Easy	93%

Only if time is left · 1.8 h

Topic	Chance	Level	Covered
A8 Numericals: generating cost per kWh	Possible 47%	Medium	99%
A6 Economic load sharing between a base load and a peak load plant (derivation)	Possible 33%	Hard	100%

Section B: Diesel, gas turbine, combined cycle, steam plant and auxiliaries

Block 1: very likely and easy · 1.2 h

Topic	Chance	Level	Covered
B1 Diesel power plant: layout, working, fuel system, applications, advantages, peak-load use, lubrication	Very likely 100%	Easy	15%
B10 Feed water: impurities, troubles in boiler, treatment methods (lime-soda, zeolite)	Very likely 93%	Easy	23%
B6 Combined cycle power plant (gas turbine + steam turbine)	Very likely 93%	Easy	31%

Block 2: very likely, medium effort · 2.2 h

Topic	Chance	Level	Covered
B4 Improving gas turbine efficiency: regeneration, intercooling, reheating (and all three together)	Very likely 100%	Medium	40%
B11 Draught: natural vs mechanical (forced, induced, balanced), chimney and its height formula	Very likely 100%	Medium	46%

B13 Cooling towers: types, induced-draught counterflow wet tower, range, approach, efficiency + numerical	Very likely 93%	Medium	53%

Block 4: likely · 2.2 h

Topic	Chance	Level	Covered
B3 Gas turbine plant: open-cycle layout, open vs closed, applications, peak-load advantages, fuel properties	Likely 60%	Easy	57%
B8 Steam power plant: layout and four circuits, working, base-load advantages, site, design activities	Likely 67%	Medium	69%
B5 Brayton cycle numericals and the maximum-work condition	Likely 67%	Hard	76%

✔ Pass line: 76% of Section B marks covered, enough to choose 3 of the 4 questions (about 5.7 h)

Only if time is left · 2.9 h

Topic	Chance	Level	Covered
B2 Diesel plant cooling (necessity, thermostat cooling) and starting methods	Possible 53%	Easy	84%
B9 Reheat Rankine cycle and binary vapour cycle	Possible 53%	Medium	90%
B12 Numericals: chimney draught and chimney height	Possible 33%	Medium	92%
B7 Exergy analysis numericals (2024): Diesel cycle and closed Brayton cycle	Possible 33%	Hard	100%

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One-shot notes: ESE 4115 Power Plant Engineering

Load curve words

Load curve = load vs time (in order). **Load duration curve (LDC)** = same loads sorted high → low. Area = energy (MWh). Ideal curve flat (LF = 1); realized curve has peaks.

Base load = always there (coal, nuclear, big hydro, combined cycle). **Peak load** = few hours (gas turbine, diesel, pumped storage).

Factors (A-M-I rule)

A = average load, M = maximum demand, I = installed capacity.

Load factor = A/M · capacity factor = A/I · utilisation = M/I · demand factor = M / connected load · diversity = Σ individual maxima / system max (**> 1**, the only one) · use factor = E / (I × hours run) · reserve = I − M.

Hours in service = 8760 × CF / use factor.

Cost per kWh recipe

1 crore = 10^7, 1 lac = 10^5. kWh/yr = kW × LF (or CF) × 8760. Fixed = (interest + depreciation)% × capital + fixed wages. Running = fuel Tk/kWh × kWh. Cost = total / kWh sent out (minus auxiliaries).

Answers: 2023 Tk 1.28 · 2020 Tk 1.06 · 2021 Tk 2.00 per kWh.

Economics words

Fixed cost: interest, depreciation, insurance, taxes, salaries. **Running**: fuel, oil, water, maintenance, output wages.

Depreciation = loss of value; causes WADOI (wear, age, deterioration, obsolescence, inadequacy). Methods: straight line $(P - S)/n$, diminishing value, sinking fund $q = (P - S)r/[(1 + r)^n - 1]$.

Economic load sharing: peak plant runs $(R_1 - R_2)/(P_2 - P_1)$ h/yr (extra fixed per kW over extra running per kWh).

Tariff

Y = DX + EZ + C (demand × kW + energy × kWh + constant). Flat demand Y = DX; straight-line meter Y = EZ; step meter (whole use at one step rate, rate falls); block rate $\Sigma E_i Z_i$; two-part (Hopkinson) DX + EZ; three-part (Doherty) DX + EZ + C; time-of-use.

Objectives: recover generation, transmission/distribution, operation costs + profit. Requirements: fair, simple, reasonable profit, encourage off-peak use and high power factor.

Load forecasting

Purposes: plan capacity, unit commitment, fuel, maintenance, reserve, tariff. Factors TWEDS-R: time, weather, economy, demography, special events, random.

Short (hour-week), medium (month-year), long (> 1 yr). Methods: similar-day, regression, ARMA/ARIMA, ANN, expert system. Additive $L = L_n + L_w + L_s + L_r$.

Peak approach (extrapolate MW) vs energy approach (forecast kWh, peak = E/(LF × 8760)).

Bangladesh

Gas about 55 % of capacity (two-thirds of energy), furnace oil 27 %, import 9 %, coal 6 %, diesel 2 %, hydro 230 MW (Kaptai), renewables about 2.7 % (mostly solar home systems). Cost: hydro < gas < coal < oil < diesel.

PSMP 2015: peak 8,921 MW (2015) → about 50,400 MW (2041), need about 57,000 MW with reserve, GDP 6 %/yr. Meet with imported coal (Payra, Rampal, Matarbari 1200-1320 MW each), LNG + gas combined cycle, nuclear Rooppur 2 × 1200 MW (Russia), imports (Bheramara, Adani), renewables, efficiency.

Variable load and siting

Effects: extra equipment, higher production cost, reserve capacity, low LF, part-load efficiency loss, peak plants needed, control/load prediction problems.

Site: fuel, water, load centre, land, transport, ash disposal, environment, labour, expansion. Grid benefits: less reserve, cheapest plants first, reliability, diversity, big units, power exchange.

Hydro plant

Catchment → reservoir → dam (+ spillway) → gate, tunnel → surge tank → penstock → turbine-generator → draft tube → tail race.

Surge tank: absorbs water hammer on load drop, supplies water on load rise. **Forebay**: basin at penstock head, temporary storage. **Spillway**: releases flood water. Dams: gravity (weight), arch (arch action, gorges), buttress (slab on buttresses), earth.

Hydrograph Q vs time; flow duration curve Q vs % time; mass curve cumulative volume vs time; run-off = rain reaching streams.

Hydro numbers and turbines

Q = area × rain × fraction / 31.536×10^6 s. **P (kW) = 9.81 Q H η**. Capacity = P / LF. Energy (J) = m g H η; 1 kWh = 3.6 MJ.

Answers: 2023 6.67 MW (13.3 MW capacity) · 2020 6.58 MW (13.2) · 2022 3.36 MW (7.47) · 2021 24.6 MW · 2024 3,567 GWh/yr.

Pelton (impulse) high head; Francis medium; Kaplan low (< 50 m). $N_s = N\sqrt{P}/H^{5/4}$; N = 120f/p. Governor: fly balls → relay valve → servomotor → spear (+ deflector).

Diesel plant

Systems FACLSG: fuel, air intake/exhaust, cooling, lubrication, starting, governing. Fuel path: storage → day tank → filter → injection pump → injector. Solid injection (no air compressor) beats air-blast.

Uses: peak, standby, mobile, small central/island, nursery and starting stations. Peak-load: starts in minutes, no standby loss, low capital.

Cooling needed: walls < 400 °C, oil film, valves, ηv. Thermostat keeps water in jacket until 70-85 °C then opens to radiator. Starting: compressed air (big), electric (small), auxiliary engine.

Gas turbine

Open: air → compressor → combustor → turbine → exhaust. Closed: heater + cooler, same fluid.

$x = r_p^{0.2857}$; T2s = T1x; T4s = T3/x; $T2 = T1 + (T2s - T1)/\eta_c$; $T4 = T3 - \eta_t(T3 - T4s)$; $w = 1.005\,\Delta T$; $\eta = w_{net}/q_{in}$; ideal $\eta = 1 - 1/x$; $bwr = w_c/w_t$. Regenerator $T5 = T2 + \varepsilon(T4 - T2)$.

Max work: $T2 = T4 = \sqrt{(T1T3)}$. Regeneration: η ↑, work same (needs T4 > T2). Intercooling: w_c ↓.

Reheat: w_t ↑. All three + regenerator: both ↑ (Ericsson).

Answers: 2020 46.6 % · 2021 27.4 % → 35.6 % · 2022 37.5 % → 68.9 %.

Combined, reheat, binary

Combined: GT exhaust → HRSG → steam turbine. $\eta = \eta_{GT} + \eta_{ST}(1 - \eta_{GT}) \approx$ 50-62 %. Gas on top, steam below.

Reheat Rankine: HP turbine → reheater → LP turbine: drier exhaust, more work, η +4-5 %.

Binary: mercury cycle on top, its condenser boils water. Not used: costly, toxic, does not wet metal.

Exergy (2024)

$X_{destroyed} = T0\,s_{gen}$. $X_{in} = q_{in}(1 - T0/T_s)$. $\eta_{II} = w_{net}/X_{in}$. Check $X_{in} - \Sigma X_d = w_{net}$.

Diesel 5(c): T2 1094.9 K, P2 5424 kPa, T4 785.5 K, w_{net} 299.1 kJ/kg, η 47.4 %, s_{gen} 0.791, X_d 237.4, η_{II} 55.7 %.

Brayton 6(a): T2 709 K, T4 1058 K, W_{net} 4.66 MW, η 36.4 %, s_{gen} 22.65 kW/K, X_d 6.64 MW, η_{II} 41.3 %.

Steam plant

Pump → boiler → turbine → condenser. Circuits CAFe Cool: coal-ash, air-flue gas, feed water-steam, cooling water. Base-load: big units, cheap fuel, long life, steady efficiency.

Design activities: load survey, site, fuel, capacity and units, main equipment, auxiliaries, layout, economics, EIA.

Feed water

Impurities: hardness salts (scale), O_2/CO_2 (corrosion), suspended solids (sludge), oil/alkali (foaming, priming, embrittlement).

Treat: mechanical (settle, coagulate, filter), thermal (deaerate, distil), chemical (lime-soda, zeolite), demineralise, blow-down. Zeolite: Na_2Z + Ca salt → CaZ + Na salt; regenerate with brine.

Draught, chimney, cooling tower

Natural (chimney, 10-20 mm water) vs mechanical: forced (FD fan at inlet), induced (ID fan near chimney), balanced (both, furnace slightly below atmospheric).

$h_w = 353\ H\ [1/T_a - ((m + 1)/m)/T_g]$ mm water; $H_g = H[(m/(m + 1))T_g/T_a - 1]$. 1 mm water = 1 kgf/m^2.

2023: 15.2 mm, 23.9 m; 2020: H = 43.6 m.

Range = $T_{in} - T_{out}$; approach = T_{out} − WBT; efficiency = range/(range + approach). 2021: T_{out} 22.4 °C, evaporated 1.03 %, range 7.6, approach 7.2 °C.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
AC	alternating current
ANN	artificial neural network
ARIMA	auto-regressive integrated moving average
ARMA	auto-regressive moving average
BHP	brake horsepower
BIFPCL	Bangladesh-India Friendship Power Company Ltd
BPDB	Bangladesh Power Development Board
CCGT	combined cycle gas turbine
CCPP	combined cycle power plant
CF	capacity factor
CHP	combined heat and power (cogeneration)
CI	compression ignition (diesel)
CPGCBL	Coal Power Generation Company Bangladesh Ltd
CV	calorific value (heat released by 1 kg of fuel)
DBT	dry bulb temperature
EGCB	Electricity Generation Company of Bangladesh
ESP	electrostatic precipitator
FBC	fluidised bed combustion
FD	forced draught
GT	gas turbine
GWh	gigawatt-hour = 10^6 kWh
HCV	higher calorific value
HFO	heavy fuel oil
HP	high pressure
HRSG	heat recovery steam generator
HSD	high speed diesel
ID	induced draught
IP	intermediate pressure
IPP	independent power producer
kW	kilowatt = 1000 W
kWh	kilowatt-hour = 1 unit of electrical energy
LDC	load duration curve
LF	load factor
LNG	liquefied natural gas
LP	low pressure
LTLF	long-term load forecasting

MTLF	medium-term load forecasting
MW	megawatt = 1000 kW
MWh	megawatt-hour = 1000 kWh
NOx	nitrogen oxides
NWPGCL	North-West Power Generation Company Ltd
PSMP	Power System Master Plan
PV	photovoltaic (solar cells)
RE	renewable energy
RH	relative humidity
RPCL	Rural Power Company Ltd
rpm	revolutions per minute
SI	spark ignition (petrol)
SOx	sulphur oxides
ST	steam turbine
STLF	short-term load forecasting
TDC	top dead centre
Tk	Taka (Bangladeshi currency)
ToU	time of use
TWL	tail water level
UF	use (utilisation) factor
WBT	wet bulb temperature

Section A: Fundamentals, load curves, economics, forecasting, Bangladesh, hydro

A10 Load forecasting: need, factors, methods (similar-day, additive, peak vs energy) Must study Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2024 Q3(b)** [11] What factors influence load forecasting? Explain the additive model.
- **2024 Q3(c)** [5] Why is load forecasting necessary?
- **2023 Q3(a)** [10] What is load forecasting? What factors affect it? What are its purposes?
- **2023 Q3(b)** [8] What are the methods of short-term load forecasting? Explain the similar-day approach.
- **2022 Q3(c)** [10] What is load forecasting? What are the factors affecting it?
- **2022 Q3(d)** [8] Describe the importance of load forecasting.
- **2021 Q3(a)** [10] What is load forecasting? What are its purposes and the factors affecting it?
- **2021 Q3(b)** [6] Distinguish between peak load approach and energy approach of load forecasting.
- **2020 Q2(b)** [11] Why is load forecasting required? Describe its importance.

★ REMEMBER IT

- Load forecasting horizons: short (hours–week: unit commitment), medium (months–year: fuel, maintenance), long (5–20 years: new plants). Factors: weather, time/day type, economy, population, prices, special events.
- Additive model: $L = L_{normal} + L_{weather} + L_{special\ event} + L_{random}$.

Before you start: what load forecasting is

Building a power plant takes 3-10 years, and fuel must be bought weeks ahead. So the utility must **predict** how much power (MW) and energy (MWh) customers will need in the next hour, day, year and decade. That prediction is **load forecasting**.

Definition, purposes and necessity (2023 3a, 2022 3d, 2024 3c, 2020 2b)

Load forecasting is the estimation of future electrical load (peak demand in MW and energy in MWh) of a power system for a given period, using past load data, weather, economic and demographic information.

Purposes / why it is necessary:

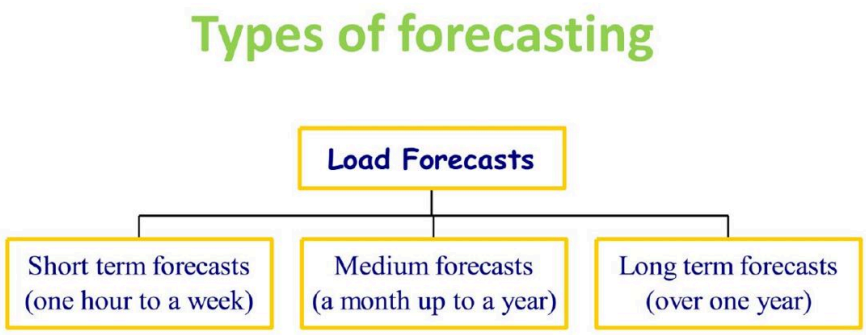
1. **Planning new capacity:** decide the size, type and time of new plants and lines (long term).
2. **Unit commitment and economic dispatch:** decide which units to start, stop and load each hour (short term) for minimum cost.
3. **Fuel purchase and storage** planning.
4. **Maintenance scheduling** of units in low-demand periods.
5. **Security and reliability:** keep enough spinning reserve; avoid load shedding.
6. **Financial planning, tariff setting** and energy trading/imports.

7. **Transmission and distribution planning.**

Factors affecting load forecasting (2024 3b, 2022 3c, 2021 3a)

Factor	How it affects load
Time factors	Hour of day, weekday vs weekend, holidays (Eid), seasons
Weather	Temperature (fans, air-conditioners), humidity, rainfall, cloud cover, wind
Economic factors	GDP growth, industrialisation, electricity price, new connections
Demographic	Population growth, urbanisation, rural electrification
Special events	Strikes, big TV events, Ramadan timings
Random disturbances	Sudden large industrial loads, faults

Types by time horizon



Short term (hour to a week), medium (month to a year), long term (over a year)

Additive model (2024 3b)

Step (write this)	Why / what it means
$L = L_n + L_w + L_s + L_r$	Total forecast load is a SUM of separate parts.
L_n = normal (base) load	Typical load for that hour and day type, from past data.
L_w = weather-sensitive load	Extra load caused by temperature, humidity (e.g. air-conditioning).
L_s = special event load	Change due to holidays, festivals, events.
L_r = random load	Unexplained residual / noise.

Side note: The **multiplicative model** is $L = L_n \times F_w \times F_s \times F_r$, where the F's are correction factors (e.g. 1.05 = 5 % more). Additive adds MW; multiplicative scales by %.

Short-term methods and the similar-day approach (2023 3b)

Methods: similar-day approach, regression methods, time series (ARMA, ARIMA (auto-regressive integrated moving average)), artificial neural network (ANN), expert systems, fuzzy logic.

Similar-day approach: search past data (usually 1-3 years) for days with **similar characteristics** to the day being forecast: same day of week, same season, similar weather (temperature, humidity) and similar date. The load curve of that similar day (or a weighted average of several similar days) is taken as the forecast, with small trend corrections. It is simple and works well for normal days.

Peak load approach vs energy approach (2021 3b)

	Peak load approach	**Energy approach**
What is forecast first	Annual peak demand (MW) directly, by extrapolating past peaks (trend)	Annual energy consumption (kWh) of each sector (domestic, industry, agriculture)
How peak is found	Directly from the trend	Peak = energy / (load factor × 8760)
Use	Simple, quick; short and medium-term	Long-term planning; links to economic growth of each sector
Weakness	Ignores causes of growth	Needs reliable load factor and sector data

> **Memory trick:** Factors: **"TWEDS-R"** = Time, Weather, Economy, Demography, Special events, Random.

A4 Effects of variable load on power plant design and operation Must study

Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2024 Q1(d)** [10] Explain the effect of variable load on a steam power plant with a schematic.
- **2023 Q2(a)** [9] What are the effects of variable load on power plant design and operation?
- **2022 Q2(a)** [9] What are the effects of variable load on power plant design and operation?
- **2021 Q2(a)** [9] What are the effects of variable load on power plant design and operation?
- **2020 Q1(b)** [10] What are the effects of variable load on power plant design and operation?

★ REMEMBER IT

- Variable load means: part-load (lower efficiency), extra reserve units, frequent start/stop (thermal stress), costly controls, more operators, and a higher cost per kWh (kilowatt-hour = 1 unit of electrical energy). Draw the load curve and say "the plant must follow this".

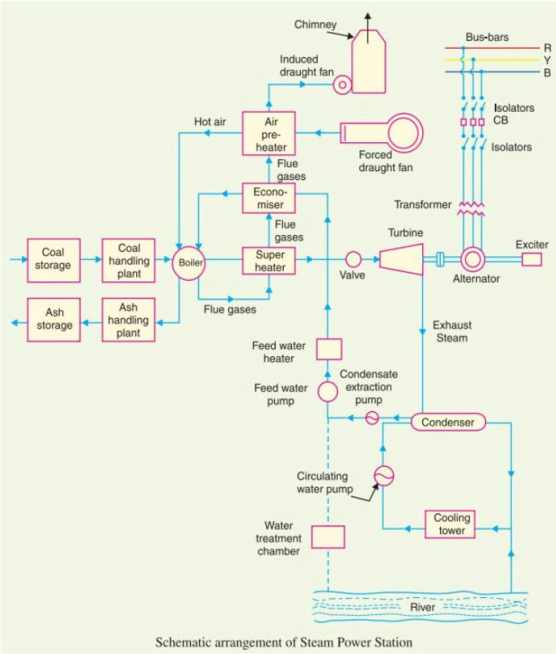
Before you start: why load variation matters

Machines are cheapest per kWh when they run steadily at about full rating. But the customers' load changes every hour (look at the Bangladesh load curve in A2). The plant must follow the load, and every effect below comes from that mismatch.

Effects of variable load on design and operation (asked every year)

1. **Need for additional equipment:** for a steam plant the supply of coal, air and water must change with load, so extra equipment (variable-speed fans, feed pumps, coal feeders, control system) is needed. Much of a modern plant exists only to follow load.
2. **Increase in production cost:** a single big machine would run at part load most of the day with poor efficiency, so several smaller units are installed so that each runs near full load. More units = more capital, more floor area, more cost per kW.
3. **Reserve (standby) capacity** must be installed above maximum demand to meet peaks and breakdowns; it earns nothing most of the time.
4. **Lower load factor** means fixed charges (interest, depreciation) are spread over fewer kWh, raising cost per unit.
5. **Part-load efficiency loss:** boilers, turbines and engines have lower efficiency at part load, so fuel per kWh rises.
6. **Peak load plants needed:** quick-start plants (gas turbine, diesel, pumped storage) are installed to take the peaks, while efficient base plants take the steady load.
7. **Operation and control problems:** load prediction is needed, governors must keep frequency (50 Hz) and voltage constant, units must be started and stopped many times, causing thermal stress and wear.
8. **Fuel and water storage** must be sized for peak consumption.

Effect on a steam plant with schematic (2024 1d)



Air, fuel and water are the raw materials for steam power. To produce power variably it follows that the supply of these materials will be required to be correspondingly varied.

Schematic of a steam power station: coal, air and water supply must all be varied when the load changes

Effects of variable load. The variable load on a power station introduces many perplexities i its operation. Some of the important effects of variable load on a power station are :

(*i*) Need of additional equipment. The variable load on a power station necessitates to hav additional equipment. By way of illustration, consider a steam power station. Air, coal an water are the raw materials for this plant. In order to produce variable power, the supply o these materials will be required to be varied correspondingly. For instance, if the powe demand on the plant increases, it must be followed by the increased flow of coal, air an water to the boiler in order to meet the increased demand. Therefore, additional equipmen has to be installed to accomplish this job. As a matter of fact, in a modern power plant, ther is much equipment devoted entirely to adjust the rates of supply of raw materials in accor dance with the power demand made on the plant.

(*ii*) Increase in production cost. The variable load on the plant increases the cost of the pro duction of electrical energy. An alternator operates at maximum efficiency near its rate capacity. If a single alternator is used, it will have poor efficiency during periods of ligh loads on the plant. Therefore, in actual practice, a number of alternators of different capaci ties are installed so that most of the alternators can be operated at nearly full load capacity However, the use of a number of generating units increases the initial cost per kW of th plant capacity as well as floor area required. This leads to the increase in production cost o energy.

Slide text: effects of variable load (need of additional equipment, increase in production cost)

In the schematic, when load rises the turbine needs more steam, so the boiler must get more coal (coal handling plant and feeders), more air (forced and induced draught fans), more feed water (feed pumps) and the condenser more cooling water. Each of these is fitted with variable-rate equipment and automatic control. If the load drops, all must be reduced together, otherwise pressure and temperature swing and efficiency falls.

> **Memory trick:** Remember **"MORE PEC"**: More equipment, Operating problems, Reserve capacity, Efficiency drop, Peak plants, Extra cost, Control.

A2 Load curve, load duration curve and the plant factors (load, demand, diversity, capacity, use) Must study Very likely · 93% Easy

PAST-PAPER QUESTIONS

- **2024 Q1(a)** [9] Define mathematically load factor, diversity factor and demand factor. Which of them are more than 1 and which less than 1?
- **2024 Q1(b)** [6] Distinguish between plant capacity factor and plant use factor.
- **2023 Q1(a)** [6] Distinguish between ideal load curve and realized load curve.
- **2022 Q1(b)** [6] Define: (i) plant use factor, (ii) demand factor.
- **2022 Q1(c)** [6] Distinguish between load curve and load duration curve.
- **2021 Q1(b)** [8] Describe a typical daily load curve of Bangladesh.
- **2021 Q1(c)** [9] Define: (i) plant capacity factor, (ii) diversity factor, (iii) demand factor.

LEARN IT FROM ZERO

Load curve and the plant factors, from zero

Plot the power the town takes from the station hour by hour: that is the **load curve**. Its highest point is the **maximum demand**; the area under it is the **energy** (MWh). Sort the same hourly loads from biggest to smallest and plot them: that is the **load duration curve** (LDC). Same area, but now you can read "for how many hours was the load above X MW".

Every factor is one ratio of three numbers: **A**verage load, **M**aximum demand, **I**nstalled capacity.

- Load factor = A / M (how flat the demand is; always ≤ 1)
- Capacity (plant) factor = A / I (how hard the plant is worked on average)
- Utilisation factor = M / I
- Plant use factor = energy / (I × hours it actually ran)
- Demand factor = M / connected load (≤ 1: not everything is switched on together)
- Diversity factor = sum of individual maximum demands / maximum demand of the whole system (**≥ 1**: people peak at different times)

★ REMEMBER IT

- **A-M-I**: LF (load factor) = A/M, CF (capacity factor) = A/I, UF (use (utilisation) factor) = M/I, so LF × UF = CF.
- Only the **diversity factor is more than 1** (D for "different times"). Load, demand, capacity factors are all less than 1.

Before you start: a load curve in one minute

Electricity cannot be stored in bulk, so a power station must produce exactly what customers use *at every moment*. The customers' total demand is the **load** (in kW or MW). It rises in the evening when lights and fans are on and falls at night.

If you plot load against time you get the **load curve**. The **area under it** (MW × hours) is the **energy** produced (MWh). Every factor below is just a ratio of two of these numbers: average load, maximum load, installed capacity, connected load.

Load curve and load duration curve (2022 1c)

Load Curve:

Load curve or chronological curve is the graphical representation of load (in kW or MW) in proper time sequence and the time in hours. It shows the variation of load on the power station.When the load curve is plotted for 24 hours a day, then it is called daily load curve. If the one year is considered then, it is called annual load curve.

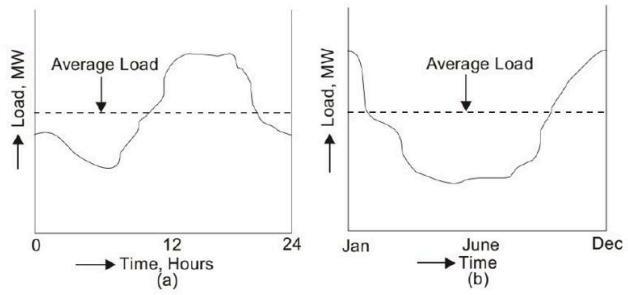
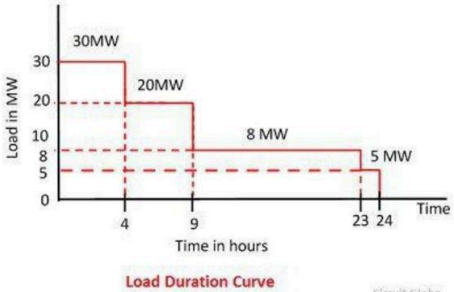


Fig. 3.6. Chronological Load Curves (*a*) Daily Load Curve (*b*) Yearly Load Curve.

Load curve (chronological): load plotted in time order over a day (a) or a year (b)

Load Duration Curve

The load duration curve is defined as the curve between the load and time in which the ordinates representing the load, plotted in the order of decreasing magnitude, i.e., with the greatest load at the left, lesser loads towards the rights and the lowest loads at the time extreme right. The load duration curve is shown in the figure below.



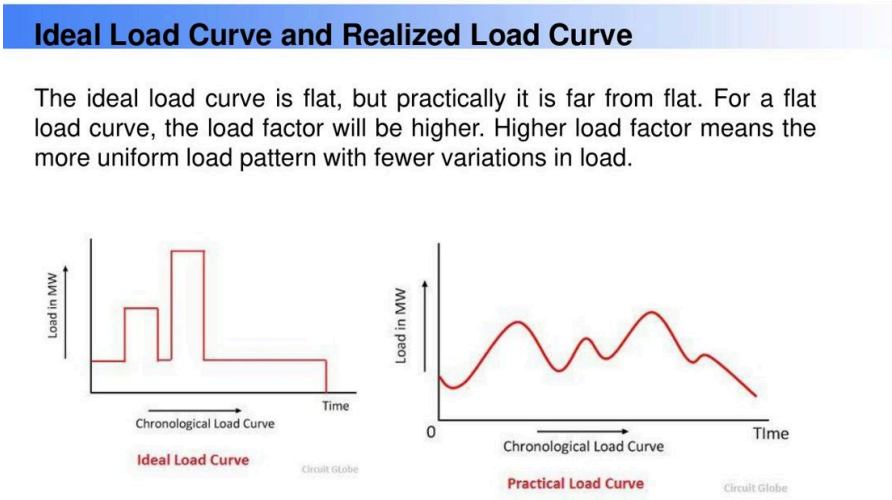
Load Duration Curve

This curve represents the same data as that of the load curve. The load duration curve is constructed by selecting the maximum peak points and connecting them by a curve.The load duration curve plotting for 24 hours of a day is called the daily load duration curve. Similarly, the load duration curve plotted for a year is called the annual load curve.

Load duration curve: the same loads re-arranged from highest at the left to lowest at the right

	Load curve	Load duration curve (LDC)
What is plotted	Load against time of day, in the actual time order	Same loads arranged in decreasing order against the duration (hours) for which each load lasts
Shape	Goes up and down (peaks in the evening)	Always falls from left to right
Shows	When the peak and off-peak occur; helps decide when to start/stop units	How many hours a year each load level is needed; helps choose base and peak plant sizes
Common to both	Area = energy generated; highest point = maximum demand	Same area and same maximum

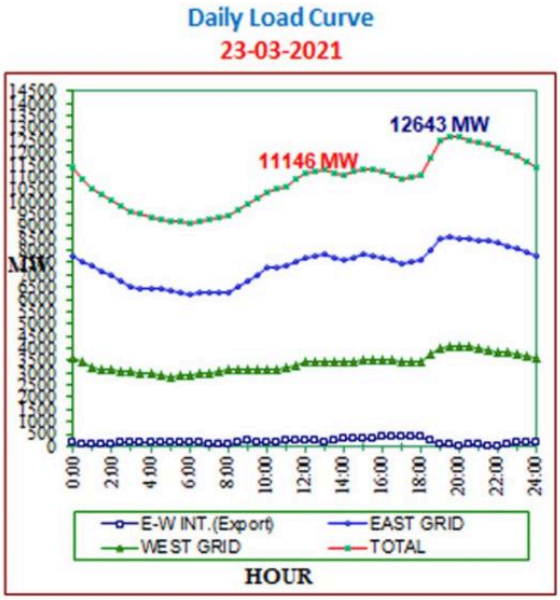
Ideal vs realized (practical) load curve (2023 1a)



Ideal load curve is flat; a realized load curve has peaks and valleys

The **ideal load curve** is **flat** (load the same all day): load factor = 1, the plant runs at its best efficiency all the time and no capacity is idle. The **realized (practical) load curve** is irregular because consumers use power at different times; it has peaks and valleys, a load factor below 1, and needs extra installed capacity for the peaks. Engineers try to flatten the real curve (off-peak tariffs, pumped storage) to approach the ideal one.

Typical daily load curve of Bangladesh (2021 1b)



Daily load curve of Bangladesh (23-03-2021): minimum after midnight to morning, rises through the day, evening peak around 18:00-21:00 (about 12,600 MW)

The national curve is lowest from about 01:00 to 07:00 (sleeping hours), rises from morning as offices, factories and irrigation pumps start, stays high in the afternoon (cooling load in summer), and reaches its **evening peak between about 18:00 and 21:00** when domestic lighting, fans and commercial loads add on top of industry. After 22:00 it falls again. The east grid carries most of the load. The evening peak is met by gas turbines, engines and peaking plants; the night valley is the base load.

The factors: definitions with formulas (2024 1a, 2022 1b, 2021 1c)

Step (write this)	Why / what it means
`Load factor = average load / maximum demand`	Same period (day, month, year). Always ≤ 1. High LF = flatter load = cheaper energy.
`Demand factor = maximum demand / connected load`	Connected load = sum of the ratings of everything plugged in. Not everything runs at full power at once, so ≤ 1.
`Diversity factor = sum of individual maximum demands / maximum demand of the whole system`	Consumers peak at different times, so the system peak is smaller than the sum: always ≥ 1.
`Plant capacity factor = average demand / plant (installed) capacity = actual energy / (capacity × total hours)`	How much of the installed plant is really used over the whole period. ≤ 1.
`Plant use factor = actual energy / (capacity × hours the plant actually ran)`	Like capacity factor, but counts only the hours in service. ≤ 1.
`Utilisation factor = maximum demand / rated capacity`	How close the peak came to the plant rating. ≤ 1.
`Reserve capacity = installed capacity - maximum demand`	Spare kept for breakdowns and growth.

(vi) Diversity factor. *The ratio of the sum of individual maximum demands to the maximum demand on power station is known as* **diversity factor** *i.e.,*

$$\text{Diversity factor} = \frac{\text{Sum of individual max. demands}}{\text{Max. demand on power station}}$$

A power station supplies load to various types of consumers whose maximum demands generally do not occur at the same time. Therefore, the maximum demand on the power station is always less than the sum of individual maximum demands of the consumers. Obviously, diversity† factor will always be greater than 1. The greater the diversity factor, the lesser‡ is the cost of generation of power.

(vii) Plant capacity factor. *It is the ratio of actual energy produced to the maximum possible energy that could have been produced during a given period i.e.,*

$$\text{Plant capacity factor} = \frac{\text{Actual energy produced}}{\text{Max. energy that could have been produced}}$$

$$= \frac{\text{Average demand} \times \text{T}^{**}}{\text{Plant capacity} \times \text{T}}$$

$$= \frac{\text{Average demand}}{\text{Plant capacity}}$$

Diversity factor and plant capacity factor (slide)

(viii) Plant use factor. *It is ratio of kWh generated to the product of plant capacity and the number of hours for which the plant was in operation i.e.*

$$\text{Plant use factor} = \frac{\text{Station output in kWh}}{\text{Plant capacity} \times \text{Hours of use}}$$

Suppose a plant having installed capacity of 20 MW produces annual output of $7{\cdot}35 \times 10^6$ kWh and remains in operation for 2190 hours in a year. Then,

$$\text{Plant use factor} = \frac{7{\cdot}35 \times 10^6}{(20 \times 10^3) \times 2190} = 0{\cdot}167 = 16{\cdot}7\%$$

$$\text{Reserve factor,} \qquad r = \frac{\text{kW}_{\text{inst}}}{\text{kW}_{\text{max}}} = \frac{m}{n} = \frac{\text{load factor}}{\text{capacity factor}}$$

Plant use factor with example, and reserve factor = load factor / capacity factor

Which are more than 1 and which less (2024 1a)? Diversity factor **> 1** (in the limit = 1 if all consumers peak together). Load factor, demand factor, capacity factor, use factor and utilisation factor are all **< 1** (they can only reach 1 in the ideal case).

Plant capacity factor vs plant use factor (2024 1b)

	Plant capacity factor	Plant use factor
Formula	$E / (C \times T_{total})$	$E / (C \times T_{in\ operation})$
Hours counted	All hours in the period (8760 h in a year), including when the plant was shut down	Only the hours the plant was actually running
Tells you	How well the investment is used over the year	How well the plant is loaded while it is running
Relation	Capacity factor ≤ use factor (equal only if the plant ran all year)	

Side note: Example from the slide: a 20 MW plant makes 7.35×10^6 kWh (kilowatt-hour = 1 unit of electrical energy) in 2190 hours of running. Use factor = $7.35 \times 10^6 / (20{,}000 \times 2190) = 16.8$ %. Capacity factor over the year = $7.35 \times 10^6 / (20{,}000 \times 8760) = 4.2$ %.

Memory trick: "Diversity Divides": diversity is the only factor where the big sum sits on top, so it is the only one > 1. For the others the bigger number is at the bottom.

A9 Tariff: objectives, requirements and types (flat, block, two-part, maximum demand, ToU) Must study Very likely · 93% Easy

PAST-PAPER QUESTIONS

- **2024 Q3(a)** [9] What are the objectives and requirements of a tariff?
- **2023 Q2(c)** [8] What is meant by meter rate? Compare straight line meter rate and step meter rate tariffs.
- **2022 Q2(d)** [8] What is tariff? What are its different types?
- **2021 Q3(c)** [12] What is meant by tariff? Write the objectives and requirements of a tariff.

★ REMEMBER IT

- Tariff types in order of fairness: flat demand → simple (flat rate) → block (step) rate → two-part (Tk/kW + Tk/kWh) → maximum demand → time-of-use. Objectives: recover cost, fair, simple, encourage off-peak use.

Tariff: meaning, objectives and requirements (2024 3a, 2021 3c)

A **tariff** is the rate (method) by which the cost of electrical energy is charged to consumers. It must recover the supplier's cost of generation, transmission and distribution plus a reasonable profit.

Objectives of a tariff (what it must recover):

1. Recovery of the **cost of producing** electricity at the station.
2. Recovery of the **cost on capital investment** in transmission and distribution.
3. Recovery of **operating and maintenance cost** of supply.
4. A suitable **profit** on the capital invested.

Desirable characteristics (requirements):

1. **Proper return:** must give the supplier a fair return on investment.
2. **Fairness:** each consumer pays according to the cost of supplying him (large consumers can get a lower rate per kWh (kilowatt-hour = 1 unit of electrical energy)).
3. **Simplicity:** easy for consumers to understand.
4. **Reasonable profit** only (public utility).
5. **Attractive:** encourage consumers to use more energy, especially in off-peak hours, to improve load factor.
6. **Encourage high power factor** (penalty for low pf).

Types of tariff (2022 2d, 2023 2c)

Step (write this)	Why / what it means
`Simple (uniform) rate: Y = EZ`	Y = total bill, Z = kWh used, E = Tk per kWh. Same price for every unit.
`Flat rate: different fixed price per kWh for each type of consumer (lighting vs power)`	Needs separate meters.
`Flat demand rate: Y = DX`	Charge on number of lamps / kW connected, no meter. X = connected load.

Straight line meter rate: $Y = EZ$	Bill directly proportional to units consumed (graph is a straight line through origin).
Step meter rate: $Y = EZ$ if $Z \le A$; $Y = E_1Z$ if $A < Z \le B$ ...	All units are charged at the rate of the step the total falls into; rate falls as consumption rises.
Block rate: $Y = E_1Z_1 + E_2Z_2 + \ldots$	First block of units at one price, next block at another (Bangladesh domestic tariff uses rising blocks).
Two-part (Hopkinson): $Y = DX + EZ$	D Tk per kW of maximum demand + E Tk per kWh. Needs a maximum demand meter; for industry.
Three-part (Doherty): $Y = DX + EZ + C$	Adds a fixed charge C per consumer (meter, billing).
Maximum demand tariff, power factor tariff, time-of-day (ToU) tariff	Peak hours priced higher to shift load.

(1) Flat Demand Rate. It is based on the number of lamps installed and a fixed number of hours of use per month or per year. The rate is expressed as a certain price per lamp or per unit of demand (kW) of the consumer. This energy rate eliminates the use of metering equipment. It is expressed by the expression.

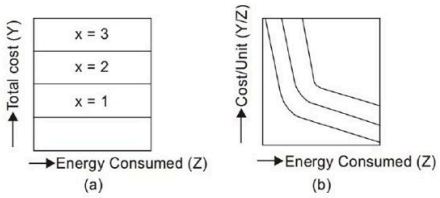


Fig. 3.1

(2) Straight Line Meter Rate. According to this energy rate the amount to be charged from the consumer depends upon the energy consumed in kWh which is recorded by a means of a kilowatt hour meter. It is expressed in the form

$$Y = EZ$$

This rate suffers from a drawback that a consumer using no energy will not pay any amount although he has incurred some expense to the power station due to its readiness to serve him. Secondly since the rate per kWh is fixed, this tariff does not encourage the consumer to use more power.

Flat demand rate and straight line meter rate (slide)

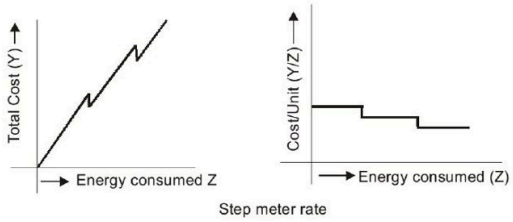
(3) Step Meter Rate. According to this tariff the charge for energy consumption goes down as the energy consumption becomes more. This tariff is expressed as follows.

$Y = EZ$ If $0 \le Z \le A$

$Y = E_1Z_1$ If $A \le Z_1 \le B$

$Y = E_2Z_2$ If $B \le Z_2 \le C$

And so on. Where E, E_1, E_2 are the energy rate per kWh and A, B and C, are the limits of energy consumption.



Step meter rate

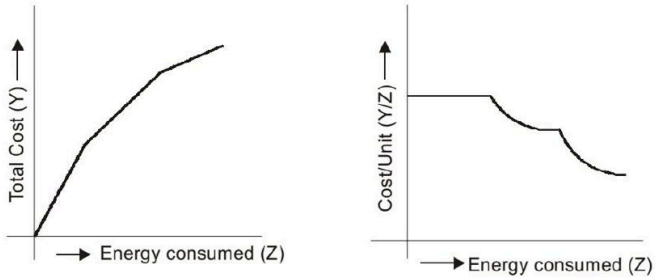
Step meter rate: cost per unit falls in steps

(4) Block Rate Tariff. According to this tariff a certain price per units (kWh) is charged for all or any part of block of each unit and for succeeding blocks of energy the corresponding unit charges decrease.

It is expressed by the expression

$$Y = E_1Z_1 + E_2Z_2 + E_3Z_3 + E_4Z_4 + ...$$

where E_1, E_2, E_3.... are unit energy charges for energy blocks of magnitude Z_1, Z_2, Z_3,.... respectively.



Block rate tariff

(5) Two Part Tariff (Hopkinson Demand Rate). In this tariff the total charges are based on the maximum demand and energy consumed. It is expressed as

$$Y = D \cdot X + EZ$$

A separate meter is required to record the maximum demand. This tariff is used for industrial loads.

(6) Three-Part Tariff (Doherty Rate). According to this tariff the customer pays some fixed amount in addition to the charges for maximum demand and energy consumed. The fixed amount to be charged depends upon the occasional increase in fuel price, rise in wages of labour etc. It is expressed by the expression

$$Y = DX + EZ + C.$$

Two-part and three-part tariffs

Straight line meter rate vs step meter rate (2023 2c)

	Straight line meter rate	Step meter rate
Formula	Y = EZ (one rate)	Y = EZ, E_1Z, E_2Z ... rate depends on the step the total consumption falls in
Rate per kWh	Constant	Falls as consumption increases ($E > E_1 > E_2$)
Graph of total bill	Straight line through origin	Saw-tooth line with drops at each step
Drawback	A consumer who uses nothing pays nothing, though supplier has fixed cost; does not encourage more use	Consumer may waste energy just to reach a lower step; total bill can drop when usage rises

Memory trick: Tariff general equation **Y = DX + EZ + C**: **D**emand charge × kW, **E**nergy charge × kWh, **C**onstant per customer. Every tariff is this equation with some terms missing.

A7 Cost of generation, fixed and operating costs, depreciation Must study

Very likely · 87% Easy

PAST-PAPER QUESTIONS

- **2024 Q2(b)** [10] Explain fixed cost and operating cost of a power plant.
- **2023 Q2(b)** [8] What is meant by depreciation of a power plant? What factors influence depreciation?
- **2022 Q2(c)** [12] Describe the costs involved in the generation of electrical energy.
- **2020 Q2(a)** [8] What is meant by depreciation? What factors influence depreciation?

Costs involved in generating electrical energy (2022 2c, 2024 2b)

Fixed cost (independent of kWh (kilowatt-hour = 1 unit of electrical energy)**)**	**Operating / running cost (depends on kWh)**
Interest on capital invested	Cost of fuel (the largest running cost)
Depreciation (loss of value of plant)	Lubricating oil, water, chemicals
Insurance, taxes, rent of land	Wages of operating staff tied to output
Salaries of management and permanent staff	Maintenance and repairs
Cost of land, buildings, plant and its installation (capital cost)	Supplies and stores used during operation

Total annual cost = fixed cost + operating cost. Fixed cost per kWh falls when the plant generates more (high load factor); operating cost per kWh is almost constant. Some books split into three: **fixed cost** (capital charges), **semi-fixed cost** (depends on maximum demand, e.g. size of plant) and **running cost** (depends on kWh). This is why the tariff has the form Y = DX + EZ + C (see A9).

Depreciation and factors affecting it (2023 2b, 2020 2a)

Depreciation is the **reduction in the value of plant and equipment with time**. Each year a sum is set aside so that, at the end of the useful life, enough money exists to replace the plant.

Factors that cause/influence depreciation:

1. **Wear and tear** from continuous use (physical deterioration).
2. **Ageing / deterioration** by weather, corrosion, erosion, even when not used.
3. **Obsolescence**: newer, more efficient technology makes the plant out of date.
4. **Inadequacy**: the plant becomes too small for the grown load.
5. **Accidents** and sudden failures.
6. **Useful life** assumed, **salvage (scrap) value** and **rate of interest**.

Methods of calculating depreciation:

Step (write this)	**Why / what it means**
`Straight line: D = (P - S)/n per year`	P = capital cost, S = salvage value, n = useful life in years. Same amount every year; simplest.
`Diminishing value: each year depreciation = fixed % of the value left at the start of that year`	Large in early years, small later; matches real resale value.

Sinking fund: $q = (P - S) \times r / [(1 + r)^n - 1]$	A fixed sum q is invested each year at interest r so that the fund grows to (P − S) after n years. Most used for power plants.

Memory trick: Depreciation factors **"WADOI"**: Wear, Age, Deterioration, Obsolescence, Inadequacy.

A5 Site selection for a power plant and the interconnected grid Must study

Very likely · 80% Easy

PAST-PAPER QUESTIONS

- **2024 Q1(c)** [10] What factors are considered when selecting the site of a power plant?
- **2023 Q1(d)** [10] What are the advantages of an interconnected grid system?
- **2021 Q2(b)** [8] What factors should be considered while selecting the site for a power plant?
- **2021 Q2(c)** [10] Describe the benefits of an interconnected grid system.
- **2020 Q1(c)** [8] What factors should be considered while selecting the site for a power plant?

Site selection for a power plant (2024 1c, 2021 2b, 2020 1c)

1. **Availability of fuel/energy source:** near coal mines, gas field or port for imported coal; hydro at a good head and flow.
2. **Water supply:** steam plants need huge amounts for condenser cooling and boiler feed, so they are put beside rivers or the sea.
3. **Distance from load centre:** close to load to reduce transmission cost and losses.
4. **Land:** cheap, enough area for the plant, fuel storage and future extension; firm ground able to carry heavy foundations.
5. **Transport facilities:** rail, road or river access for fuel and heavy machinery.
6. **Ash and waste disposal** space (coal plants).
7. **Environment and public health:** away from crowded towns because of smoke, noise, heat; wind direction considered.
8. **Labour** availability, **water drainage**, **climate**, security, and **future expansion**.

Interconnected grid and its advantages (2023 1d, 2021 2c)

An **interconnected grid** links many power stations and load centres by high-voltage transmission lines, so that all stations feed a common network and share the total load.

1. **Reduced reserve capacity:** one reserve serves the whole system instead of one per station.
2. **Economical operation:** the most efficient, cheapest plants run as base load; costly ones only at peaks.
3. **Greater reliability:** if one station trips, others supply its load, so fewer blackouts.
4. **Use of diversity:** regions peak at different times, so the combined peak is smaller and less capacity is needed overall.
5. **Larger, more efficient units** can be installed because the grid can absorb their output.
6. **Exchange of surplus power** between regions (and countries, e.g. import from India via Bheramara).
7. **Better use of hydro** in rainy season and thermal in dry season; easier maintenance scheduling.

Side note: Disadvantages (for a full answer): a fault can spread through the grid (cascading blackout, Bangladesh 2014 and 2022), need for complex protection and control, high cost of transmission lines.

Memory trick: Site: **"FW-LLT-ADE"** = Fuel, Water, Load centre, Land, Transport, Ash, Disposal/Environment, Expansion.

A12 Hydro-electric plant: layout, essential features, dam, spillway, surge tank, forebay, hydrology terms, classification

Must study · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

- **2024 Q4(a)** [9] Write the essential features of a hydro plant. What are the functions of surge tank and forebay?
- **2024 Q4(b)** [8] What is the function of a dam? Describe the various types of dam.
- **2023 Q4(a)** [9] Define: (i) surge tank, (ii) spillway, (iii) hydrograph.
- **2023 Q4(b)** [8] Classify hydro-electric power plants.
- **2022 Q4(a)** [10] Describe the working principle of a hydro-electric plant with a neat sketch. Write the functions of surge tank and forebay.
- **2022 Q4(b)** [9] Define: (i) run-off, (ii) hydrograph, (iii) specific speed.
- **2021 Q4(a)** [10] Describe the essential features of a hydro-electric power plant with a neat sketch.
- **2021 Q4(c)** [4] Define hydrograph and mass curve.
- **2020 Q3(b)** [6] Why is hydropower called a renewable source of energy?

★ REMEMBER IT

- Water's path: **reservoir → dam → intake/forebay → penstock → turbine → draft tube → tailrace**. Surge tank sits on the penstock and absorbs pressure surges (water hammer) when the load is suddenly cut. Spillway lets flood water pass safely.

Before you start: how a hydro plant makes power

Water held high behind a dam has **potential energy** = m g H. When it falls through a pipe (penstock) it speeds up and hits a turbine, which turns a generator. Power P = ρ g Q H η, where Q = flow (m^3/s), H = head (height of fall, m), η = efficiency.

So hydro needs two things: **lots of water (Q)** and **a big height (H)**.

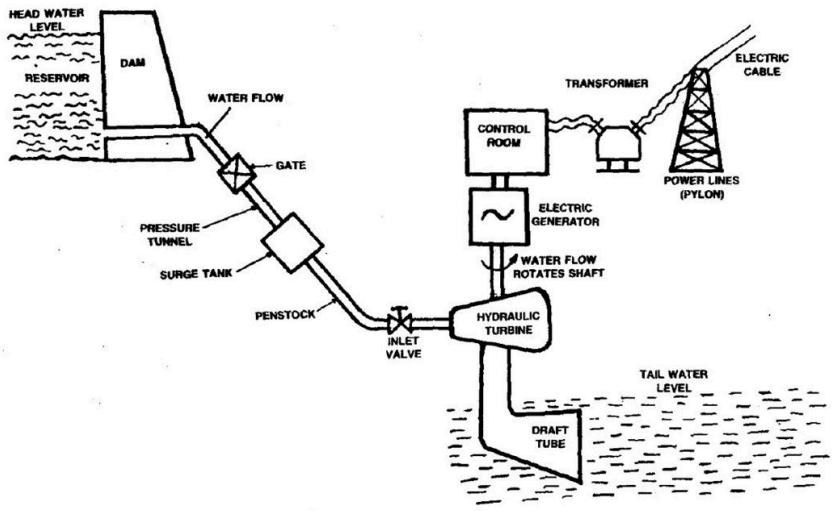


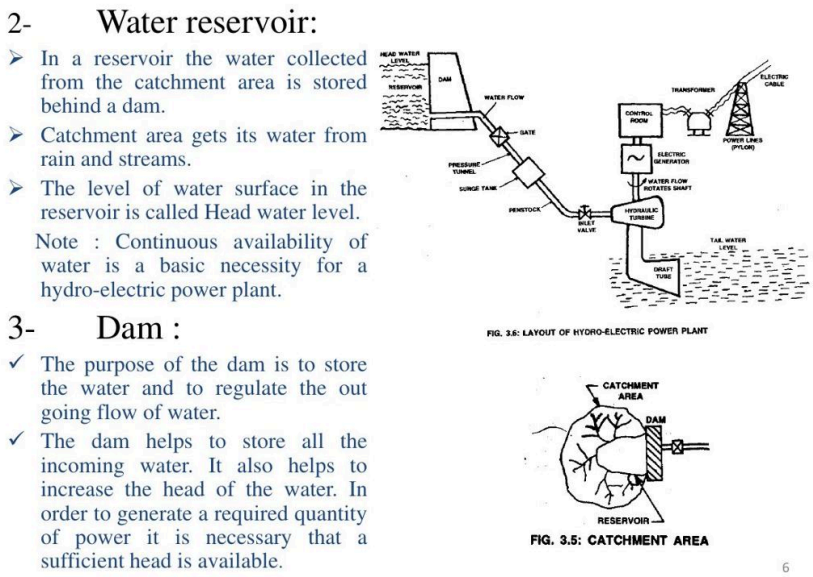
FIG. 3.6: LAYOUT OF HYDRO-ELECTRIC POWER PLANT

Layout of a hydro-electric plant: reservoir, dam, gate, pressure tunnel, surge tank, penstock, turbine, generator, draft tube, tail race

Essential features / components and working (2024 4a, 2022 4a, 2021 4a)

Step (write this)	Why / what it means
1. Catchment area	Whole area behind the dam that drains rain water into the river/reservoir.
2. Reservoir	Stores water collected from the catchment; its surface is the head water level.
3. Dam (+ spillway)	Stores water, raises the head; spillway discharges surplus water safely in floods.
4. Water ways: intake + gate, pressure tunnel, surge tank, forebay, penstock	Carry water from reservoir to turbine. Penstock = steel pipe from surge tank/forebay to turbine.
5. Power house	Contains turbines, generators, governors, valves, transformers, switchgear, cranes.
6. Draft tube and tail race	Draft tube recovers leftover kinetic energy and lets the turbine sit above tail water; tail race carries water back to the river.

Working: water from the reservoir passes through the intake gate and pressure tunnel to the surge tank, then down the penstock to the turbine. The pressure (potential) energy becomes kinetic energy that turns the turbine runner; the coupled generator produces electricity. Water leaves through the draft tube to the tail race. Governors adjust the flow to keep speed constant as load changes.



2- Water reservoir:

- In a reservoir the water collected from the catchment area is stored behind a dam.
- Catchment area gets its water from rain and streams.
- The level of water surface in the reservoir is called Head water level.

Note : Continuous availability of water is a basic necessity for a hydro-electric power plant.

3- Dam :

- ✓ The purpose of the dam is to store the water and to regulate the out going flow of water.
- ✓ The dam helps to store all the incoming water. It also helps to increase the head of the water. In order to generate a required quantity of power it is necessary that a sufficient head is available.

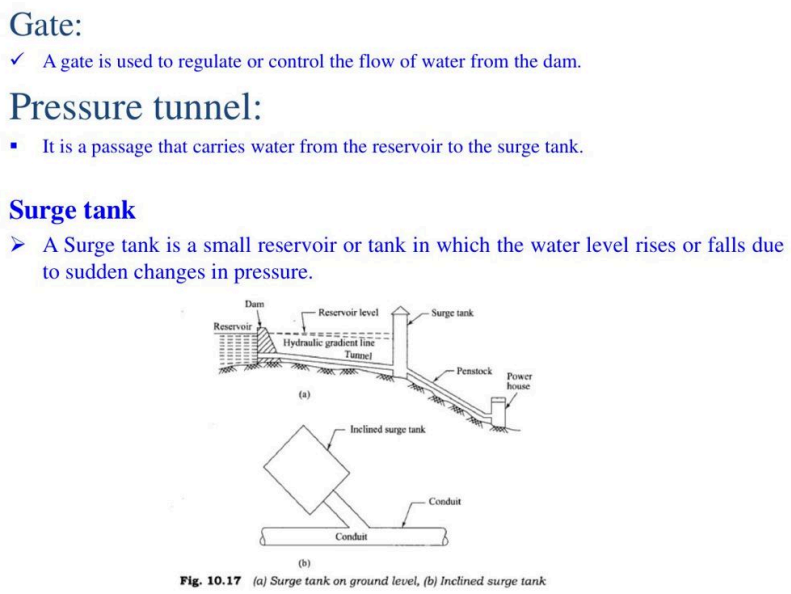
FIG. 3.6: LAYOUT OF HYDRO-ELECTRIC POWER PLANT

FIG. 3.5: CATCHMENT AREA

6

Reservoir and catchment area

Surge tank and forebay (2024 4a, 2022 4a, 2023 4a)



Gate, pressure tunnel and surge tank

Surge tank: a small open tank/reservoir placed between the pressure tunnel and the penstock, close to the power house, in which the water level rises or falls with sudden changes of load. Functions:

- When load drops and the turbine gates close quickly, the moving water rushes up the surge tank instead of hammering the pipe: it **absorbs water hammer** (pressure surge) and protects the tunnel/penstock.
- When load rises suddenly, the surge tank **supplies extra water** to the turbine until the flow in the long tunnel speeds up.
- It shortens the length of pipe that feels water hammer, so the tunnel can be thinner/cheaper.

> **Side note:** Water hammer = a sudden rise in pipe pressure when flowing water is stopped quickly, like the bang you hear when a tap is shut fast.

Forebay: an enlarged basin/pond at the head of the penstocks (at the end of an open canal) that temporarily stores water. It **distributes water to the penstocks**, acts as a small reservoir that meets sudden load increases, absorbs flow fluctuations when load falls, and lets trash racks remove floating debris. It works as a surge tank for canal-fed plants.

Dam: function and types (2024 4b)

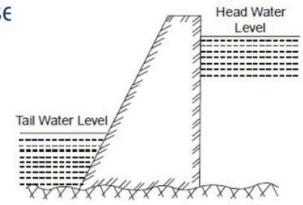
Function: stores water to supply a steady flow through dry periods, **creates/raises the head**, regulates outflow, and controls floods.

Types of Dam:

1. Masonry Dams.
2. Earth Dams.

The masonry dams are of three major classe

a) Gravity dam.
b) Buttress dam.
c) Arched dam.



d) Gravity dam:

Resist the pressure of water by its weight.

Construction of material used for his dam, is solid masonry or concrete.

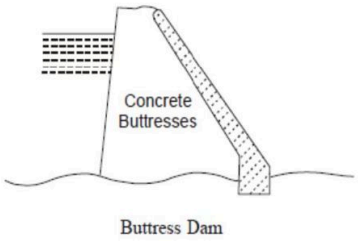
Types of dam: masonry (gravity, buttress, arch) and earth dams; gravity dam resists water by its own weight

b) Arch dam:

It resist the pressure of water partly due to its weight and partly due to arch action.

c) Buttress dam:

- Buttress supporting a flat slab.
- When cost of reinforced concrete is high such type of dam is selected.

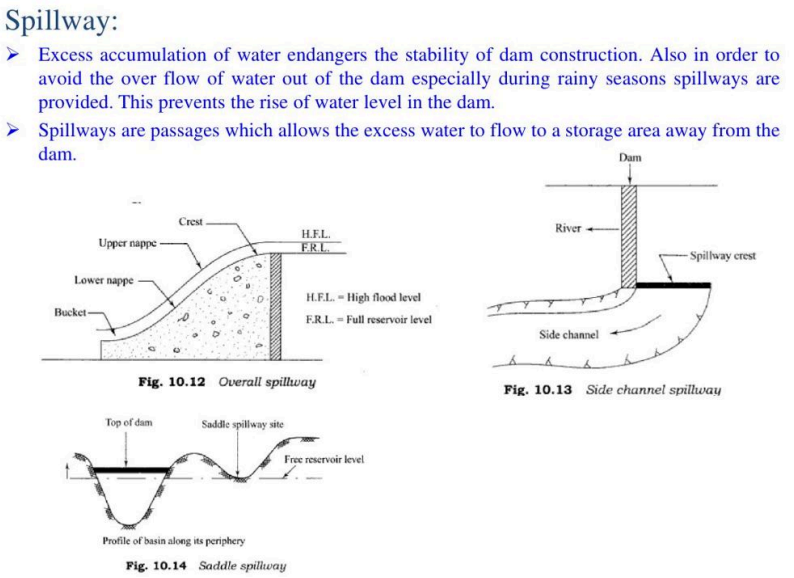


Arch dam (weight + arch action) and buttress dam (slab supported on buttresses)

Type	How it resists water pressure	Where used
Gravity dam	Its own huge weight (solid masonry/concrete)	Any good rock foundation; most common
Arch dam	Partly weight, mostly arch action that carries thrust to the valley sides	Narrow rocky gorges
Buttress dam	Inclined flat slab supported by a series of buttresses	Where concrete is costly; less material
Earth / rock-fill dam	Mass of compacted earth/rock with impervious core	Wide valleys, weak foundation (Kaptai is an earth-fill dam)

Dams are also classified by function (storage, diversion, detention), shape (trapezoidal, arch), material (earth, rock, masonry, concrete) and design (gravity, arch, buttress).

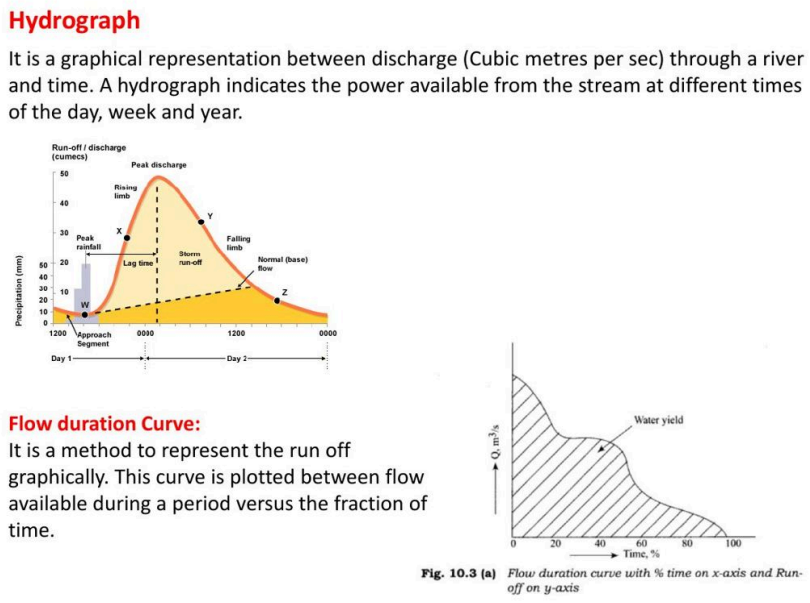
Spillway (2023 4a)



Overfall (ogee) spillway, side channel spillway and saddle spillway

A **spillway** is a passage that safely releases **excess (flood) water** from the reservoir downstream, so the water level never rises over the top of the dam and the dam stays stable in heavy rain. Types: overfall (ogee), chute, side channel, siphon, shaft, saddle spillways.

Hydrology terms (2023 4a, 2022 4b, 2021 4c)



Hydrograph (discharge vs time) and flow duration curve (discharge vs % of time)

Mass curve

A mass cure is a plot of accumulated flow against time, made from the records of mean monthly flows of a stream.

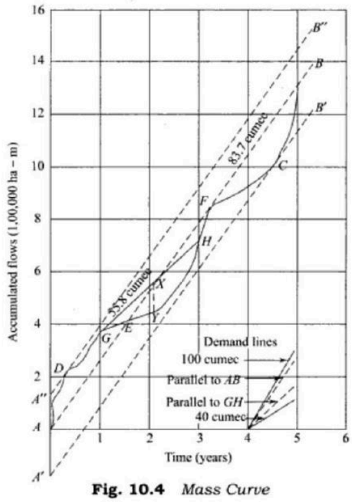


Fig. 10.4 *Mass Curve*

Mass curve: accumulated flow volume vs time; slope = flow rate

Step (write this)	Why / what it means
`Run-off`	Part of rainfall (precipitation) that flows over/under the ground into streams, rivers and lakes; = rain – evaporation – infiltration losses.
`Hydrograph`	Graph of river discharge (m^3/s) against time (day, week, month, year). Shows the power available at different times.
`Flow duration curve`	Discharge against the % of time that flow is equalled or exceeded. Like an LDC (load duration curve) for water.
`Mass curve`	Accumulated (cumulative) flow volume against time; its slope is the flow. Used to size the reservoir storage.
`Specific speed` $N_s = NVP / H^{5/4}$	Speed of a geometrically similar turbine that develops 1 kW (or 1 hp) under 1 m head. Classifies the turbine type (see A13).

Classification of hydro plants (2023 4b)

Classification of hydro-Electric power plant

The classification of hydro electric power plant depend on the following factors:

1) Quantity of water:

It is following types.

i. Run of river plant.

ii. Storage plant.

iii. Pumped storage.

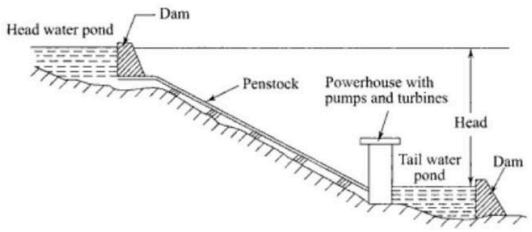


Fig. 10.22 *Pumped storage plant*

2) Availability of Head of Water:

a) Low head plant. Operating head < 15m.

b) Medium head plant. Operating head 15 to 50m.

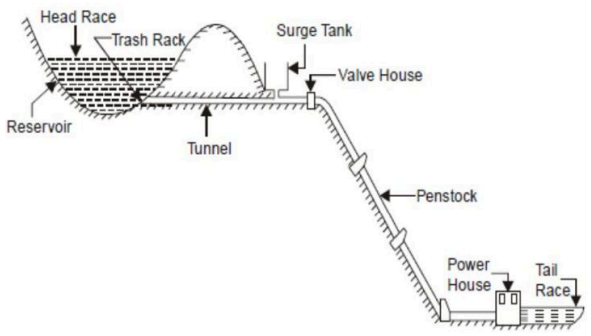
c) High head plants Operating head > 50m.

Classification by quantity of water (run-of-river, storage, pumped storage) and by head

Basis	Types
Head	Low head (< 15 m, Kaplan), medium head (15-50 m, Francis, with forebay), high head (> 50 m, Pelton, with surge tank)
Quantity of water	Run-of-river (no storage or small pondage), storage (reservoir) plant, pumped storage plant
Load supplied	Base load plant, peak load plant
Capacity	Micro (up to 5 MW), medium (5-100 MW), high (101-1000 MW), super (> 1000 MW)
Topography	Low land, hilly area, mountainous region
Turbine	High specific speed, medium/low specific speed

a) High head plant

- Operating head exceed 50m.
- Pelton turbines.
- surge tank is attached to the penstock to reduce water hammer effect on the penstock.



High head plant with tunnel, surge tank and penstock

Why hydropower is renewable (2020 3b)

Hydropower uses water that is continuously replenished by the **hydrologic cycle**: the sun evaporates water from the sea, it falls as rain, collects in rivers and reservoirs, passes through the turbine **without being consumed or polluted**, and returns to the sea to be evaporated again. The source therefore never runs out (unlike coal or gas), no fuel is burnt, and no CO_2 is emitted in operation.

> **Memory trick:** Water's path: **"Can Really Do Water Power Delivery Today"** = Catchment, Reservoir, Dam, Water ways, Power house, Draft tube, Tail race.

A14 Numericals: hydro power from catchment, rainfall and head

Must study

Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2024 Q4(d)** [10] Reservoir 250 m above the turbine, annual replenishment 70 × 10^{11} kg, head lost 30 m, overall efficiency 85 %. Find the energy available at the bus-bars per year.
- **2023 Q4(d)** [11] Catchment 800 km², rainfall 185 cm/year, head 35 m, infiltration + evaporation loss 42 %, turbine efficiency 84 %, generator efficiency 85 %, load factor 0.5. Find power developed and suggest turbines.
- **2022 Q4(d)** [11] Rainfall 800 mm, catchment 150 km², head 200 m, yield factor 60 %, overall efficiency 75 %, annual load factor 45 %. Find (i) average power, (ii) plant capacity.
- **2021 Q4(d)** [15] Catchment 14 × 10^8 m², rainfall 125 cm/year, 70 % available, head 40 m, 5 % head lost in penstock, turbine efficiency 85 %, generator 90 %, annual load factor 45 %. Find the maximum plant capacity.
- **2020 Q3(d)** [15] Catchment 800 km², rainfall 180 cm/year, head 32 m, loss 40 %, turbine efficiency 85 %, generator 90 %, load factor 0.5. Find power developed and suggest turbines.

LEARN IT FROM ZERO

Hydro power from rainfall, from zero

Rain falls on the catchment (the land that drains into the river). Some soaks in or evaporates; the rest flows to the dam. Spread that yearly volume over the seconds in a year and you get the average **flow Q** (m³/s). Each kg of water dropping through height **H** gives gH joules. So power = (mass per second) × g × H × efficiency:

$$\mathbf{P = \rho g Q H \eta = 9.81 \times Q \times H \times \eta\ kW} \text{ (with } \rho = 1000 \text{ kg/m}^3\text{)}$$

That is the **average** power. The installed capacity must cover the peak: **capacity = average power / load factor**. Then pick the turbine from the head.

Symbols used here

Symbol	Name and meaning	Unit
A	Catchment area (1 km² = 10^6 m²)	m²
r	Annual rainfall (cm ÷ 100 = m)	m
k	Fraction available (run-off / yield factor) = 1 − losses	–
Q	Average flow = A r k / 31.536 × 10^6	m³/s
H	Net head (gross minus losses)	m
η	Overall efficiency = $\eta_{turbine} \times \eta_{generator}$	–
LF (load factor)	Load factor	–

CALCULATOR **Hydro plant from catchment and rainfall**

Change any number; every step is recalculated and shown with the numbers put in.

1. Water available per year: **800** × 10^6 × **185** / 100 × (1 - **42** / 100) = **8.584 × 10^8** m^3
2. Average flow: Q = V / (365 × 24 × 3600) = **8.584 × 10^8** / 31.536×10^6 = **27.22** m^3/s
3. Net head: **35** × (1 - **0** / 100) = **35.00** m
4. Average power: P = 9.81 Q H η_t η_g = 9.81 × **27.22** × **35** × **84** / 100 × **85** / 100 = **6673** kW
5. Installed capacity: Capacity = P / LF = **6673** / **0.5** = **13346** kW

Head under 50 m → Kaplan; 50–250 m → Francis; above 250 m → Pelton.

Before you start: the only formula you need

P = ρ g Q H η watts. With ρ = 1000 kg/m^3 for water: **P (kW) = 9.81 × Q × H × η**.

Q (m^3/s) = (catchment area in m^2 × rainfall in m × fraction available) / seconds in a year (365 × 24 × 3600 = 31.536 × 10^6 s).

Units: 1 km^2 = 10^6 m^2; 185 cm = 1.85 m; 800 mm = 0.8 m. η overall = $\eta_{turbine}$ × $\eta_{generator}$ (× $\eta_{penstock}$ if given).

Plant capacity (installed) = average power / load factor.

Example 10.18 The following data refers to a proposed hydroelectric power plant:

Available head = 27 m, Catchment area 430 sq. km, Rainfall = 150 cm/year, Percentage of total rainfall utilized = 65%, Penstock efficiency = 95%, Turbine efficiency = 80%, Generator efficiency = 86% and Load factor = 0.45.

(a) Calculate the power developed.

(b) Suggest suitable turbines for the plant.

Solution

Quantity of water available per year

$$= (430 \times 10^6)\ m^2 \times 1.50\ m \times 0.65$$
$$= 419.25 \times 10^6\ m^3$$

Quantity of water available per second

$$Q = \frac{419.25 \times 10^6}{365 \times 24 \times 3600} = 13.29\ m^3$$

Power developed

$$P = h_p \times h_t \times \eta_G\ \rho Q g H$$
$$= 0.95 \times 0.8 \times 0.86 \times 1000 \times 13.29 \times 9.81 \times 27 \times 10^{-3}$$
$$= 2300\ kW \qquad \textit{Ans.}\ (a)$$

$$\text{Load factor} = \frac{\text{Average load}}{\text{Peak load}} = 0.45$$

$$\therefore \quad \text{Peak load capacity} = \frac{2300}{0.45} = 5111\ kW$$

If two machines of equal capacity are provided,

$$\text{Capacity of each unit} = \frac{5111}{2 \times 0.86} = 2971.5\ kW$$

As the available head is low, Kaplan turbines are suggested. Two such turbines, each of 3000 kW capacity, may be installed.

Slide example 10.18: same steps for 430 km^2, 150 cm, 27 m

2023 4(d): 800 km^2, 185 cm, 35 m, loss 42 %, η_t 84 %, η_g 85 %, LF 0.5

Step (write this)	Why / what it means
`Volume/year = 800 × 10^6 × 1.85 × (1 - 0.42) = 8.584 × 10^8 m^3`	42 % lost to infiltration and evaporation, 58 % reaches the plant.
`Q = 8.584 × 10^8 / 31.536 × 10^6 = 27.22 m^3/s`	Average flow.
`P = 9.81 × 27.22 × 35 × 0.84 × 0.85 = 6,673 kW ≈ 6.67 MW (average power developed)`	

Plant capacity = 6,673 / 0.5 = **13,346 kW ≈ 13.35 MW**	Peak = average / load factor.
Turbine: head 35 m (low-medium) → **Kaplan** (or Francis); e.g. 2 units of about 7 MW	Head below 50 m suits Kaplan/propeller.

2020 3(d): 800 km^2, 180 cm, 32 m, loss 40 %, η_t 85 %, η_g 90 %, LF 0.5

Step (write this)	Why / what it means
Volume = 800 × 10^6 × 1.80 × 0.60 = 8.64 × 10^8 m^3/year	
Q = 8.64 × 10^8 / 31.536 × 10^6 = 27.40 m^3/s	
P = 9.81 × 27.40 × 32 × 0.85 × 0.90 = **6,579 kW**	
Capacity = 6,579 / 0.5 = **13,159 kW ≈ 13.2 MW**; low head 32 m → **Kaplan turbines**	

2022 4(d): 800 mm, 150 km^2, 200 m, yield factor 60 %, η 75 %, LF 45 %

Step (write this)	Why / what it means
Volume = 150 × 10^6 × 0.8 × 0.60 = 7.2 × 10^7 m^3/year	Yield factor = fraction of rain that becomes usable run-off.
Q = 7.2 × 10^7 / 31.536 × 10^6 = 2.283 m^3/s	
(i) Average power = 9.81 × 2.283 × 200 × 0.75 = **3,360 kW**	
(ii) Capacity = 3,360 / 0.45 = **7,467 kW ≈ 7.5 MW**	Head 200 m, small flow → Francis (or Pelton).

2021 4(d): 14 × 10^8 m^2, 125 cm, 70 % available, 40 m with 5 % head loss, η_t 85 %, η_g 90 %, LF 45 %

Step (write this)	Why / what it means
Volume = 14 × 10^8 × 1.25 × 0.70 = 1.225 × 10^9 m^3/year	
Q = 1.225 × 10^9 / 31.536 × 10^6 = 38.84 m^3/s	
Net head = 40 × 0.95 = 38 m	5 % of head is lost by friction in the penstock.
Average power = 9.81 × 38.84 × 38 × 0.85 × 0.90 = 11,078 kW	
Maximum plant capacity = 11,078 / 0.45 = **24,617 kW ≈ 24.6 MW**	Low head → Kaplan.

2024 4(d): reservoir 250 m above turbine, 70 × 10^{11} kg/year, head loss 30 m, overall η 85 %

Step (write this)	Why / what it means
Net head H = 250 - 30 = 220 m	Subtract friction head loss.
Energy = m g H n = 70 × 10^{11} × 9.81 × 220 × 0.85 = 1.284 × 10^{16} J	Potential energy m g H, times efficiency.

In kWh (kilowatt-hour = 1 unit of electrical energy): $1.284 \times 10^{16} / 3.6 \times 10^{6} = \mathbf{3.567 \times 10^{9}}$ **kWh/year (≈ 3,567 GWh)**	1 kWh = 3.6×10^{6} J.
Average power = $3.567 \times 10^{9} / 8760 = 407$ MW	Extra line; head 220 m → Francis turbines.

Memory trick: "9.81 QHE" (kW). And **31.5 million seconds** in a year.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2023 4(d) Given: A = 800 km², rainfall 185 cm, losses 42 %, H = 35 m, $\eta_t = 0.84$, $\eta_g = 0.85$, LF = 0.5

Volume per year: A × r × (1 − loss)

= $800 \times 10^{6} \times 1.85 \times 0.58$

= **8.584×10^{8} m³**

Average flow: Q = V / (365 × 24 × 3600)

= $8.584 \times 10^{8} / 31.536 \times 10^{6}$

= **27.22 m³/s**

Power developed: $P = 9.81\ Q\ H\ \eta_t\eta_g$

= $9.81 \times 27.22 \times 35 \times 0.84 \times 0.85$

= **6,673 kW**

Plant capacity: P / LF

= 6,673 / 0.5

= **13.35 MW**

Head 35 m (low) → Kaplan turbines, e.g. 2 units of about 7 MW.

A3 Numericals on load curve, load factor, capacity factor, use factor Must study Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2024 Q2(c)** [15] The annual load duration curve of a station is a straight line from 20 MW to 4 MW. Installed: 2 × 10 MW + 1 × 5 MW. Find installed capacity, plant factor, units generated per year, load factor and utilisation factor.
- **2023 Q1(e)** [9] Maximum demand 25 MW, load factor 60 %, plant capacity factor 50 %, plant use factor 74 %. Find (i) reserve capacity, (ii) daily energy produced, (iii) maximum energy that could be produced daily if the plant ran fully loaded as per its schedule.
- **2022 Q1(d)** [11] Daily load cycle: 300 MW for 8 h, 150 MW for 6 h, 130 MW for 3 h, 100 MW for 7 h. Station has 2 sets of 200 MW. Find (i) daily load factor, (ii) plant capacity factor, (iii) daily fuel needed if CV (calorific value (heat released by 1 kg of fuel)) = 10,000 kcal/kg and heat rate = 25,000 kcal/kWh.
- **2021 Q1(d)** [12] Daily load: 6-8 h 15 MW, 8-12 h 35 MW, 12-16 h 65 MW, 16-20 h 25 MW, 20-24 h 60 MW, 24-6 h 20 MW. Draw the load curve and LDC (load duration curve) and find energy generated per day.
- **2020 Q1(d)** [8] A power station has use factor 45 % and capacity factor 35 %. For how many hours in a year was it not in service?

UNDERSTAND **Read this to understand; you do not need to write it**

Every factor numerical is: find **energy** (area under the curve), divide by hours to get the **average**, then take ratios with the maximum and installed capacity.

A straight-line LDC from P_{max} to P_{min} is a trapezium: average = $(P_{max} + P_{min})/2$. One year = 8760 h.

Symbols used here

Symbol	Name and meaning	Unit
E	Energy generated (area under the load curve)	MWh or kWh (kilowatt-hour = 1 unit of electrical energy)
P_{avg}	Average load = E / hours	MW
P_{max}	Maximum demand	MW
C	Installed (plant) capacity = sum of machine ratings	MW
t	Hours the plant actually ran	h
LF (load factor), CF (capacity factor), UF (use (utilisation) factor)	Load factor, capacity factor, utilisation factor	– or %

CALCULATOR **Straight-line LDC: every factor (2024 2(c) type)**

Change any number; every step is recalculated and shown with the numbers put in.

1. Average load: (**20** + **4**) / 2 = **12.00** MW
2. Energy in the period: E = P_{avg} × T = **12** × **8760** = **105120** MWh
3. Plant capacity factor: P_{avg}/C = **12** / **25** × 100 = **48.0** %
4. Load factor: P_{avg}/P_{max} = **12** / **20** × 100 = **60.0** %
5. Utilisation factor: P_{max}/C = **20** / **25** × 100 = **80.0** %
6. Reserve capacity: **25** - **20** = **5.0** MW

CALCULATOR **Hours not in service from use factor and capacity factor (2020 1(d))**

Change any number; every step is recalculated and shown with the numbers put in.

1. Hours in service: t = 8760 × CF / use factor = 8760 × **35** / **45** = **6813** h
2. Hours not in service: 8760 - **6813** = **1947** h

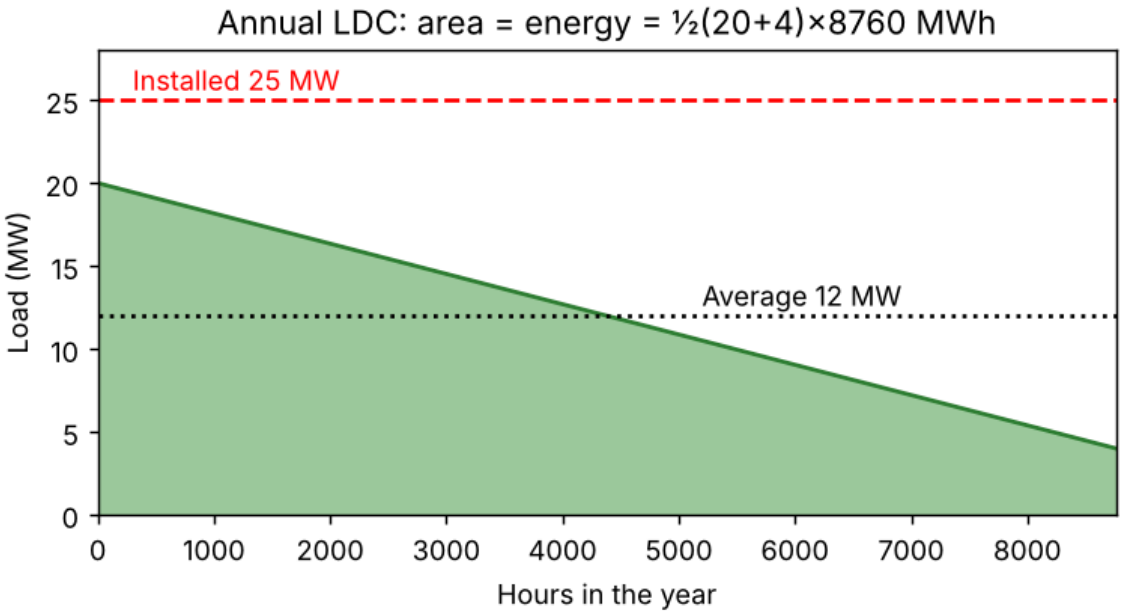
Before you start: the three numbers every factor problem uses

Energy (E) = load × hours, added up = area under the load curve (MWh or kWh).

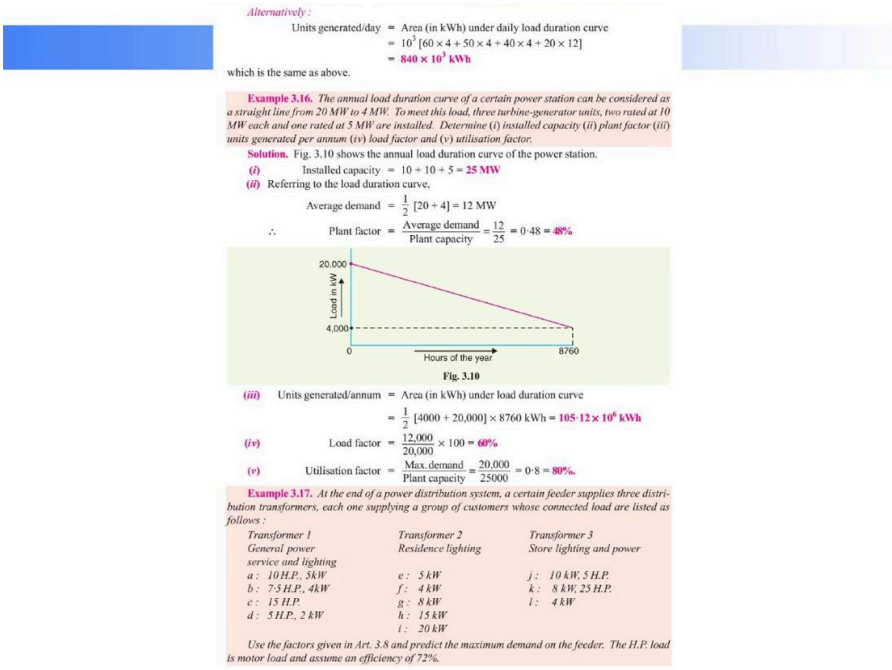
Average load = E / total hours. **Maximum demand** = the highest load. **Installed capacity** = sum of machine ratings.

Then: LF = average / maximum; CF = average / installed; UF (utilisation) = maximum / installed; use factor = E / (installed × hours run). Write these four lines first and the rest is division.

2024 2(c): straight-line LDC from 20 MW to 4 MW, units 2 × 10 + 1 × 5 MW



Annual LDC is a straight line; the area under it is a trapezium



Alternatively :

Units generated/day = Area (in kWh) under daily load duration curve
$= 10^3 [60 \times 4 + 50 \times 4 + 40 \times 4 + 20 \times 12]$
$= 840 \times 10^3$ kWh

which is the same as above.

Example 3.16. *The annual load duration curve of a certain power station can be considered as a straight line from 20 MW to 4 MW. To meet this load, three turbine-generator units, two rated at 10 MW each and one rated at 5 MW are installed. Determine (i) installed capacity (ii) plant factor (iii) units generated per annum (iv) load factor and (v) utilisation factor.*

Solution. Fig. 3.10 shows the annual load duration curve of the power station.

(i) Installed capacity = 10 + 10 + 5 = **25 MW**

(ii) Referring to the load duration curve,

Average demand = $\frac{1}{2}$ [20 + 4] = 12 MW

∴ Plant factor = $\frac{\text{Average demand}}{\text{Plant capacity}} = \frac{12}{25} = 0{\cdot}48 =$ **48%**

Fig. 3.10

(iii) Units generated/annum = Area (in kWh) under load duration curve
$= \frac{1}{2}$ [4000 + 20,000] × 8760 kWh = **105·12 × 10^6 kWh**

(iv) Load factor = $\frac{12{,}000}{20{,}000} \times 100 =$ **60%**

(v) Utilisation factor = $\frac{\text{Max. demand}}{\text{Plant capacity}} = \frac{20{,}000}{25000} = 0{\cdot}8 =$ **80%.**

Example 3.17. *At the end of a power distribution system, a certain feeder supplies three distribution transformers, each one supplying a group of customers whose connected load are listed as follows :*

Transformer 1 General power service and lighting	Transformer 2 Residence lighting	Transformer 3 Store lighting and power
a : 10 H.P., 5kW	e : 5 kW	j : 10 kW, 5 H.P.
b : 7·5 H.P., 4kW	f : 4 kW	k : 8 kW, 25 H.P.
c : 15 H.P.	g : 8 kW	l : 4 kW
d : 5 H.P., 2 kW	h : 15 kW	
	i : 20 kW	

Use the factors given in Art. 3.8 and predict the maximum demand on the feeder. The H.P. load is motor load and assume an efficiency of 72%.

Same example in the slide (Example 3.16)

Step (write this)	Why / what it means
`Installed capacity = 10 + 10 + 5 =` **`25 MW`**	Add the ratings of all machines.
`Maximum demand = 20 MW (left end of LDC); minimum = 4 MW`	An LDC starts at the highest load and ends at the lowest.

Average load = (20 + 4)/2 = 12 MW	For a straight line the average is the mid-value.
Energy per year = 12 MW × 8760 h = 105,120 MWh = **105.12×10^6 kWh**	Area of trapezium = ½(20 + 4) × 8760. 1 year = 365 × 24 = 8760 h.
Plant (capacity) factor = 12 / 25 = **0.48 = 48 %**	Average load over installed capacity.
Load factor = 12 / 20 = **0.60 = 60 %**	Average over maximum demand.
Utilisation factor = 20 / 25 = **0.80 = 80 %**	Maximum demand over installed capacity.

2023 1(e): max demand 25 MW, LF 60 %, CF 50 %, use factor 74 %

Step (write this)	Why / what it means
Average load = LF × max demand = 0.60 × 25 = 15 MW	From LF = average / maximum.
Plant capacity = average / CF = 15 / 0.50 = 30 MW	From CF = average / capacity.
(i) Reserve capacity = 30 - 25 = **5 MW**	Installed minus maximum demand.
(ii) Daily energy = 15 MW × 24 h = **360 MWh**	Average load × hours in a day.
Use factor = E / (C × t_{run}) ⇒ t_{run} = 360 / (30 × 0.74) = 16.2 h	The plant actually runs about 16.2 h a day.
(iii) Max energy if fully loaded during its schedule = 30 × 16.2 = **486.5 MWh/day**	Same as E / use factor = 360 / 0.74.

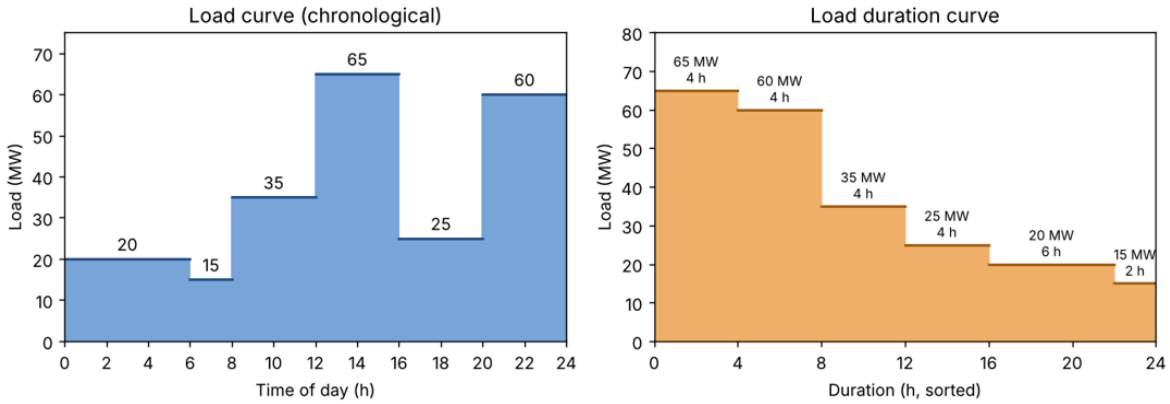
Side note: If the examiner meant "running all 24 h at full load", that would be 30 × 24 = 720 MWh. Write both lines to be safe; the "as per schedule" answer is 486.5 MWh.

2022 1(d): load cycle 300/150/130/100 MW, 2 × 200 MW sets, CV 10,000 kcal/kg, heat rate 25,000 kcal/kWh

Step (write this)	Why / what it means
E = 300×8 + 150×6 + 130×3 + 100×7 = 2400 + 900 + 390 + 700 = **4390 MWh**	Hours add to 8 + 6 + 3 + 7 = 24, a full day.
Average load = 4390 / 24 = 182.9 MW; max demand = 300 MW	Largest load in the cycle.
(i) Daily LF = 182.9 / 300 = **61 %**	
(ii) Capacity factor = 182.9 / 400 = **45.7 %**	Installed = 2 × 200 = 400 MW.
Heat needed = 4390 × 10^3 kWh × 25,000 kcal/kWh = 1.0975 × 10^{11} kcal	Heat rate = heat input per kWh generated.
(iii) Fuel = 1.0975 × 10^{11} / 10,000 = 1.0975 × 10^7 kg = **10,975 tonnes/day**	Fuel mass = heat needed / calorific value.

Side note: A heat rate of 25,000 kcal/kWh means only 860/25,000 = 3.4 % efficiency, which is unrealistic (1 kWh = 860 kcal). The textbook version uses 2,500 kcal/kWh, giving **1,097.5 t/day**. Solve with the paper's number and add this one-line remark.

2021 1(d): draw load curve and LDC, energy per day



Left: chronological load curve. Right: the same blocks sorted from highest to lowest = load duration curve

Step (write this)	Why / what it means
`Blocks: 24-6 h 20 MW (6 h); 6-8 h 15 MW (2 h); 8-12 h 35 MW (4 h); 12-16 h 65 MW (4 h); 16-20 h 25 MW (4 h); 20-24 h 60 MW (4 h)`	Write duration of each block first; they must add to 24 h.
`E = 20×6 + 15×2 + 35×4 + 65×4 + 25×4 + 60×4`	Area under the load curve, block by block.
`= 120 + 30 + 140 + 260 + 100 + 240 =` **`890 MWh/day`**	
`LDC order: 65 (4 h), 60 (4 h), 35 (4 h), 25 (4 h), 20 (6 h), 15 (2 h)`	Sort loads high to low and lay their durations end to end.
`Extra: average = 890/24 = 37.1 MW, LF = 37.1/65 = 57 %`	Not asked, but quick marks.

2020 1(d): use factor 45 %, capacity factor 35 %, hours not in service

Step (write this)	Why / what it means
`CF = E / (C × 8760) and use factor = E / (C × t)`	Same E and C in both.
`Divide: CF / use factor = t / 8760 ⇒ t = 8760 × 0.35 / 0.45 = 6813 h`	E and C cancel.
`Not in service = 8760 - 6813 =` **`1947 hours per year`**	About 81 days.

Memory trick: "A-M-I": Average, Maximum, Installed. LF = A/M, CF = A/I, UF = M/I. So LF × UF = CF.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2023 1(e) Given: P_{max} = 25 MW, LF = 0.60, CF = 0.50, use factor = 0.74

Average load: LF × P_{max}

= 0.60 × 25

= **15 MW**

Plant capacity: P_{avg} / CF

= 15 / 0.50

= **30 MW**

(i) Reserve capacity: $C - P_{max}$

= 30 − 25

= **5 MW**

(ii) Daily energy: $P_{avg} \times 24$

= 15 × 24

= **360 MWh**

(iii) Max energy on schedule: E / use factor

= 360 / 0.74

= **486.5 MWh/day**

(If running 24 h at full load were meant: 30 × 24 = 720 MWh. Write both lines.)

A13 Hydraulic turbines: selection, specific speed, governing of impulse (Pelton) turbine

Must study | Very likely · 100% | Medium

PAST-PAPER QUESTIONS

- **2024 Q4(c)** [8] Explain the governing of an impulse turbine.
- **2023 Q4(c)** [7] Which factors should be considered while selecting turbines for a hydro plant?
- **2022 Q4(c)** [7] Which factors should be considered while selecting turbines for a hydro plant? Explain.
- **2021 Q4(b)** [7] Explain the governing of an impulse turbine with a neat sketch.
- **2020 Q3(a)** [7] Which factors should be considered while selecting turbines for a hydro plant?
- **2020 Q3(c)** [7] Why is governing of a hydro-electric power plant required?

★ REMEMBER IT

- Head decides the turbine: **high head (> 250 m) Pelton** (impulse), **medium (50–250 m) Francis**, **low (< 50 m) Kaplan/propeller**. Pelton governing: a spear moves in the nozzle to change the jet; a deflector swings in first for sudden load drops so the penstock pressure does not jump.

Before you start: impulse vs reaction turbine

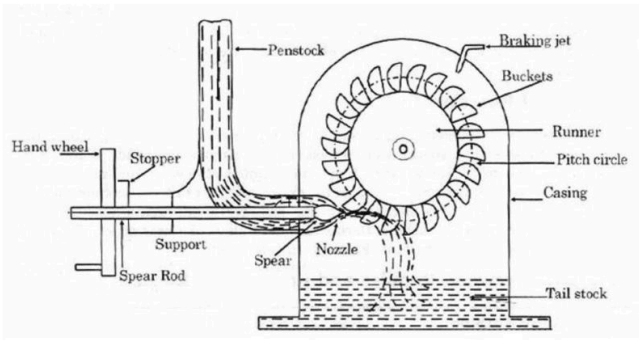
Impulse turbine (Pelton wheel): all the pressure of water is turned into a high-speed jet in a nozzle; the jet hits buckets at atmospheric pressure. Good for very high head, small flow.

Reaction turbine (Francis, Kaplan): the runner is full of water under pressure; water pushes the blades by both pressure and velocity. Good for medium and low head, large flow.

Impulse Turbine:

The passages are not completely filled, water acting on a wheel buckets is at atmospheric pressure and is supplied at few points at the periphery of wheel & kinetic energy is supplied to the wheel.

Casing
Penstock
Nozzle
Runner
Buckets
Needle Valve
shaft



Pelton Wheel

Pelton wheel (impulse turbine)

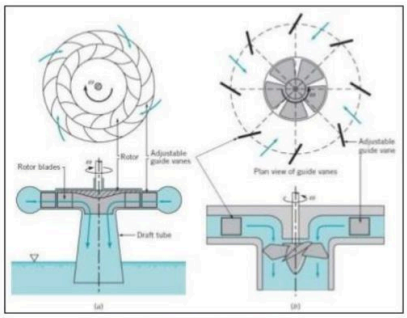
Reaction Turbine:

Water passages are completely filled with water, water acting on wheel vanes is under pressure greater than atmospheric, water enter all around the periphery of wheel and energy is in the form of both pressure & kinetic energy is utilized by the wheel.

Essential parts:

Spiral casing
Guide wheel
Runner
Draft tube

Reaction Turbine



(a) Typical radial-flow Francis turbine
(b) Typical axial-flow Kaplan turbine.

Reaction turbines: Francis (radial/mixed flow) and Kaplan (axial flow)

Factors for selecting turbines (2023 4c, 2022 4c, 2020 3a)

1. **Operating head:** Kaplan/propeller for heads up to about 50 m (low), Francis for about 50-400 m (medium), Pelton above about 400 m (high). (Slide figures: low < 15 m, medium 15-50 m, high > 50 m.)
2. **Specific speed:** choose the turbine whose specific speed range suits; higher specific speed = smaller turbine, generator and power house, so cheaper. Pelton 10-35 (single jet), Francis 60-300, Kaplan 300-1000 (metric).
3. **Rotational speed:** must match a synchronous generator speed N = 120f/p.
4. **Performance at part load:** if the plant runs at part load often, choose a turbine with a flat efficiency curve (Kaplan, Pelton) rather than Francis/propeller.
5. **Height of installation / cavitation:** install as high above tail water level as possible without cavitation (reaction turbines with draft tube).
6. **Size and number of units:** larger units are cheaper per kW, but at least two units so one is available during breakdown.
7. **Quantity of water available** and its variation, and **cost**.

Step (write this)	Why / what it means
$N_s = NvP / H^{5/4}$ (P in kW, metric)	N in rpm, P in kW, H in m.
Low N_s (< 35) → Pelton; 60-300 → Francis; 300-1000 → Kaplan	Big head + small flow gives low N_s.
N = 120 f / p (f = 50 Hz, p = number of poles)	E.g. 20 poles: N = 120 × 50 / 20 = 300 rpm.

Substituting the value of Q in the above equation, we get

$$P \propto \frac{\rho}{75}\left[\frac{(H)^{3/2}}{N^2}\right] H \propto \frac{(H)^{5/2}}{N^2}$$

$$N^2 \propto \frac{(H)^{5/2}}{P}$$

$$N \propto \frac{(H)^{5/4}}{\sqrt{P}} = C\frac{(H)^{5/4}}{\sqrt{P}}$$

where C is constant depending upon the type of the turbine.

If the turbine develops 1 B.H.P. under one meter head then

$$C = N = N_S$$

where N_s is the specific speed as per the definition.

Substituting the value of C in the above equation, we get

$$N_s = \frac{N\sqrt{P}}{(H)^{5/4}} \text{ when } P \text{ is in H.P.}$$

$$= \frac{1.165\, N\sqrt{KW}}{(H)^{5/4}} \text{ when the power is in kW.}$$

Derivation of specific speed $N_S = N\sqrt{P} / H^{5/4}$

Why governing is needed (2020 3c)

The turbine is directly coupled to the generator, whose speed must stay at the **synchronous speed N = 120f/p** so that frequency stays at 50 Hz. If load falls and water flow stays the same, the turbine speeds up (frequency rises); if load rises, it slows down. **Governing** automatically adjusts the quantity of water entering the runner in proportion to the load so that the speed stays constant at all loads.

Governing of an impulse (Pelton) turbine (2024 4c, 2021 4b)

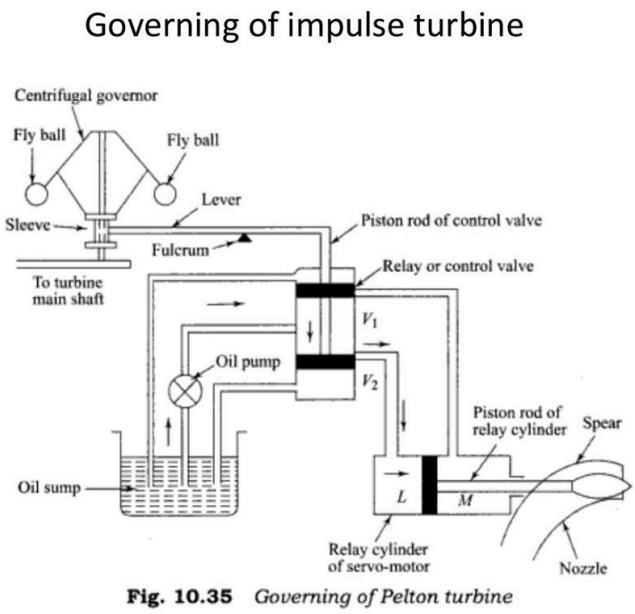


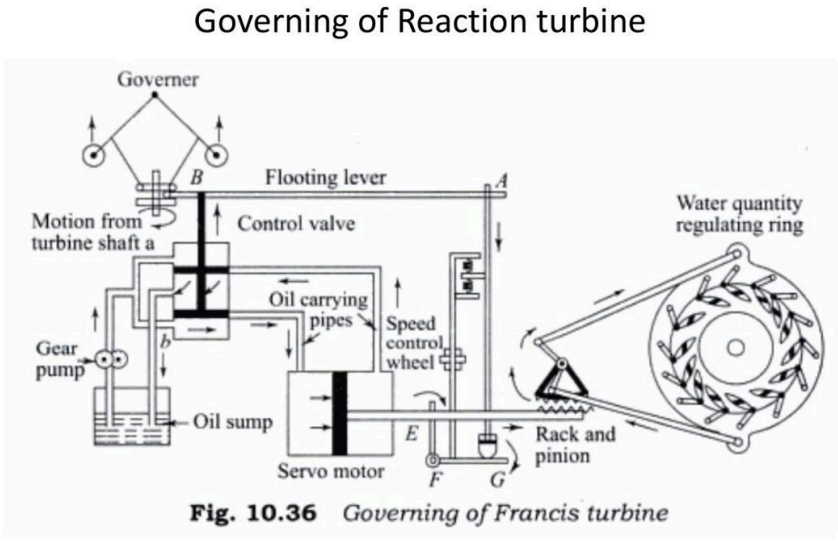
Fig. 10.35 *Governing of Pelton turbine*

Oil-pressure governor of a Pelton turbine: centrifugal governor, relay (control) valve, servomotor (relay cylinder) and spear in the nozzle

Step (write this)	Why / what it means
`1. Load on generator decreases → turbine speeds up`	More water energy than needed.

2. Fly balls of the centrifugal governor (driven by turbine shaft) fly outward and lift the sleeve	Speed sensor.
3. Sleeve moves the lever, which shifts the piston of the relay (control) valve	Amplifier: small force opens an oil path.
4. Oil from the oil pump flows to one side of the servomotor (relay cylinder) piston	Servomotor gives the big force needed.
5. Servomotor piston pushes the spear (needle) forward into the nozzle	Nozzle opening shrinks, so less water in the jet.
6. Less water → less power → speed falls back to normal; the reverse happens when load increases	Closed-loop control.

Deflector: closing the spear quickly could cause water hammer in the long penstock, so on a sudden large load drop a **jet deflector** first swings into the jet and diverts part of it away from the buckets; the spear then closes slowly. Some designs use a relief valve on the penstock instead.



Governing of a reaction (Francis) turbine: servomotor turns the guide vanes through a regulating ring

Memory trick: Governor chain **"Fly-Relay-Servo-Spear"**: fly balls sense, relay valve directs oil, servomotor pushes, spear closes.

A11 Bangladesh power sector: fuel mix, generation cost, renewables, PSMP 2015, nuclear and coal plans

Must study | Very likely · 93% | Medium

PAST-PAPER QUESTIONS

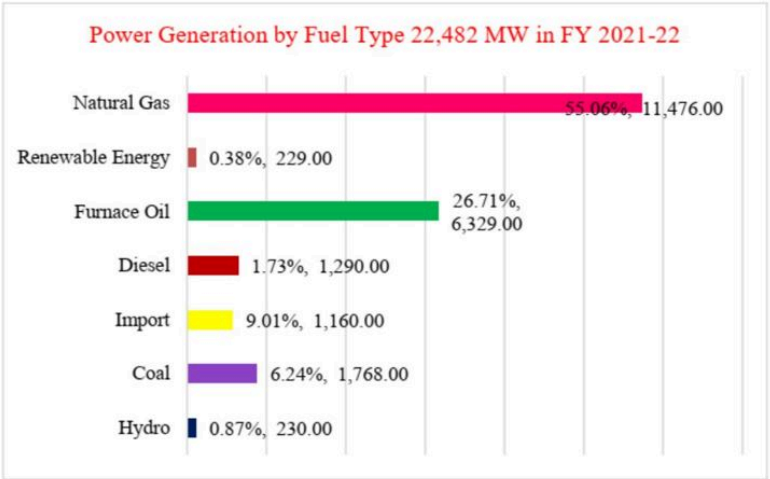
- **2024 Q3(d)** [10] Describe the future plan for nuclear and coal based generation according to PSMP (Power System Master Plan).
- **2023 Q3(c)** [9] Explain the present situation of power generation in Bangladesh from renewable energy sources.
- **2023 Q3(d)** [11] Explain the future power demand of Bangladesh according to PSMP 2015. How can the demand be met?
- **2022 Q3(a)** [10] Which fuels are widely used for power generation in Bangladesh? Compare the generation cost of BPDB (Bangladesh Power Development Board)'s own plants by fuel type.
- **2022 Q3(b)** [11] Explain the future power demand of Bangladesh according to PSMP 2015. How can it be met?
- **2021 Q3(d)** [11] Explain the future power demand of Bangladesh according to PSMP 2015. How can it be met?

Before you start: names you will see

PSMP 2015 = Power System Master Plan 2015, prepared for Bangladesh with JICA (Japan International Cooperation Agency) and TEPCO. It forecasts demand up to 2041 and lists the plants needed.

BPDB = Bangladesh Power Development Board (the main state utility). IPP (independent power producer) = independent (private) power producer.

Fuels used and generation cost of BPDB plants (2022 3a)



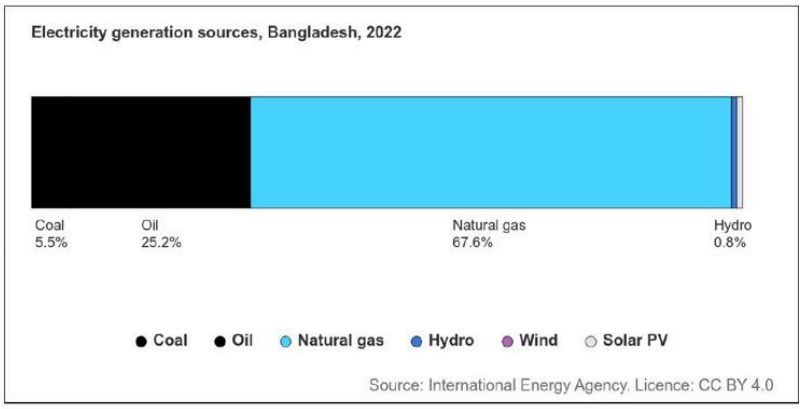
[Source: Power Division Annual Report 2021-22]

Figure 14: Power Generation by Fuel Type (2021-22)

Power generation capacity by fuel in FY 2021-22: natural gas 55 %, furnace oil 27 %, import 9 %, coal 6 %, diesel 2 %, hydro 0.9 %, renewables 0.4 %

Energy Sources for electricity generation (2022)

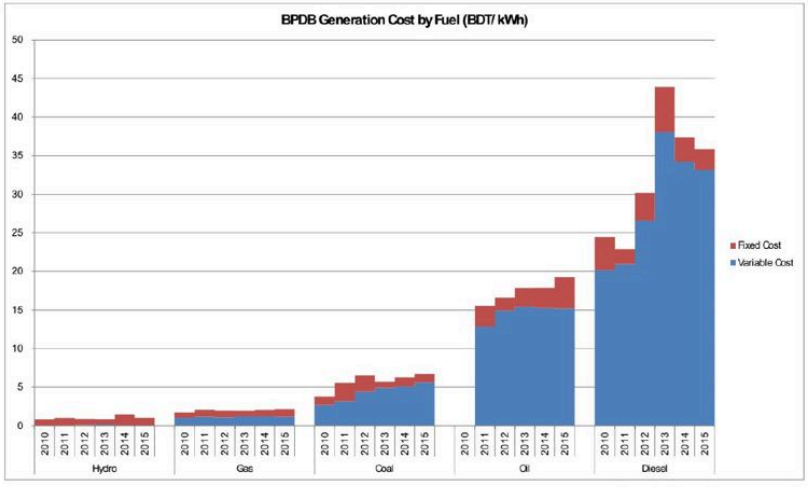
Source	Electricity (GWh)
Coal	5663
Oil	25810
Natural gas	69214
Hydro	788
Solar PV	865
Wind	5



Electricity generated by source in 2022 (GWh): gas 69,214, oil 25,810, coal 5,663, solar 865, hydro 788, wind 5

Natural gas is the most widely used fuel (about 55 % of capacity and about two-thirds of energy), followed by **furnace oil (HFO)**, **imported power**, **coal**, **diesel (HSD)**, **hydro** (Kaptai 230 MW) and **solar**.

Cost of Generation



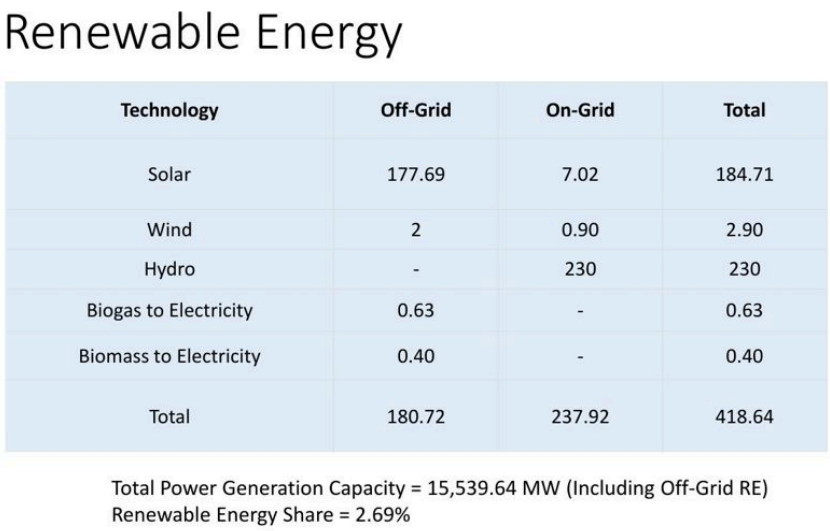
*per kWh of net generation
Source: BPDB Annual Reports

Figure 19-29 Generation Cost of BPDB's Own Plants (by Fuel)

Generation cost of BPDB's own plants by fuel (Tk/kWh), 2010-2015: hydro and gas cheapest, coal moderate, oil and diesel very costly

Fuel	Approx. generation cost (Tk/kWh, BPDB 2010-15)	Why
Hydro	about 1	No fuel cost; only fixed cost
Natural gas	about 1.5-2	Cheap subsidised domestic gas
Coal	about 4-7	Coal price + higher capital cost
Furnace oil	about 15-19	Imported oil, high fuel price
Diesel	about 25-44	Most expensive fuel; used only for peaking/emergency

Present renewable situation (2023 3c)



Renewable Energy

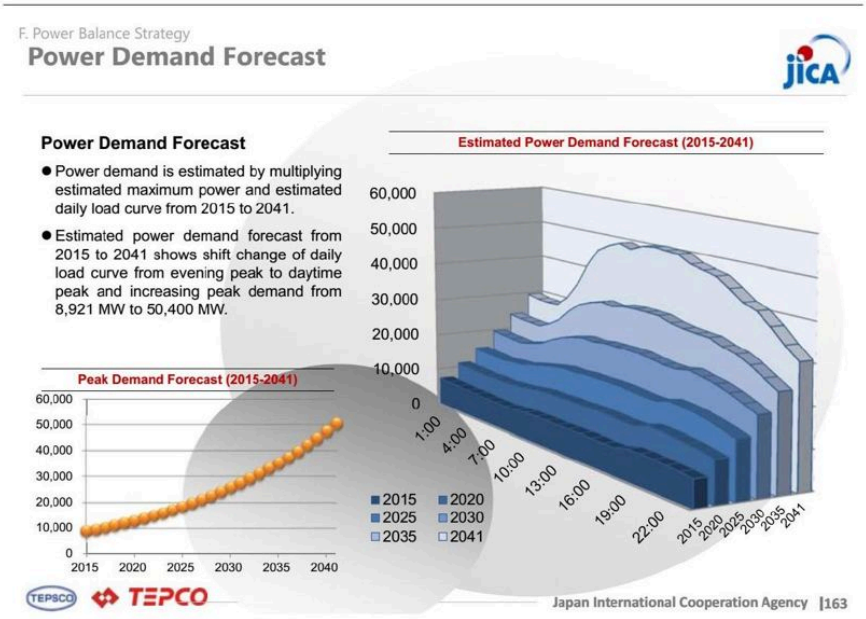
Technology	Off-Grid	On-Grid	Total
Solar	177.69	7.02	184.71
Wind	2	0.90	2.90
Hydro	-	230	230
Biogas to Electricity	0.63	-	0.63
Biomass to Electricity	0.40	-	0.40
Total	180.72	237.92	418.64

Total Power Generation Capacity = 15,539.64 MW (Including Off-Grid RE)
Renewable Energy Share = 2.69%

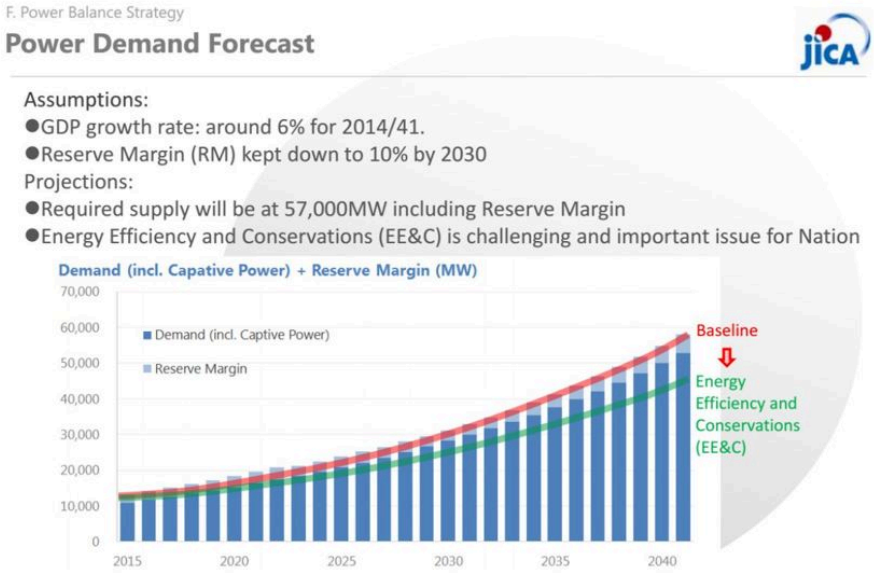
Renewable capacity: solar 184.7 MW (mostly off-grid solar home systems), wind 2.9 MW, hydro 230 MW, biogas 0.63 MW, biomass 0.4 MW; total 418.6 MW = 2.69 % of capacity

- Renewables were only about **2.7 %** of total generation capacity (418.6 MW of 15,540 MW), and much less of energy.
- **Hydro**: Kaptai (230 MW) is the only large hydro plant; flat land gives little more potential.
- **Solar**: millions of solar home systems (off-grid), solar irrigation pumps, rooftop and solar parks (e.g. Mymensingh, Teknaf) now several hundred MW; solar is the main growth area.
- **Wind**: small (Kutubdia, Feni, newer Mongla/Cox's Bazar projects), because wind speeds are low.
- **Biomass/biogas**: small plants on rice husk, poultry litter, municipal waste.
- SREDA (Sustainable and Renewable Energy Development Authority, 2012) promotes RE (renewable energy); targets of about 3,200 MW were set but not met, due to land shortage, high cost and intermittency.

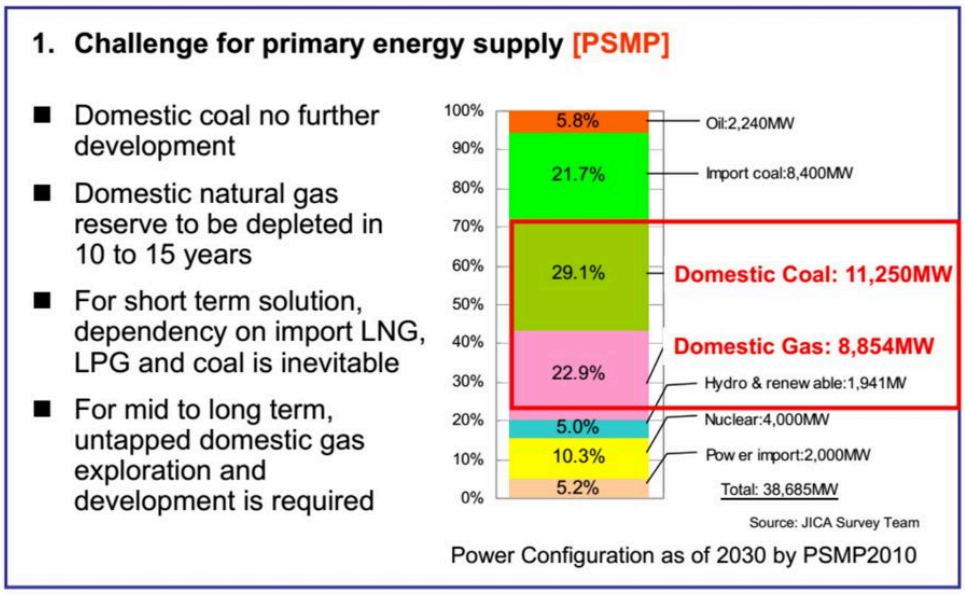
Future demand by PSMP 2015 and how to meet it (2023 3d, 2022 3b, 2021 3d)



PSMP 2015 demand forecast: peak demand rises from about 8,921 MW (2015) to about 50,400 MW (2041)



Assumptions: GDP growth about 6 %/year; reserve margin down to 10 % by 2030; required supply about 57,000 MW including reserve



Power configuration planned in PSMP: mix of domestic gas, domestic and imported coal, nuclear, power import, oil, hydro and renewables

Demand: PSMP 2015 forecasts peak demand growing from about **9,000 MW in 2015** to about **19,000 MW by 2020-21, 33,000 MW by 2030 and about 50,000-52,000 MW by 2041** (base case; high case about 60,000 MW), driven by about 6-7 % GDP growth. With a reserve margin, the **required capacity is about 57,000 MW by 2041**.

How the demand can be met (fuel diversification):

1. **Coal** (about 35 %): imported-coal ultra-supercritical plants at the coast: Payra 1320 MW, Rampal 1320 MW, Matarbari 1200 MW, plus Barapukuria domestic coal.
2. **Natural gas and LNG** (liquefied natural gas) (about 35 %): new combined cycle plants; LNG import terminals at Moheshkhali because domestic gas is running out.
3. **Nuclear**: Rooppur 2 × 1200 MW, later more units.
4. **Power import** from India, Nepal, Bhutan (Bheramara HVDC 1000 MW, Tripura 160 MW, Adani 1496 MW).
5. **Renewables**: solar parks, rooftop, wind; hydro import.
6. **Energy efficiency and conservation** to cut the baseline demand; better transmission and a strong grid.

Future nuclear and coal plan (2024 3d)

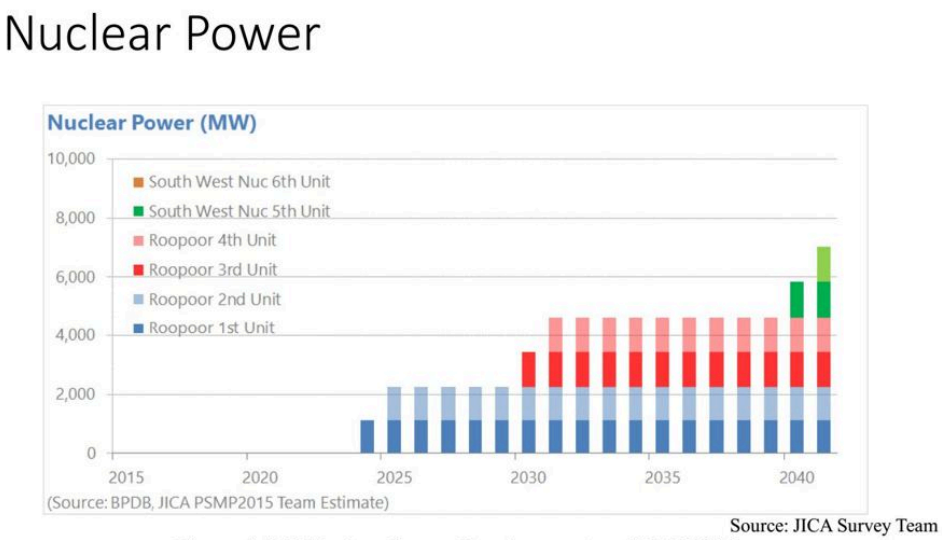
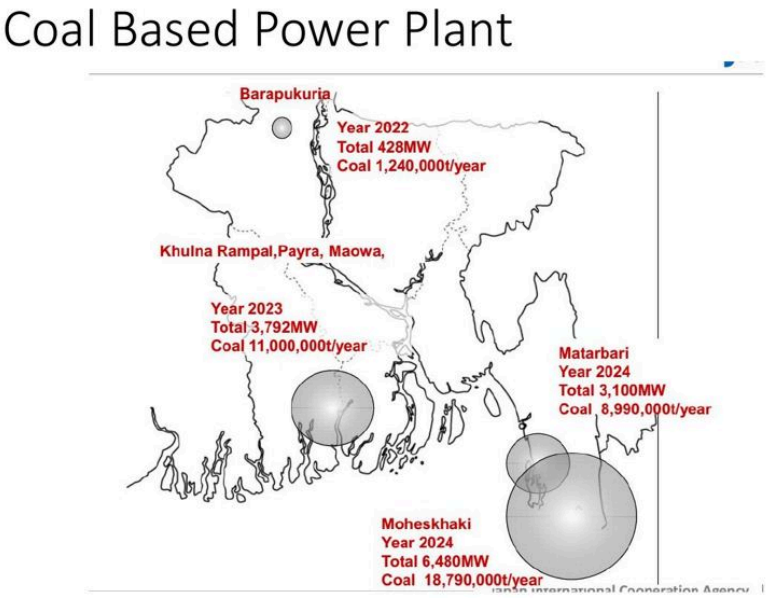


Figure 14-11 Nuclear Power Development on PSMP2016

Nuclear power slide: Rooppur Nuclear Power Plant



Coal-based power plant plans

- **Rooppur Nuclear Power Plant** (Ishwardi, Pabna, on the Padma, 160 km north-west of Dhaka): 2 × 1200 MW VVER-1200 reactors built with Russia (Rosatom); agreement 2011; first unit commissioning 2024-25. PSMP envisaged about 4,000 MW (some say 7,200 MW) nuclear by 2041.
- **Payra** 2 × 660 MW (BCPCL, China JV) running since 2020.
- **Rampal (Maitree)** 2 × 660 MW supercritical (BIFPCL, BPDB-NTPC India JV, 50:50), cost about USD 1.68 billion.
- **Matarbari** 2 × 600 MW ultra-supercritical (CPGCBL, JICA funded) with a deep-sea coal port.
- **Barapukuria** (domestic coal, 525 MW), plus planned coastal hubs at Moheshkhali and Patuakhali.
- Under construction in 2021: 16,479 MW in 41 plants, of which coal 7,619 MW (46 %) and nuclear 2,400 MW (15 %).

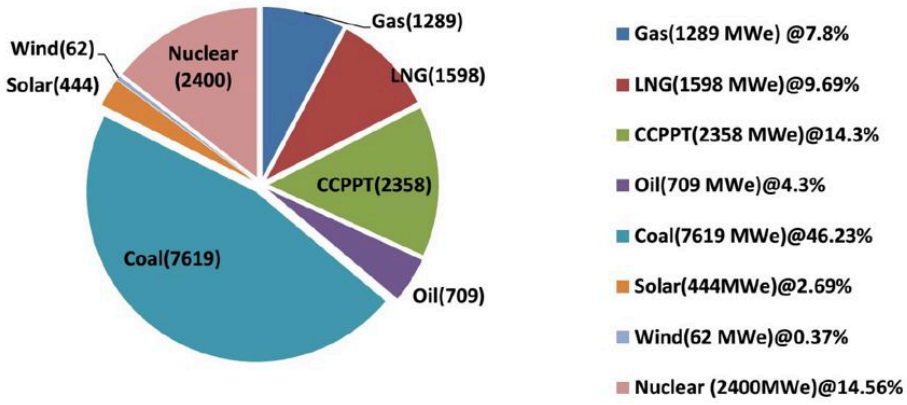


Fig. 4: Total under construction power generation from 41 plants : 16,479 MWe as of February 2021

Sources: MoPEMR & BPDB

Under-construction capacity by fuel (Feb 2021): coal 46 %, nuclear 15 %, combined cycle 14 %

Memory trick: PSMP numbers: **"9 to 50 by 41, need 57"**: 9 GW in 2015, about 50 GW peak by 2041, 57 GW capacity needed.

A1 Power plant basics: classification, energy sources, base vs peak plant, choosing capacity

Must study · Likely · 67% · Easy

PAST-PAPER QUESTIONS

- **2023 Q1(b)** [9] What is meant by base load power plant and peak load power plant? What types of power plant are suitable for peak load power plant?
- **2023 Q1(c)** [6] What factors should be considered while selecting the capacity of a power plant?
- **2022 Q1(a)** [12] What is meant by power plant engineering? Describe the classification of power plant. What are the major sources of energy widely used in power generation?
- **2022 Q2(b)** [6] What factors should be considered while selecting the capacity of a power plant?
- **2021 Q1(a)** [10] What is meant by power plant engineering? Classify power plants. What are the basic sources of energy?
- **2020 Q1(a)** [10] What is meant by power plant engineering? Describe the classification of power plant.

★ REMEMBER IT

- **Base load** plant runs all day, every day: cheap fuel, costly to build, slow to start (steam, nuclear, big hydro). **Peak load** plant runs a few hours: cheap to build, starts in minutes, fuel costly (gas turbine, diesel, pumped storage).

Before you start: what a power plant does

A power plant turns some **primary energy** (chemical energy in coal, gas or oil; potential energy of high water; nuclear energy; sunlight; wind) into **electrical energy**. Almost every plant does it the same way: the energy spins a **prime mover** (a turbine or engine), and the prime mover spins an **alternator** (generator).

Power (kW, MW) is the rate of making energy; **energy** (kWh) is power × time. 1 kWh (kilowatt-hour = 1 unit of electrical energy) = 1 kW running for 1 hour = one "unit" on your electricity bill.

What is power plant engineering? (2022 1a, 2021 1a, 2020 1a)

Power plant engineering is the branch of engineering that deals with the **design, selection of equipment, construction, operation, maintenance and economics** of plants that generate electrical energy from available energy sources, so that power is produced **reliably, at the lowest cost per kWh and with acceptable environmental impact**.

Classification of power plants

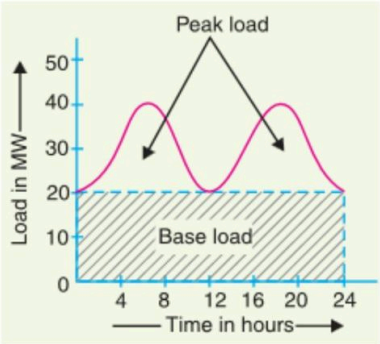
Basis	Types
Energy source / fuel	Conventional: steam (coal), gas turbine, diesel, hydro-electric, nuclear. Non-conventional: solar, wind, tidal, geothermal, biomass, fuel cell
Nature of load served	Base load plant, intermediate load plant, peak load plant, standby plant
Prime mover	Steam turbine, gas turbine, IC engine (diesel), water turbine, wind turbine
Location / ownership	Central station (feeds the grid), captive / isolated plant (one factory or island), mobile plant
Fuel used (thermal)	Coal-fired, oil-fired, gas-fired, nuclear-fuelled, multi-fuel

Major energy sources used for power generation

- **Fossil fuels:** coal, natural gas, oil (furnace oil, diesel). In Bangladesh natural gas is the largest source.
- **Water (hydro):** potential energy of water stored at height (Kaptai 230 MW in Bangladesh).
- **Nuclear fuel:** uranium-235 fission (Rooppur 2 × 1200 MW).
- **Renewables:** solar, wind, biomass/biogas, tidal, geothermal.

Base load plant vs peak load plant (2023 1b)

Base Load and Peak Load on Power Station



Base load: The unvarying load which occurs almost the whole day on the station is known as
base load.
Peak load: The various peak demands of load over and above the base load of the station is known as peak load.

Base load is the steady part of the daily load that is there almost all day; peak load is the extra on top during a few hours (slide)

	Base load plant	Peak load plant
Meaning	Supplies the steady, always-present part of the load, runs nearly 24 h at near full load	Supplies only the peaks above the base, runs a few hours a day
Wanted features	High efficiency, low fuel (running) cost; high capital cost is acceptable	Quick start, quick load pick-up, low capital cost; high fuel cost is acceptable
Examples	Large coal steam plant, nuclear, run-of-river or big storage hydro, combined cycle	Gas turbine, diesel plant, pumped storage, small storage hydro

Suitable peak load plants: gas turbine (starts in 5-15 minutes), diesel engine plant (starts in minutes), pumped storage hydro (pump water up at night, generate at the peak), storage hydro with variable gate opening. They are suitable because they **start and stop quickly, follow load changes fast** and have **low fixed cost**, so running only a few hours a year is still economical.

> **Side note:** Why not run the coal plant at the peak? A steam boiler needs hours to warm up and suffers thermal stress if loaded and unloaded quickly; it also has a high fixed cost that needs many running hours to pay back.

Factors for selecting the capacity of a power plant (2023 1c, 2022 2b)

1. **Maximum demand and load curve:** installed capacity must exceed the expected maximum demand.
2. **Future growth of load:** add a margin for 5-10 years of demand growth, or leave space for extra units.
3. **Reserve capacity:** spare units to cover breakdown and maintenance (usually the size of the largest unit).
4. **Number and size of units:** several units of different sizes so that each can run near full load as the load varies.
5. **Load factor and diversity factor:** a low load factor means capacity sits idle; size to keep capacity factor reasonable.
6. **Type of plant and fuel availability** (base vs peak duty), **capital available**, **standard unit sizes** on the market.
7. **Interconnection:** if connected to a grid, less reserve is needed.

> **Memory trick:** Base = **B**ig, **B**oring, **B**urns cheap fuel all day. Peak = **P**rompt start, **P**art-time, **P**ricey fuel.

A8 Numericals: generating cost per kWh Must study Possible · 47% Medium

PAST-PAPER QUESTIONS

- **2023 Q2(d)** [11] Plant 100 MW, annual load factor 45 %, capital cost Tk 105 crore, annual wages Tk 40 lac, fuel + maintenance Tk 1/kWh, interest 4 %, depreciation 6 %. Find generating cost per kWh (kilowatt-hour = 1 unit of electrical energy).
- **2021 Q2(d)** [12] Installed 100 MW, capital cost Tk 25,000/kW, interest + depreciation 10 %, annual load factor 50 %, annual capacity factor 45 %, annual running charge Tk 500 × 10^6, auxiliaries 5 %. Find generating cost per kWh.
- **2020 Q2(c)** [13] Plant 100 MW, annual load factor 45 %, capital cost Tk 150 crore, annual wages Tk 1 crore, fuel Tk 0.5/kWh, interest 6 %, depreciation 8 %. Find generating cost per kWh.

UNDERSTAND **Read this to understand; you do not need to write it**

Yearly cost = **fixed** (interest + depreciation on the capital, plus fixed wages) + **running** (fuel, maintenance per kWh). Divide by the kWh sold. Do all money in Tk first: 1 crore = 10^7, 1 lac = 10^5.

Energy per year = capacity × load factor × 8760 (if a capacity factor is given, use it for energy). Subtract auxiliary use if asked for "kWh sent out".

CALCULATOR **Generating cost per kWh**

Change any number; every step is recalculated and shown with the numbers put in.

1. kWh generated per year: E = P(kW) × LF × 8760 = **100** × 1000 × **0.45** × 8760 = **3.942×10^8** kWh
2. kWh sent out: **3.942×10^8** × (1 - **0** / 100) = **3.942×10^8** kWh
3. Fixed charges: **10** / 100 × **105** × 10^7 + **0.4** × 10^7 = **1.090×10^8** Tk
4. Running charges: **1** × **3.942×10^8** = **3.942×10^8** Tk
5. Cost per kWh: c = (fixed + running) / kWh sent out = (**1.090×10^8** + **3.942×10^8**) / **3.942×10^8** = **1.277** Tk/kWh

Before you start: the recipe for cost per kWh

1 crore = 10^7 Tk; 1 lac (lakh) = 10^5 Tk. Convert everything to Tk first.

Step 1: kWh per year = plant capacity (kW) × load factor × 8760 h (use capacity factor if given for energy, see 2021).

Step 2: fixed cost = (interest % + depreciation %) × capital cost + wages (if fixed).

Step 3: running cost = fuel Tk/kWh × kWh.

Step 4: cost per kWh = (fixed + running) / kWh sent out.

Example 4.7. *Estimate the generating cost per kWh delivered from a generating station from the following data :*

Plant capacity = 50 MW ;Annual load factor = 40%

Capital cost = 1·2 crores ; annual cost of wages, taxation etc. = Rs 4 lakhs ; cost of fuel, lubrication, maintenance etc. = 1·0 paise/kWh generated. Interest 5% per annum, depreciation 6% per annum of initial value.

Solution : The maximum demand on the station may be assumed equal to the plant capacity *i.e.*, 50 MW.

Annual fixed charges

Interest and depreciation = Rs $120 \times 10^5 \times (5+6)/100$ = Rs $13{\cdot}2 \times 10^5$

Wages and taxation = Rs 4×10^5

Total annual fixed charges = Rs $(13{\cdot}2 \times 10^5 + 4 \times 10^5)$ = Rs $17{\cdot}2 \times 10^5$

Annual running charges

Units generated/annum = Max. demand × L.F. × Hours in a year

= $(50 \times 10^3) \times (0{\cdot}4) \times (8760)$ kWh

= 1752×10^5 kWh

Cost of fuel, lubrication etc. = Rs $1752 \times 10^5 \times 0{\cdot}01$ = Rs $17{\cdot}52 \times 10^5$

Total annual charges = Rs $(17{\cdot}2 \times 10^5 + 17{\cdot}52 \times 10^5)$ = Rs $34{\cdot}72 \times 10^5$

$\therefore$ Cost per kWh = Rs $\dfrac{34{\cdot}72 \times 10^5}{1752 \times 10^5}$ = Re 0·02 = **2 paise**

Slide example: same four steps (50 MW, LF (load factor) 40 %)

2023 2(d): 100 MW, LF 45 %, capital Tk 105 crore, wages Tk 40 lac, fuel + maintenance Tk 1/kWh, interest 4 %, depreciation 6 %

Step (write this)	Why / what it means
kWh/year = 100,000 kW × 0.45 × 8760 = 394.2 × 10^6 kWh	Max demand taken as plant capacity; average = 0.45 × that.
Interest + depreciation = (4 + 6)% × 105 × 10^7 = Tk 10.5 × 10^7	10 % of capital every year.
Wages = Tk 0.4 × 10^7	40 lac = 40 × 10^5 = 0.4 crore.
Fuel + maintenance = 1 × 394.2 × 10^6 = Tk 39.42 × 10^7	
Total = 10.5 + 0.4 + 39.42 = Tk 50.32 × 10^7 per year	
Cost per kWh = 50.32 × 10^7 / 394.2 × 10^6 = **Tk 1.28 per kWh**	

2020 2(c): 100 MW, LF 45 %, capital Tk 150 crore, wages Tk 1 crore, fuel Tk 0.5/kWh, interest 6 %, depreciation 8 %

Step (write this)	Why / what it means
kWh/year = 394.2 × 10^6	Same as above.
Interest + depreciation = 14 % × 150 crore = Tk 21 crore	
Wages = Tk 1 crore; fuel = 0.5 × 394.2 × 10^6 = Tk 19.71 crore	
Total = 21 + 1 + 19.71 = Tk 41.71 crore = 41.71 × 10^7	
Cost per kWh = 41.71 × 10^7 / 394.2 × 10^6 = **Tk 1.06 per kWh**	

2021 2(d): 100 MW, Tk 25,000/kW, interest + depreciation 10 %, LF 50 %, CF (capacity factor) 45 %, running Tk 500 × 10^6, auxiliaries 5 %

Step (write this)	Why / what it means
`Capital = 25,000 × 100,000 kW = Tk 2500 × 10^6`	Cost per kW × kW installed.
`Fixed charges = 10 % × 2500 × 10^6 = Tk 250 × 10^6`	
`Total yearly cost = 250 + 500 = Tk 750 × 10^6`	Running charge is given directly.
`kWh generated = 100,000 × 0.45 × 8760 = 394.2 × 10^6 kWh`	Use CAPACITY factor: energy = installed × CF × 8760. (LF only tells max demand = 45/0.5 = 90 MW.)
`kWh sent out = 0.95 × 394.2 × 10^6 = 374.5 × 10^6 kWh`	5 % is used inside the plant (pumps, fans), not sold.
`Cost per kWh = 750 × 10^6 / 374.5 × 10^6 =` **`Tk 2.00 per kWh`**	

Side note: Also quick marks: maximum demand = average / LF = 45 / 0.5 = 90 MW, so reserve capacity = 100 − 90 = 10 MW.

Memory trick: "K-F-R-D": kWh, Fixed, Running, Divide.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2023 2(d) Given: 100 MW, LF 45 %, capital Tk 105 crore, wages Tk 40 lac, fuel + maintenance Tk 1/kWh, interest 4 %, depreciation 6 %

Energy per year: P × LF × 8760

= 100,000 kW × 0.45 × 8760

= **394.2 × 10^6 kWh**

Interest + depreciation: 10 % × capital

= 0.10 × 105 × 10^7

= **Tk 10.5 × 10^7**

Wages = 40 × 10^5 = Tk 0.4 × 10^7; fuel + maintenance = 1 × 394.2 × 10^6 = Tk 39.42 × 10^7

Total yearly cost: 10.5 + 0.4 + 39.42

= Tk 50.32 × 10^7

= **Tk 50.32 crore**

Cost per kWh: total / kWh

= 50.32 × 10^7 / 394.2 × 10^6

= **Tk 1.28 per kWh**

A6 Economic load sharing between a base load and a peak load plant (derivation)

Skip if short · Possible · 33% · Hard

PAST-PAPER QUESTIONS

- **2024 Q2(a)** [10] Derive the condition for economic load sharing between a base load plant and a peak load plant.

LEARN IT FROM ZERO

Economic load sharing, from zero

Imagine two cars. A **diesel bus** costs a lot to buy but very little per km. A **cheap taxi** costs little to buy but a lot per km. If you need a vehicle every day, the bus is cheaper overall; for a trip once a month, the taxi wins. There is one number of km per year where both cost the same.

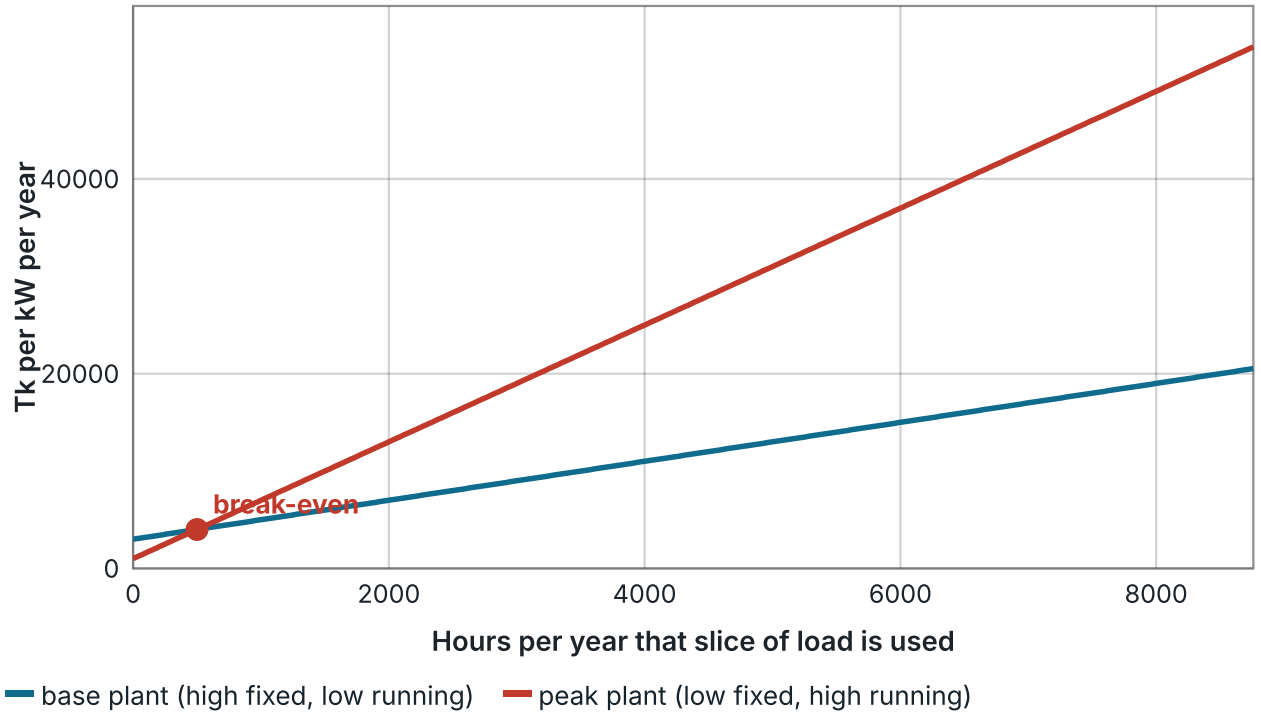
Power plants are the same. Each plant's yearly cost per kW of capacity = fixed part R (Tk/kW/year) + running part P (Tk/kWh) × hours used. A slice of load used for **h** hours a year costs $\mathbf{R_1 + P_1 h}$ on the base plant and $\mathbf{R_2 + P_2 h}$ on the peak plant. They are equal when

$$h^* = (R_1 - R_2) / (P_2 - P_1)$$

Read h* on the LDC (load duration curve): every load that lasts **more** than h* hours goes on the base plant, the top part that lasts less goes on the peak plant. The slide derivation reaches the same line by setting $dC/dM_1 = 0$.

INTERACTIVE **Yearly cost per kW vs hours used: where base and peak plants break even**

Move the sliders and watch the curve change.



Break-even hours h*: **500** h/yr

Loads lasting longer than h* belong on the base plant; the short peak above it on the peak plant.

Before you start: the idea behind the derivation

Each plant's yearly cost has two parts: a **fixed part** paid per kW installed (interest, depreciation) and a **running part** paid per kWh (kilowatt-hour = 1 unit of electrical energy) generated (fuel). A base plant has high fixed cost but cheap running; a peak plant is the opposite.

For a load that lasts only a few hours a year, the cheap-to-build peak plant wins. For a load that lasts many hours, the cheap-to-run base plant wins. The derivation finds the **break-even number of hours**.

2.10 Economic Load Sharing between Base Load and Peak Load Power Stations

Consider a system having load duration curve shown in Fig. 2.5. The load to be supplied by a base load power plant and peak load power plant.

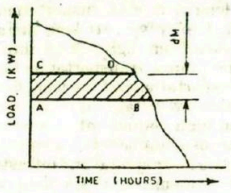


Fig. 2.5

Let C_1 = Annual cost of base load power station

C_2 = Annual cost of peak load power station

$C_1 = R_1$ kW + P_2 kWh

$C_2 = R_2$ kW + P_1 kWh

Assuming $R_1 > R_2$ and $P_1 < P_2$

Let M = Maximum demand of the system

E = Total number of unit (kWh) generated

M_1 = Peak load of base load station

E_1 = No. of units (kWh) generated by base load station

M_2 = Peak load on peak load station = $M - M_1$

E_2 = Units generated by peak load station = $E - E_1$

$C_1 = R_1M_1 + P_1E_1$

$C_2 = R_2 (M - M_1) + P_2 (E - E_1)$

P = Total annual cost of operation of the system

Slide: LDC split into base load (below line AB) and peak load (above); cost equations

$$C = C_1 + C_2$$

$$C = R_1M_1 + P_1E_1 + R_2 (M - M_1) + P_2 (E - E_1)$$

Now for minimum cost

$$\frac{dC}{dM_1} = 0$$

This gives $$\frac{dC}{dM_1} = 0 = R_1 + P_1 \frac{dE_1}{dM_1} - R_2 - P_2 \frac{dE_1}{dM_1}$$

$$\frac{dE_1}{dM_1} = \frac{R_1 - R_2}{P_2 - P_1} \text{ hours.}$$

E_1 the number of units generated by the base load station is represented by area under straight line AB

$\therefore$ dE_1 = Area of strip $ABCD = dM_1 \times AB$

$\therefore$ $$AB = \frac{DE_1}{dM_1} = \frac{R_1 - R_2}{P_2 - P_1}$$

This indicates that for economic load sharing the peak load power station should run for $\frac{R_1 - R_2}{P_2 - P_1}$ hours in one year.

Slide: differentiate the total cost with respect to M_1 and set to zero

Step (write this)	Why / what it means
$C_1 = R_1M_1 + P_1E_1$	Base plant yearly cost. M_1 = its peak (kW), E_1 = its kWh. R_1 = Tk per kW (fixed), P_1 = Tk per kWh (running).
$C_2 = R_2(M - M_1) + P_2(E - E_1)$	Peak plant takes the rest: M = system maximum demand, E = total kWh.
$R_1 > R_2$ and $P_1 < P_2$	Base plant: costly to build, cheap to run. Peak plant: the opposite.
$C = C_1 + C_2$; for minimum cost $dC/dM_1 = 0$	We choose how big the base plant (M_1) should be; minimum is where the slope is zero.
$dC/dM_1 = R_1 + P_1(dE_1/dM_1) - R_2 - P_2(dE_1/dM_1) = 0$	M and E are fixed by the customers, so their derivatives are zero.
$dE_1/dM_1 = (R_1 - R_2)/(P_2 - P_1)$	Rearrange.
dE_1 = area of thin strip ABCD = $dM_1 \times AB \Rightarrow AB = dE_1/dM_1$	Raising the base plant by dM_1 adds a thin horizontal strip of the LDC whose length AB is in hours.
$AB = (R_1 - R_2)/(P_2 - P_1)$ hours	Result.

Conclusion: draw a horizontal line on the LDC at the height where the load lasts exactly **$(R_1 - R_2)/(P_2 - P_1)$ hours** a year. Load below the line goes to the base plant; load above it goes to the peak plant. In words: the peak load plant should run for $(R_1 - R_2)/(P_2 - P_1)$ hours in a year.

> **Side note:** Sanity check with numbers: base plant Tk 10,000/kW-yr extra fixed cost, saves Tk 5/kWh in fuel. Break-even = 10,000/5 = 2000 h. A kW needed for more than 2000 h a year is cheaper on the base plant.

> **Memory trick: "Extra fixed over extra fuel"**: hours = (difference in fixed cost per kW) / (difference in running cost per kWh).

Section B: Diesel, gas turbine, combined cycle, steam plant and auxiliaries

B1 Diesel power plant: layout, working, fuel system, applications, advantages, peak-load use, lubrication

Must study · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2024 Q5(a)** [5] What are the advantages of a diesel power plant as a peak load plant?
- **2024 Q5(b)** [5] Why is the solid injection fuel system more widely used in diesel engines?
- **2023 Q5(a)** [12] With a schematic, describe the working principle of a diesel engine power plant.
- **2023 Q5(b)** [10] What are the applications of diesel power plants? Explain.
- **2022 Q5(a)** [11] Draw the typical layout of a diesel engine power plant. Explain its fuel supply system.
- **2022 Q5(b)** [9] Write the advantages and disadvantages of a diesel power plant.
- **2021 Q5(a)** [20] With a neat sketch, describe the working principle of a diesel engine power plant.
- **2021 Q5(b)** [6] Write the importance of the lubrication system in a diesel power plant.
- **2021 Q5(c)** [9] Explain the fuel supply system of a diesel plant.
- **2020 Q4(a)** [10] Is a diesel power plant suitable as a peak load plant? Describe the working of a diesel plant.

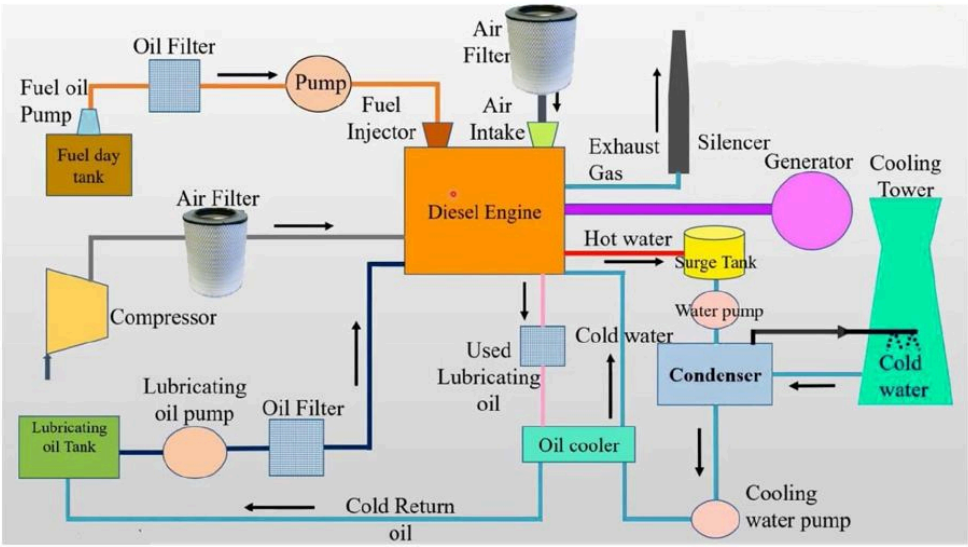
★ REMEMBER IT

- Diesel plant = engine + 5 systems: **F-A-C-L-S**: Fuel, Air intake (and exhaust), Cooling, Lubrication, Starting. Good peak plant: starts in minutes, no standby losses, compact; bad for base: costly fuel, small units.
- Solid injection (high-pressure pump pushes fuel directly through the nozzle) replaced air injection because it needs no big air compressor: simpler, lighter, cheaper.

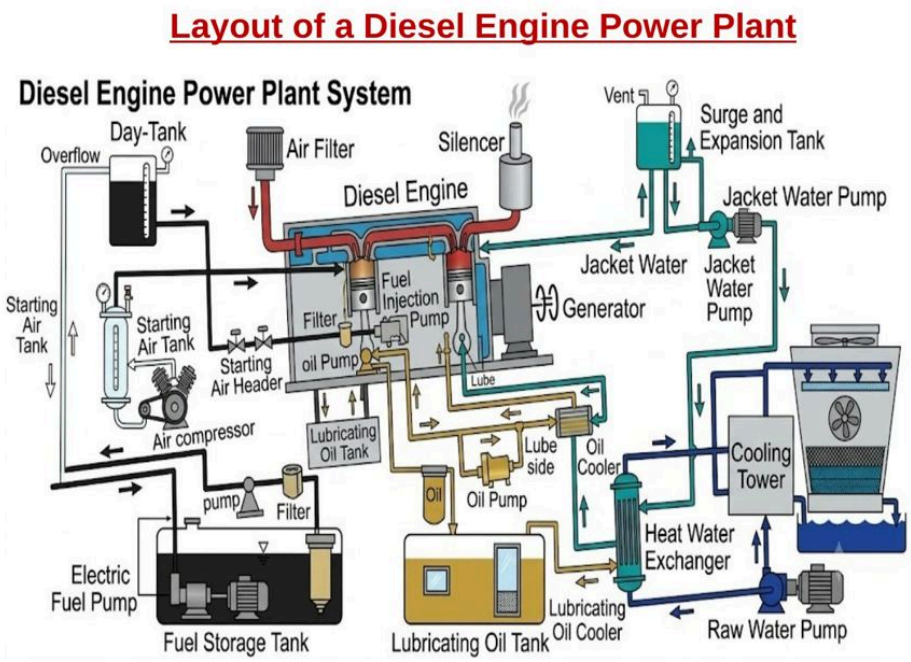
Before you start: what a diesel power plant is

A diesel engine burns fuel inside its cylinders: air is squeezed so hard (compression ratio 14-22) that it gets hotter than about 500 °C, and fuel sprayed in ignites by itself. This is compression ignition (CI). The engine crankshaft is coupled to an **alternator**. Output 2-50 MW per plant.

Layout and working principle (2023 5a, 2021 5a, 2022 5a)

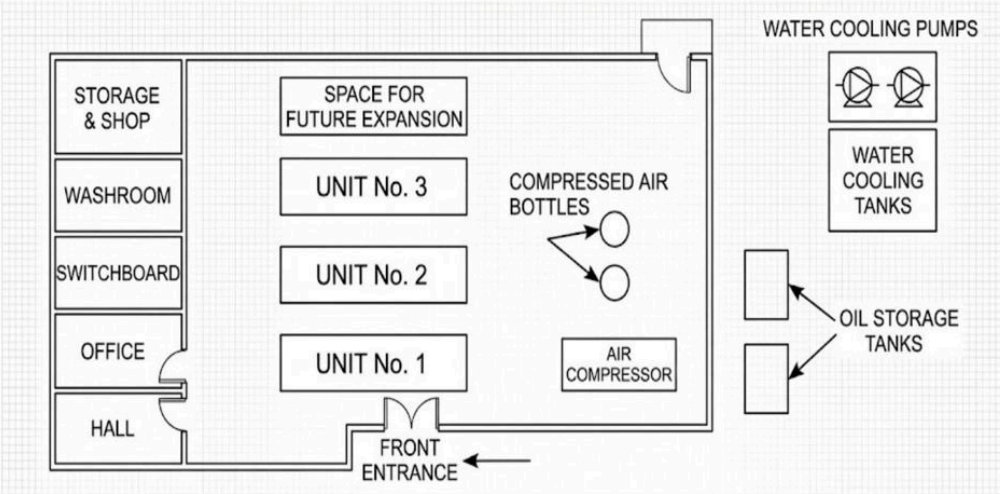


Layout of a diesel engine power plant: fuel, air intake and exhaust, cooling, lubrication and starting systems around the engine-generator



Detailed diesel plant system: day tank, fuel pump, filters, air filter, silencer, jacket water, cooling tower, lubrication oil tank and cooler, starting air tank

Layout of a Diesel Engine Power Plant

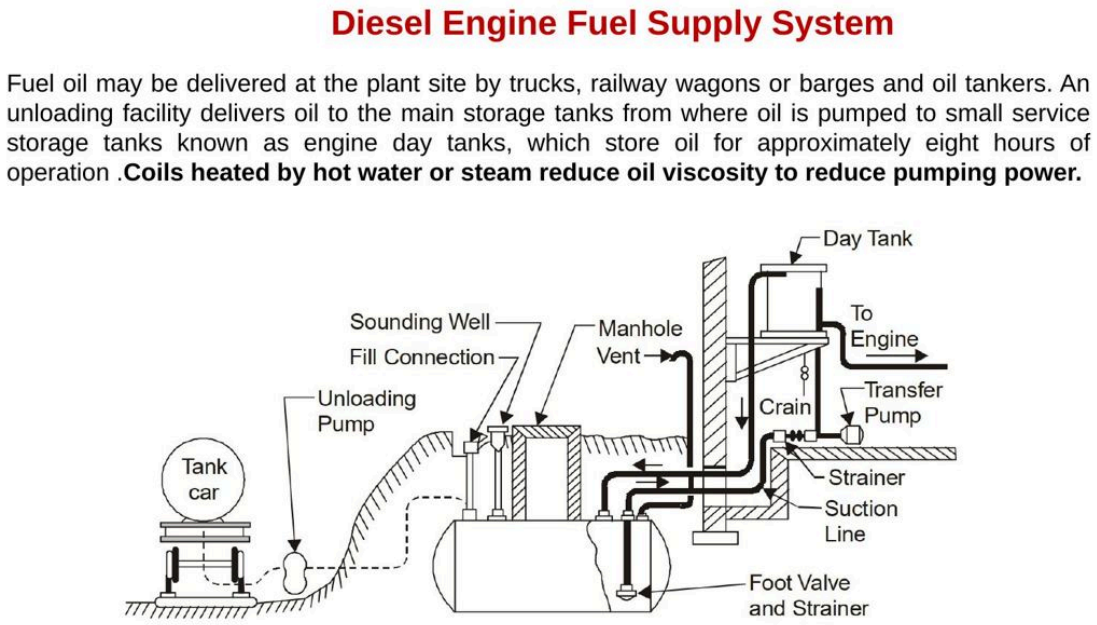


Plan of a diesel station: units 1-3, compressed air bottles, water cooling pumps and tanks, oil storage tanks

Working: fuel oil from the main storage tank is pumped through a strainer to a small **day tank**, then through filters to the **fuel injection pump**, which raises the pressure and sends a measured quantity to the **injector** at the right moment. Air enters through the **air filter** (and supercharger, if fitted), is compressed in the cylinder, and the injected fuel burns. The hot gases push the piston; the crankshaft turns the **alternator**, which supplies power to the bus-bars. Exhaust gases leave through the exhaust manifold and **silencer** to the stack. The **cooling system** circulates water through the cylinder jackets and rejects heat in a radiator/cooling tower; the **lubrication system** feeds oil to bearings and pistons and cools it in an oil cooler. A **starting system** (compressed air) turns the engine at start, and a **governor** keeps the speed constant as load changes.

Step (write this)	Why / what it means
`1. Fuel supply system`	Storage tank → transfer pump → day tank → filter → injection pump → injector.
`2. Air intake and exhaust system`	Air filter, (supercharger), intake manifold; exhaust manifold, silencer, stack.
`3. Cooling system`	Removes 25-35 % of fuel heat from cylinder walls (see B2).
`4. Lubrication system`	Oil pump, filter, oil cooler; reduces friction and wear.
`5. Starting system`	Compressed air, electric motor or auxiliary engine (see B2).
`6. Governing system`	Varies fuel quantity per stroke to keep speed (frequency) constant.

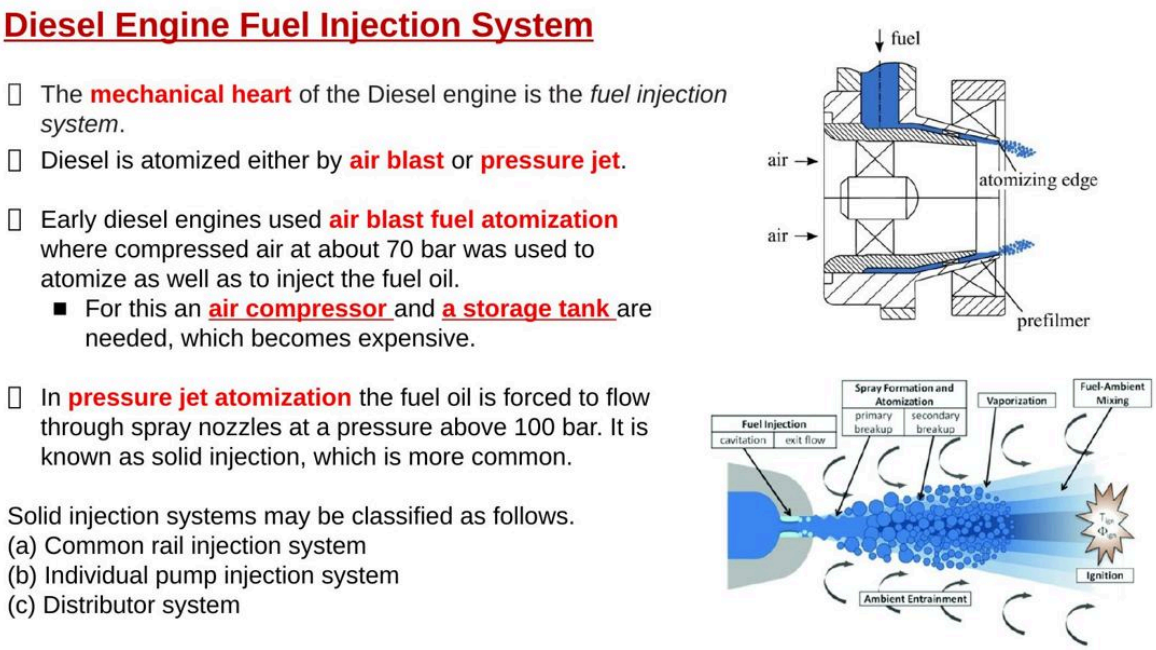
Fuel supply system (2022 5a, 2021 5c)



Diesel engine fuel supply system: tank car unloading, main storage, transfer pump, day tank, strainer

Fuel oil is delivered by road tanker, rail wagon or barge and **unloaded** into large main storage tanks. From these it is pumped to small **day tanks** that hold about 8 hours' supply and feed the engines by gravity. Steam or hot-water coils may heat heavy oil to reduce viscosity. Strainers and filters remove dirt. The **fuel injection system** then (1) delivers oil from the day tank, (2) raises its pressure (100-200 bar or more) for atomisation, (3) meters the exact amount per cycle, (4) controls the timing of injection, and (5) sprays it into the cylinder as fine droplets for complete mixing.

Why solid injection is more widely used (2024 5b)

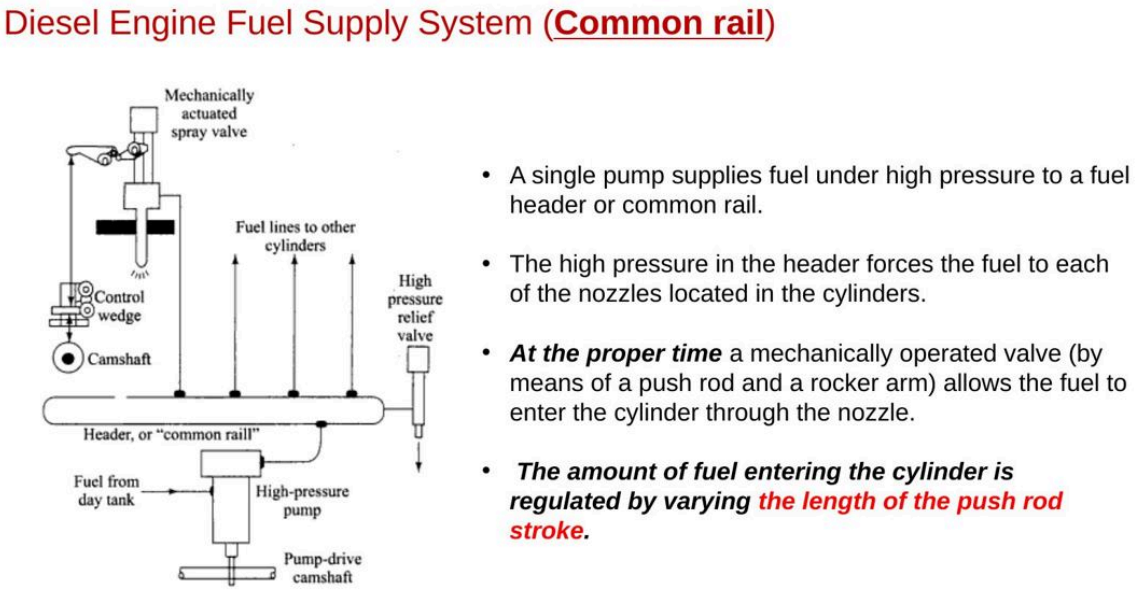


Air blast injection (needs an air compressor and storage tank) vs pressure (solid) injection

In **solid (airless) injection** the fuel alone is raised to high pressure by a pump and forced through the nozzle, which atomises it. In the older **air-blast injection**, high-pressure air (about 70 bar) from a separate compressor blows the fuel in. Solid injection is preferred because:

1. No multi-stage **air compressor** and air storage tank are needed → simpler, lighter, cheaper engine.
2. The compressor would consume 6-10 % of engine power, so solid injection gives **higher net power and efficiency**.
3. Easier, more accurate **metering and timing** of fuel; good response to load changes.
4. Less maintenance, less space, fewer parts that can fail.

Types of solid injection: **common rail** (one high-pressure pump feeds a header for all cylinders), **individual pump** (one pump per cylinder), **distributor** (one pump meters and a distributor sends fuel to cylinders in firing order).



Common rail system

Applications (2023 5b)

1. **Peak load plant** along with thermal or hydro stations.
2. **Standby / emergency plant**: hospitals, telephone exchanges, radio stations, cinemas, industries (load shedding).
3. **Mobile plants** on trailers for temporary or emergency supply and construction sites.
4. **Central station** for small towns or islands where demand is small (5-10 MW) and no grid exists.
5. **Starting stations** to run auxiliaries of large steam plants; **nursery station** for a growing town before the grid arrives.
6. Transport: locomotives, ships.

Advantages and disadvantages (2022 5b, 2020 4b)

Advantages	Disadvantages
Simple design and layout; small space	High cost of fuel (diesel)
Quick start and pick-up of load	High lubrication and maintenance cost
Can be located near the load centre	Limited capacity (up to about 50 MW)
Little cooling water needed; no ash	Cannot work under sustained overload
No standby losses; easy fuel handling and storage	Noisy; exhaust pollution
Low capital cost per kW; quick to build	Life shorter than steam plant (about 2-5 years less)
Higher thermal efficiency than a small steam plant	Repair costs high

Is diesel suitable as a peak load plant? Advantages (2024 5a, 2020 4a)

Yes. A diesel plant is well suited to peak load because:

1. It can be **started and loaded within a few minutes** from cold; no boiler warm-up.
2. **No standby losses**: when stopped it burns no fuel.
3. **Low capital cost**, so it is economical even though it runs only a few hours a day.
4. Good efficiency at part load; responds quickly to load changes.
5. Small size and can be placed near the load; little water needed.

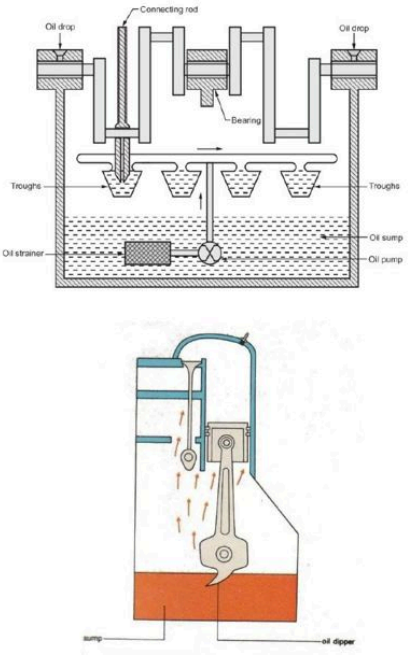
Importance of lubrication (2021 5b)

- Reduces friction and wear between moving parts (piston-liner, bearings) by forming a thin oil film.
- Carries away heat from bearings and pistons (cooling).
- Seals the gap between piston rings and cylinder so gas does not leak.
- Cleans: carries away carbon and metal particles to the filter.
- Absorbs shocks and reduces noise; protects against corrosion.

Diesel Engine Lubrication System

Wet sump lubrication system

Splash system: This system is used on some ***small four stroke stationary engines***. In this case the caps on the big end bearings of connecting rods are provided with scoops which, **when the connecting rod is in the lowest position, just dip into oil troughs and thus directs the oil through holes in the caps to the big end bearings.** Due to splash of oil it reaches the lower portion of the cylinder walls, crankshaft and other parts requiring lubrication. Surplus oil eventually flows back to the oil sump. Oil level in the troughs is maintained by means of a oil pump which takes oil from sump, through a filter. *Splash system is suitable for low and medium speed engines having moderate bearing load pressures.* For high performance engines, which normally operate at high bearing pressures and rubbing speeds this system does not serve the purpose.



Wet sump splash lubrication

Memory trick: Six systems: **"FACLSG"** → "Fresh Air Cools Lazy Sleepy Guys" = Fuel, Air, Cooling, Lubrication, Starting, Governing.

B10 Feed water: impurities, troubles in boiler, treatment methods (lime-soda, zeolite) Must study Very likely · 93% Easy

PAST-PAPER QUESTIONS

- **2024 Q8(a)** [6] Discuss the adverse effects of using raw water in a boiler.
- **2024 Q8(b)** [12] What methods are used to treat water in a power plant? Explain one commonly used method.
- **2023 Q8(a)** [11] Why is treatment of raw water necessary? Explain the zeolite process of softening feed water.
- **2022 Q8(a)** [12] What impurities may be present in natural water? What problems do they cause in a power plant?
- **2021 Q8(a)** [15] What are the methods of feed water treatment? Describe the chemical treatment method with a sketch.

★ REMEMBER IT

- Raw water troubles: **S-C-F-C** = Scale, Corrosion, Foaming/priming, Caustic embrittlement. Zeolite: Na_2Z swaps its sodium for the Ca/Mg that cause hardness; regenerate with brine (NaCl).

Before you start: why water quality matters

A boiler boils off pure steam and leaves every dissolved salt behind. Over months these salts build up into hard scale on the hot tubes, just like the white crust in a kettle, but at 500 °C the result is a burst tube.

Impurities in natural water and their problems (2022 8a, 2024 8a)

Troubles Caused by Impurities in Boiler Water

Boiler feed water contains dissolved and suspended impurities that create serious operating problems in steam generators and associated equipment. The principal troubles caused by impurities are:

- **Scale formation**
- **Corrosion**
- **Carryover**
- **Embrittlement**

These problems may reduce heat transfer, damage boiler tubes and drums, lower plant efficiency, contaminate steam, and in severe cases lead to failure of major compon





Troubles caused by impurities: scale formation, corrosion, carry-over, embrittlement

Impurity	Problem in a power plant
Dissolved salts of Ca and Mg (carbonates, sulphates, chlorides): hardness	Scale on boiler tubes → poor heat transfer, overheating, tube failure; more fuel
Dissolved gases: O_2, CO_2	Corrosion and pitting of boiler, economiser, pipes
Suspended matter: mud, silt, organic matter	Sludge, choking of tubes, foaming
Oil, soap, alkalis (NaOH in excess)	Priming and foaming (water carried over with steam), caustic embrittlement
Mineral acids, silica	Corrosion; silica deposits on turbine blades

Adverse effects of raw water in a boiler (2024 8a): (1) **scale formation**, lowering heat transfer and overheating tubes until they burst; (2) **corrosion** of metal by dissolved O_2, CO_2 and acids; (3) **priming and foaming** causing **carry-over** of water and solids with steam, damaging superheater and turbine blades; (4) **caustic embrittlement** (cracking of boiler plates at riveted joints by concentrated NaOH); (5) **sludge** blocking tubes; all lead to lower efficiency, more fuel, frequent shutdowns and danger of explosion.

Methods of treatment (2024 8b, 2021 8a)

Step (write this)	Why / what it means
`1. Mechanical: sedimentation, coagulation (alum), filtration`	Removes suspended solids.
`2. Thermal: deaeration (heating to about 110 °C to drive out` O_2`), evaporation/distillation`	Removes dissolved gases; pure distilled make-up.
`3. Chemical: lime-soda process, zeolite (ion exchange), internal treatment (phosphates)`	Removes hardness.
`4. Demineralisation (cation + anion exchangers)`	Removes almost all dissolved salts; used for high-pressure boilers.
`5. Blow-down`	Drains part of concentrated boiler water periodically.

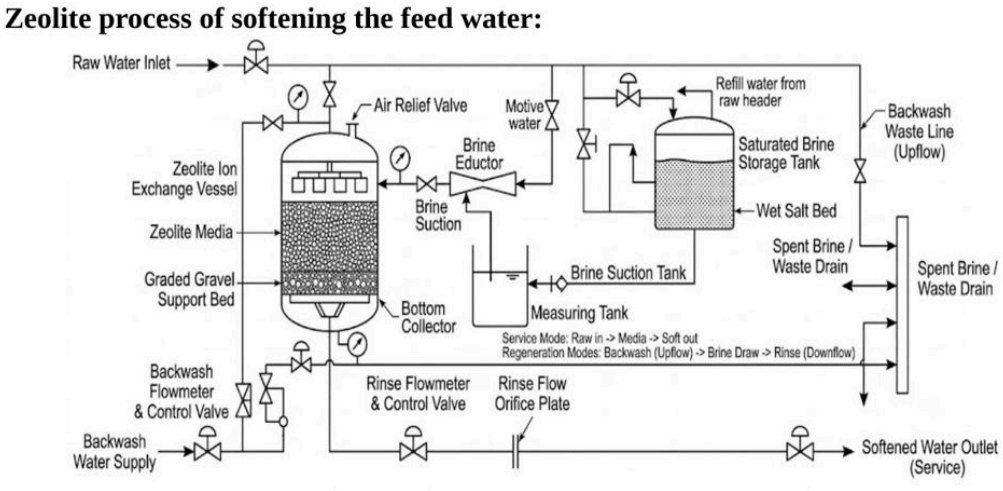
Chemical treatment: lime-soda process (2021 8a)

> **Side note:** Lime = $Ca(OH)_2$, soda = Na_2CO_3. They turn soluble hardness salts into insoluble $CaCO_3$ and $Mg(OH)_2$ that settle out.

$$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3\downarrow + 2H_2O$$
$$MgCl_2 + Ca(OH)_2 \rightarrow Mg(OH)_2\downarrow + CaCl_2$$
$$CaCl_2 + Na_2CO_3 \rightarrow CaCO_3\downarrow + 2NaCl$$
$$CaSO_4 + Na_2CO_3 \rightarrow CaCO_3\downarrow + Na_2SO_4$$

Raw water and measured doses of lime and soda are mixed in a reaction tank (hot process at about 95 °C is faster), the precipitate settles in a sedimentation tank, and clear water is filtered. Residual hardness is about 10-30 ppm, so it is often followed by zeolite softening.

Zeolite (ion exchange) process (2023 8a, commonly used method for 2024 8b)



As shown in figure, the raw water enters the zeolite softener at the top. It flows downwards through the bed although the system can be by passed while back washing, if necessary. The figure also shows arrangements for the periodic back washing with brine needed by this system.

Zeolite softener: raw water flows down through the zeolite bed; brine tank for regeneration; backwash connections

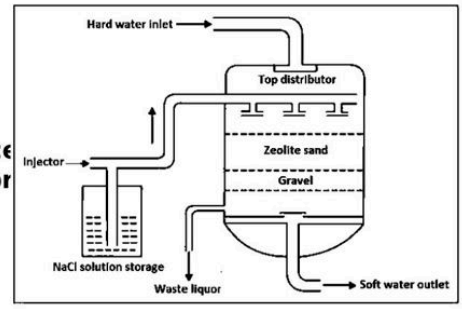
Ion Exchange Process

The ion exchange process is a water treatment method used to remove undesirable dissolved ions from water and replace them with other harmless ions.
In boiler water treatment, the main purpose is usually to remove **hardness-causing ions**, mainly:

- **Calcium ions (Ca^{2+})**
- **Magnesium ions (Mg^{2+})**

These are replaced by either:

- **Sodium ions (Na^{+})** in the **sodium zeolite proce**
- **Hydrogen ions (H^{+})** in the **hydrogen zeolite pr**



Sodium zeolite process
In this process, the zeolite contains **sodium ions**.
When hard water passes through the zeolite bed:

- calcium and magnesium ions from water are retained by zeolite
- sodium ions are released into water

So the hardness is r

$$Ca(HCO_3)_2 + Na_2Z = CaZ + 2NaHCO_3 \quad \ldots$$
$$CaSO_4 + Na_2Z = CaZ + Na_2SO_4 \quad \ldots$$
$$CaCl_2 + Na_2Z = CaZ + 2NaCl \quad \ldots$$

(Z is the symbol used for zeolite).

So:

- **Ca^{2+} is removed**
- **Na^{+} is added**

Water becomes **soft**, because sodium salts do not form hard scale like calcium and magnesium salts do.

Ion exchange reactions

Zeolite is a sodium alumino-silicate (Na_2Z) that swaps its sodium ions for calcium and magnesium ions.

Step (write this)	Why / what it means
Softening: $Ca(HCO_3)_2 + Na_2Z \rightarrow CaZ + 2NaHCO_3$	Hard water flows down through the zeolite bed; Ca and Mg stay in the bed.
$CaSO_4 + Na_2Z \rightarrow CaZ + Na_2SO_4$; $MgCl_2 + Na_2Z \rightarrow MgZ + 2NaCl$	Sodium salts do not form scale, so water is soft (near 0 hardness).
Regeneration: $CaZ + 2NaCl \rightarrow Na_2Z + CaCl_2$	When the bed is exhausted, brine (10 % NaCl) is passed through; $CaCl_2$ is washed to drain.
Backwash and rinse, then back in service	Loosens the bed and removes dirt.

Advantages: water of almost zero hardness, compact, automatic, no sludge. **Limitations:** does not remove dissolved gases or bicarbonate alkalinity; turbid water clogs the bed; sodium salts remain.

> **Memory trick:** Zeolite = **"swap shop"**: gives Na, takes Ca/Mg; salt water (brine) recharges it.

B6 Combined cycle power plant (gas turbine + steam turbine) Must study

Very likely · 93% Easy

PAST-PAPER QUESTIONS

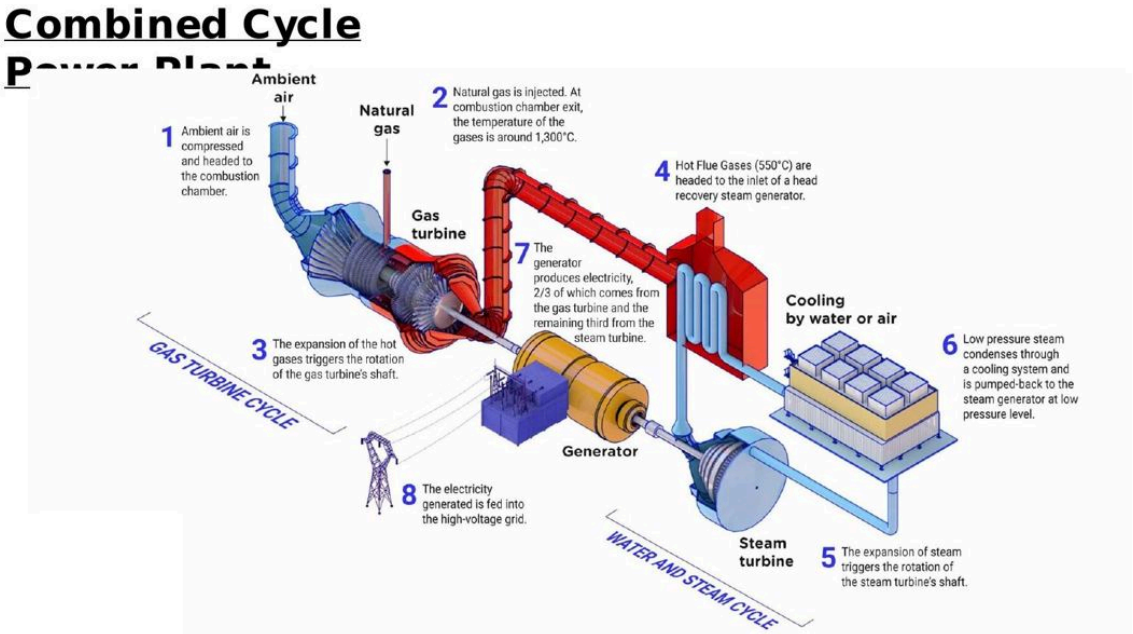
- **2024 Q7(c)** [10] What are the advantages of a combined cycle power plant? Draw the P-v and T-s diagrams.
- **2023 Q6(c)** [10] What is meant by a combined cycle power plant? What are its advantages?
- **2022 Q6(c)** [14] Explain the combined working of gas turbine and steam power plant.
- **2021 Q6(a)** [15] What is a combined cycle power plant? Explain with a diagram.

★ REMEMBER IT

- Combined cycle = **gas turbine on top, steam turbine below**. The 500–600 °C GT (gas turbine) exhaust boils water in a heat recovery steam generator (HRSG) that drives a steam turbine. One fuel, two power outputs: about 55–60 % efficiency vs 35–40 % for either alone.

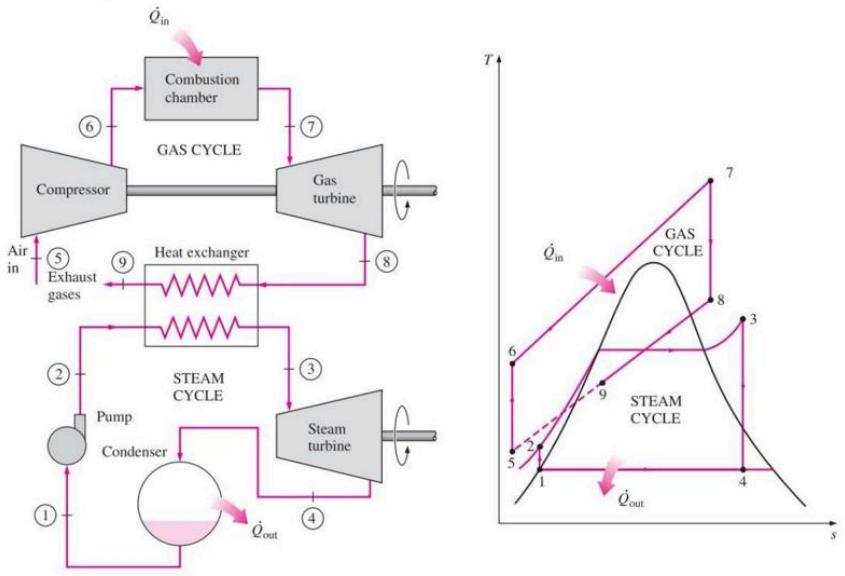
Before you start: the idea in one line

A gas turbine throws out exhaust at about 500-600 °C. That is hot enough to boil water. So put a boiler (a heat recovery steam generator) in the exhaust and run a steam turbine too: two turbines, one fuel.



Combined cycle plant: gas turbine cycle (1-4), HRSG (heat recovery steam generator) makes steam (5), steam turbine (6-7), condenser and cooling

Combined Cycle Power



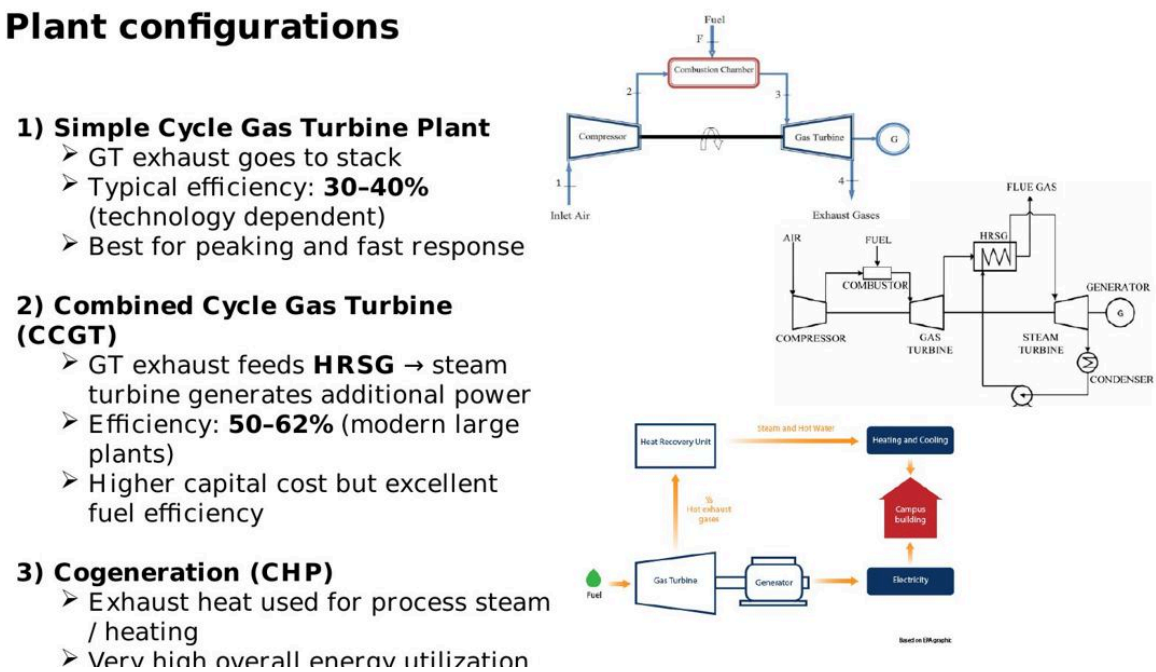
Combined gas-steam cycle: layout and T-s diagram with the Brayton cycle above the Rankine cycle

Meaning and combined working (2023 6c, 2022 6c, 2021 6a)

A **combined cycle power plant** uses a **gas turbine (Brayton) cycle as the topping cycle** and a **steam turbine (Rankine) cycle as the bottoming cycle**. Air is compressed, fuel (natural gas) is burned and the gases expand in the gas turbine, driving a generator. The exhaust (about 550 °C) passes through a **heat recovery steam generator (HRSG)**, where it heats feed water into superheated steam. This steam expands in a steam turbine that drives a second generator, is condensed in the condenser, and pumped back to the HRSG. Supplementary firing in the HRSG can boost steam output. Typical split: two-thirds power from GT, one-third from ST (steam turbine).

Step (write this)	Why / what it means
`Overall n = n_GT + n_ST(1 - n_GT)`	Steam cycle runs on the heat the GT rejects.
`E.g. 0.38 + 0.35 × (1 - 0.38) = 0.38 + 0.217 = 0.597 ≈ 60 %`	Either cycle alone: 35-40 %.

Advantages (2024 7c, 2023 6c)



Plant configurations: simple cycle GT (30-40 %), combined cycle (50-62 %), cogeneration

1. **High overall efficiency** (50-62 %), the highest of fossil plants, so less fuel per kWh (kilowatt-hour = 1 unit of electrical energy).
2. **Lower CO_2, NOx (nitrogen oxides) and SOx (sulphur oxides) per kWh** when burning natural gas.
3. **Lower capital cost and shorter construction** than coal plants; can be built in phases (GT first, steam part later).
4. **Quick start of the GT part**; good load-following.
5. Less cooling water than a steam plant of same output (only the steam part needs a condenser).
6. Smaller land area; can supply process steam (cogeneration); reliable mature technology.

P-v and T-s diagrams (2024 7c)

T-s: the Brayton cycle sits at high temperature (1 → 2 compression, 2 → 3 heat addition up to about 1400 °C, 3 → 4 expansion, 4 → 1 exhaust heat rejection). The Rankine cycle sits below it (pump, HRSG heating to about 550 °C, steam turbine expansion, condensation). The exhaust line 4 → 1 of the gas cycle supplies the heat for the steam cycle's heating line. **P-v:** draw the Brayton loop (two constant-pressure lines joined by two isentropic curves) and, separately, the Rankine loop with its liquid line and vapour dome; the shared heat transfer is the HRSG.

> **Memory trick: "Gas on top, steam at the bottom, HRSG in the middle"**. Efficiency rule: 0.4 + 0.35 × 0.6 = 0.6.

B4 Improving gas turbine efficiency: regeneration, intercooling, reheating (and all three together) Must study Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2024 Q6(b)** [10] With P-v and T-s diagrams, analyse the performance of a gas turbine plant when an intercooler, a reheater and a regenerator are used together.
- **2023 Q7(a)** [12] Explain the effect of regeneration on the work output and efficiency of a gas turbine plant.
- **2022 Q6(b)** [16] Explain the methods used to improve the thermal efficiency of a gas turbine plant. Is it possible to increase efficiency using exhaust gas?
- **2021 Q7(a)** [15] What are the purposes of intercooling and reheating in a gas turbine plant? Describe.
- **2020 Q5(a)** [12] Explain the various methods used for improvement of thermal efficiency of a gas turbine power plant.

LEARN IT FROM ZERO

Regeneration, intercooling and reheating, from zero

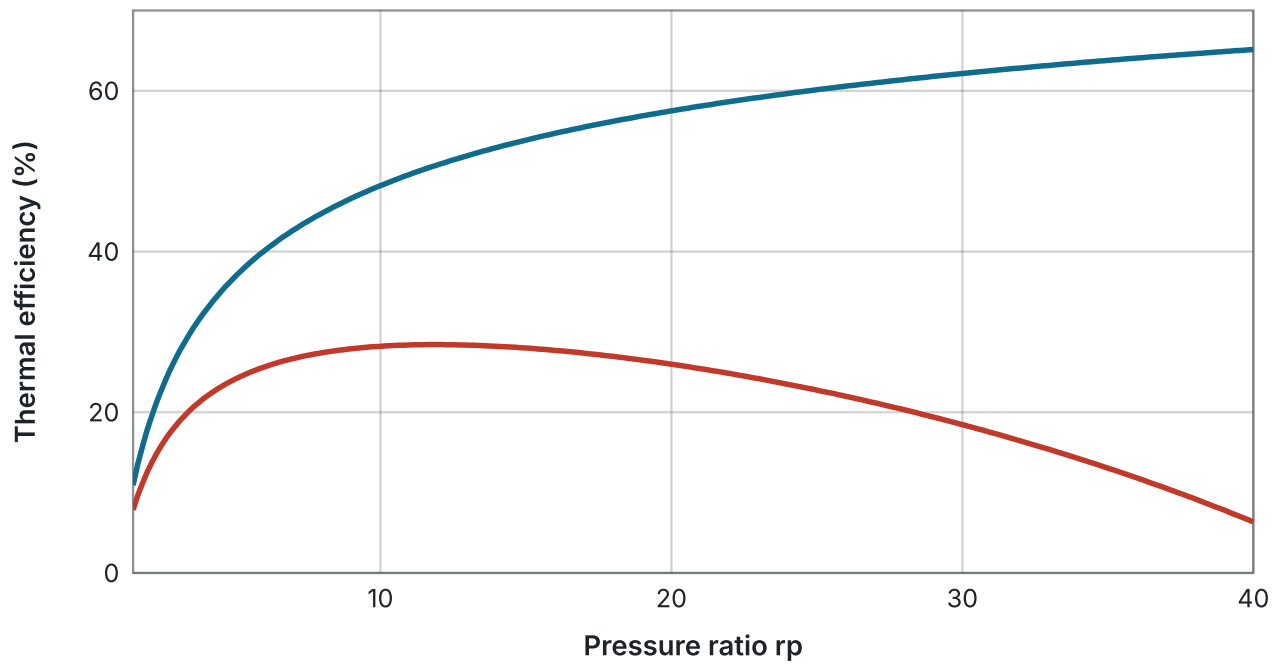
A simple gas turbine has two weak points. (1) The **exhaust leaves very hot** (500–800 °C): heat thrown away. (2) The **compressor eats 40–70 %** of the turbine's work.

- **Regenerator** fixes (1): a heat exchanger uses the hot exhaust to preheat the compressed air before the combustor, so less fuel is burnt for the same work. Efficiency up, work the same. It only helps when the exhaust is hotter than the compressor exit (low pressure ratio).
- **Intercooling** fixes (2): compress in two stages and cool the air in between. Cold air is easier to compress, so compressor work falls and net work rises. Alone it lowers efficiency (more fuel to heat the colder air) unless combined with a regenerator.
- **Reheating**: expand in two stages and add heat in between. The turbine gives more work. Alone it also wastes hotter exhaust, so again pair it with a regenerator.

All three together approach the **Ericsson cycle**, the gas-turbine version of the ideal Carnot efficiency.

INTERACTIVE **Brayton cycle: efficiency and net work vs pressure ratio**

Move the sliders and watch the curve change.

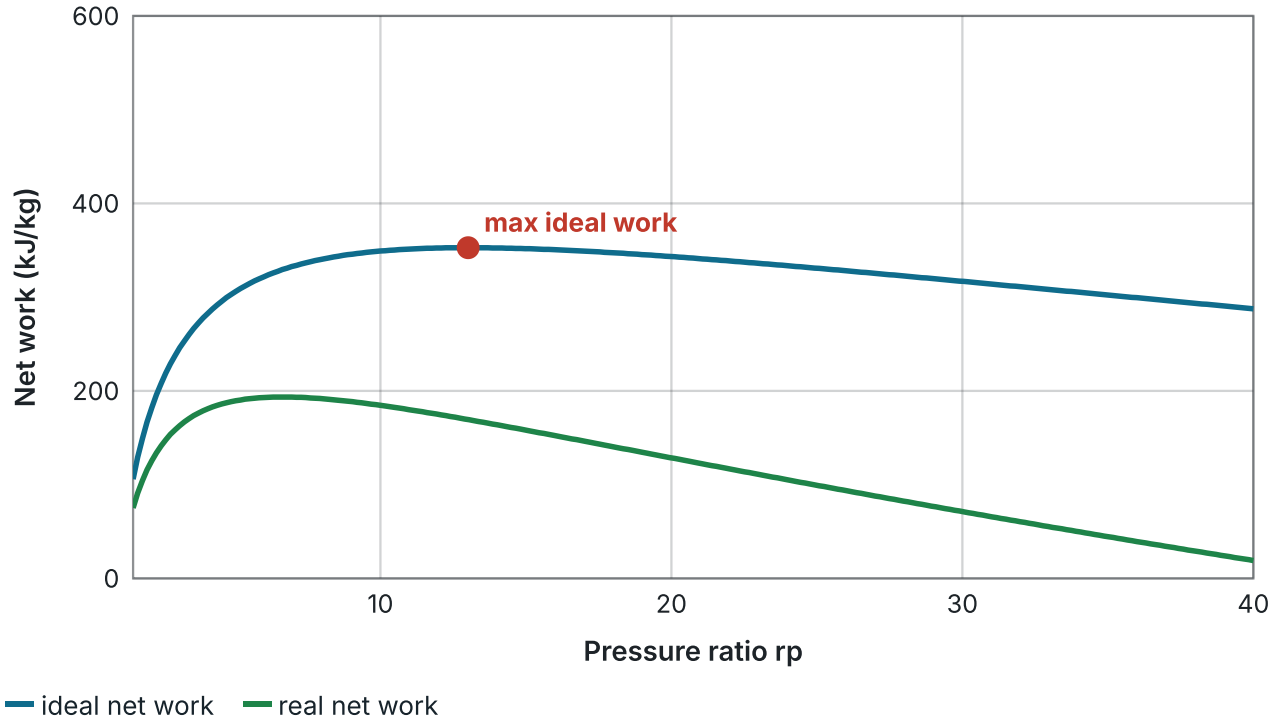


Optimum r_p for max ideal work: **13.0**

Ideal efficiency only rises with r_p; real efficiency peaks and then falls because compressor losses grow. Hotter T_3 moves everything up.

INTERACTIVE **Brayton cycle: net work vs pressure ratio**

Move the sliders and watch the curve change.

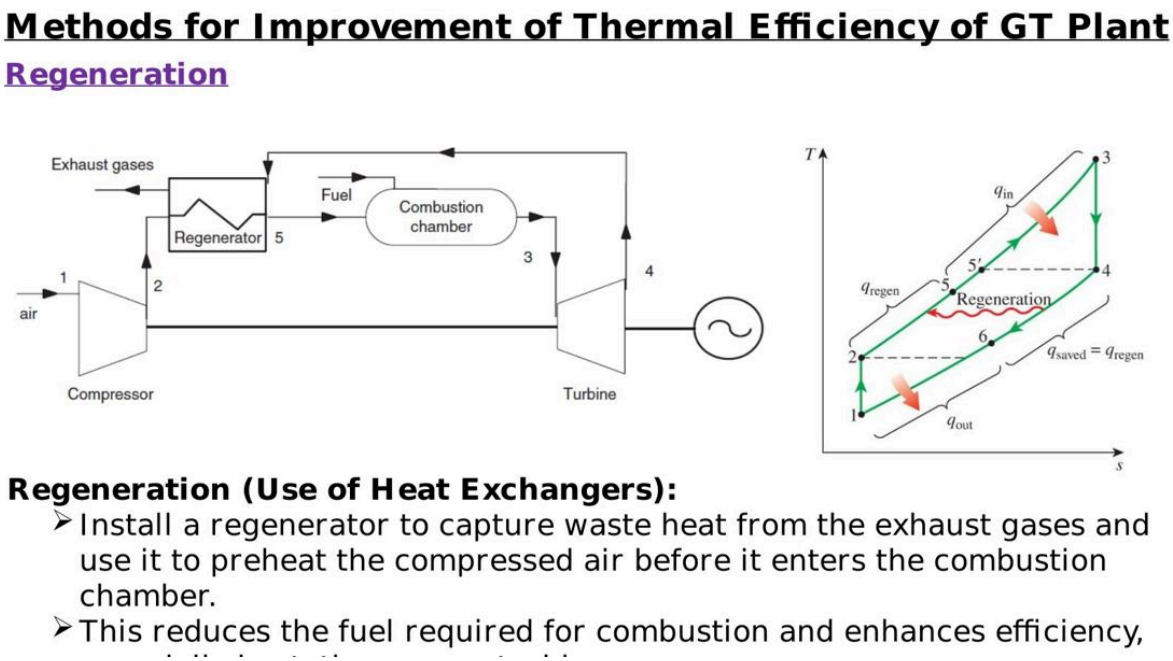


The ideal peak is where $T_2 = T_4$ (2023 6d).

Before you start: why a simple gas turbine is inefficient

Two big losses: (1) the exhaust leaves very hot (450-600 °C), carrying heat away; (2) the compressor eats about 50-60 % of the turbine work. Every method below attacks one of these. On the T-s diagram, the area enclosed = net work, and the efficiency = net work / heat added.

1. Regeneration (2023 7a, 2022 6b)



Regenerator: turbine exhaust (4) heats compressed air (2 → 5) before the combustion chamber; T-s diagram

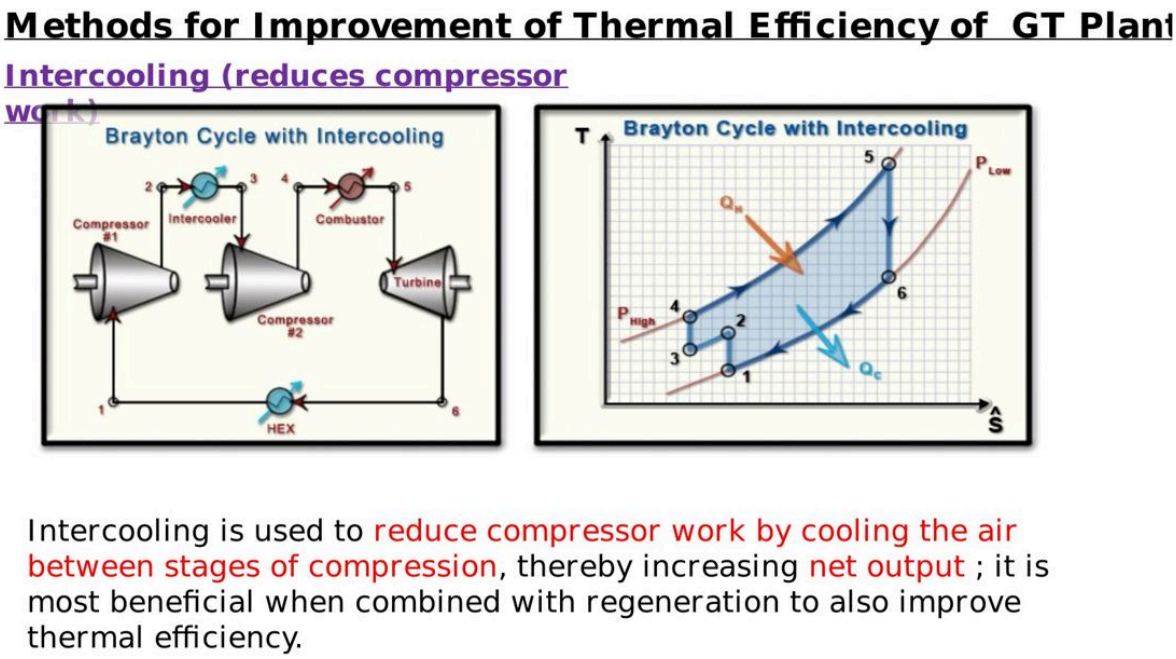
A heat exchanger (**regenerator**) uses the hot turbine exhaust to preheat the air leaving the compressor before it enters the combustion chamber. So part of the heat that would be thrown away is reused.

Step (write this)	Why / what it means
Heat added without regenerator: $q_{in} = cp(T3 - T2)$	Fuel must heat the air from T2 to T3.
With regenerator: $q_{in} = cp(T3 - T5)$, $T5 > T2$	Air arrives already warm at T5, so less fuel.
Work: $w_{net} = cp(T3 - T4) - cp(T2 - T1)$: unchanged	Compressor and turbine are the same.
$n = w_{net} / q_{in}$ increases	Same work, less heat.
Effectiveness $\varepsilon = (T5 - T2)/(T4 - T2)$	1 for an ideal regenerator, 0.6-0.85 in practice.

Effect on work output and efficiency (2023 7a): regeneration **does not change the net work** per kg (it may slightly reduce it due to pressure drop) but **raises efficiency** by cutting fuel. It is possible only when the exhaust temperature T4 is higher than compressor exit T2, which happens at **low pressure ratios**; at high pressure ratios T2 > T4 and regeneration is useless. **So yes, efficiency can**

be increased by using the exhaust gas, either in a regenerator or in a waste-heat boiler of a combined cycle (B6).

2. Intercooling (2021 7a)

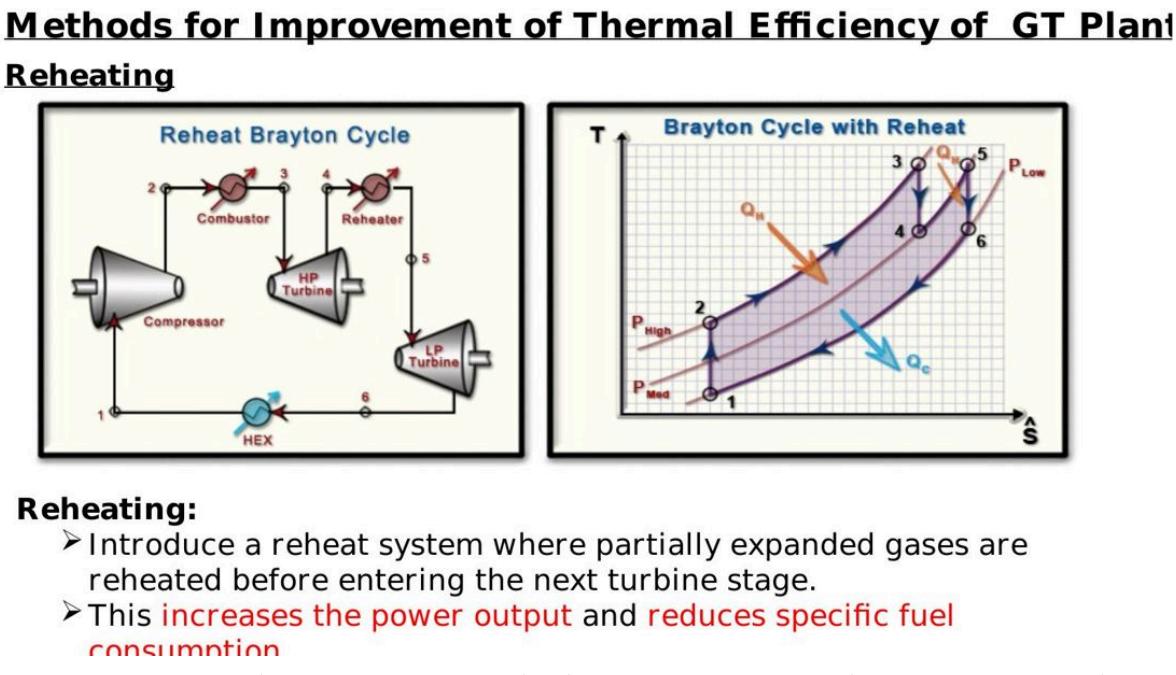


Two-stage compression with intercooler; T-s diagram

Compression is split into two (or more) stages, and the air is cooled back to (near) the inlet temperature in an **intercooler** between them. Cool air is denser and easier to compress, so **compressor work falls** and **net work rises**. Alone, it lowers efficiency a little (air enters the combustor cooler, so more fuel), but combined with regeneration it raises efficiency because the extra fuel is supplied by exhaust heat.

> **Side note:** Purpose in one line: intercooling reduces compressor work (moves compression toward isothermal).

3. Reheating (2021 7a)

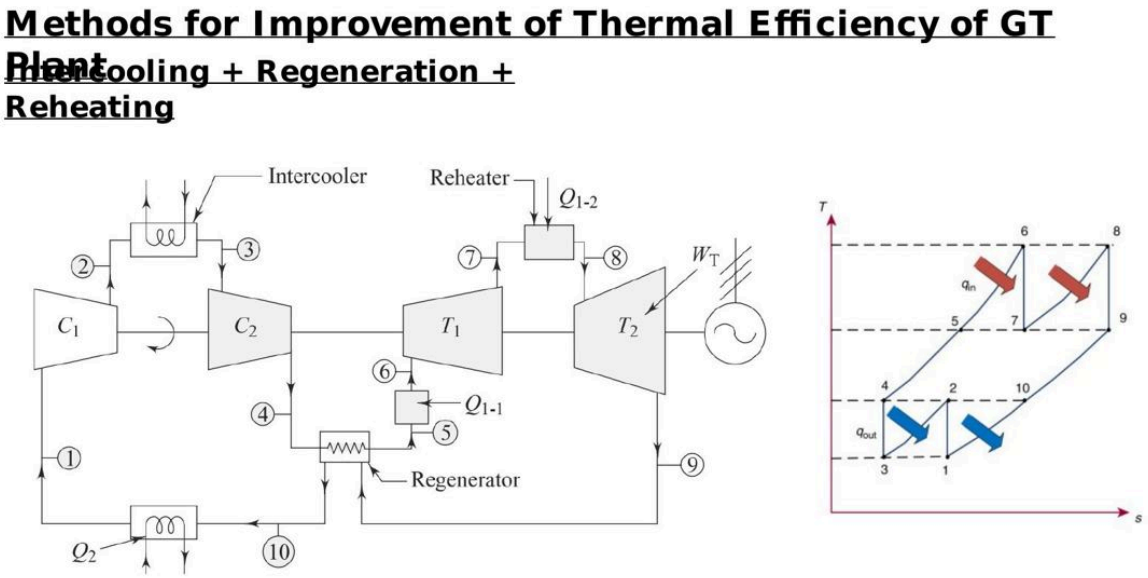


Reheat: gas expands in HP (high pressure) turbine, is reheated, then expands in LP (low pressure) turbine

Expansion is split into two turbines; after the high-pressure turbine the gas is **reheated** in a second combustor (there is plenty of excess air) back to about the maximum temperature, then expanded in the low-pressure turbine. **Turbine work increases**, so net work increases. Alone it raises exhaust temperature and lowers efficiency, but with a regenerator that hot exhaust is reused.

> **Side note:** Purpose in one line: reheating increases turbine work (moves expansion toward isothermal).

All three together (2024 6b)



Intercooling + regeneration + reheating: layout and T-s diagram

Step (write this)	Why / what it means
`1→2 LP compressor; 2→3 intercooler (cooled back to T1); 3→4 HP compressor`	Two-stage compression with intercooling: compressor work ↓.
`4→5 regenerator (heated by exhaust 9→10)`	Free heat from exhaust: fuel ↓.
`5→6 combustion chamber; 6→7 HP turbine; 7→8 reheater; 8→9 LP turbine`	Two-stage expansion with reheat: turbine work ↑.
`9→10 exhaust gives heat in regenerator, then leaves`	Exhaust leaves cooler.
`Result: $w_{net}$ ↑ and n ↑; with many stages the cycle approaches the Ericsson cycle with Carnot efficiency`	Heat is added and rejected nearly at the highest and lowest temperatures.

P-v diagram: two compression curves at the low-pressure end separated by a constant-pressure cooling line, and two expansion curves separated by a constant-pressure reheat line; the enclosed area (net work) is larger than the simple Brayton loop. **T-s diagram:** compression stays at low temperature (zig-zag near T1), expansion stays at high temperature (zig-zag near T_{max}), and the regenerator transfers heat horizontally from the exhaust to the compressed air (shown in the figure).

> **Memory trick: "Re-Cycle, Inter-Cool, Re-Heat"**: Regeneration saves fuel (η ↑, work same); Intercooling cuts compressor work; Reheat adds turbine work. Only regeneration raises efficiency on its own.

B11 Draught: natural vs mechanical (forced, induced, balanced), chimney and its height formula

Must study | Very likely · 100% | Medium

PAST-PAPER QUESTIONS

- **2024 Q8(d)** [5] Why are draught and chimney used in a power plant?
- **2023 Q8(b)** [8] What is draught? Compare natural and mechanical draught.
- **2022 Q8(b)** [8] What is a balanced draught system? Describe its working with a neat sketch.
- **2021 Q8(b)** [10] Write the purpose of a chimney. Derive an expression for the height of a chimney.
- **2020 Q6(a)** [8] What is draught? Write its importance. Compare natural and mechanical draught.

UNDERSTAND **Read this to understand; you do not need to write it**

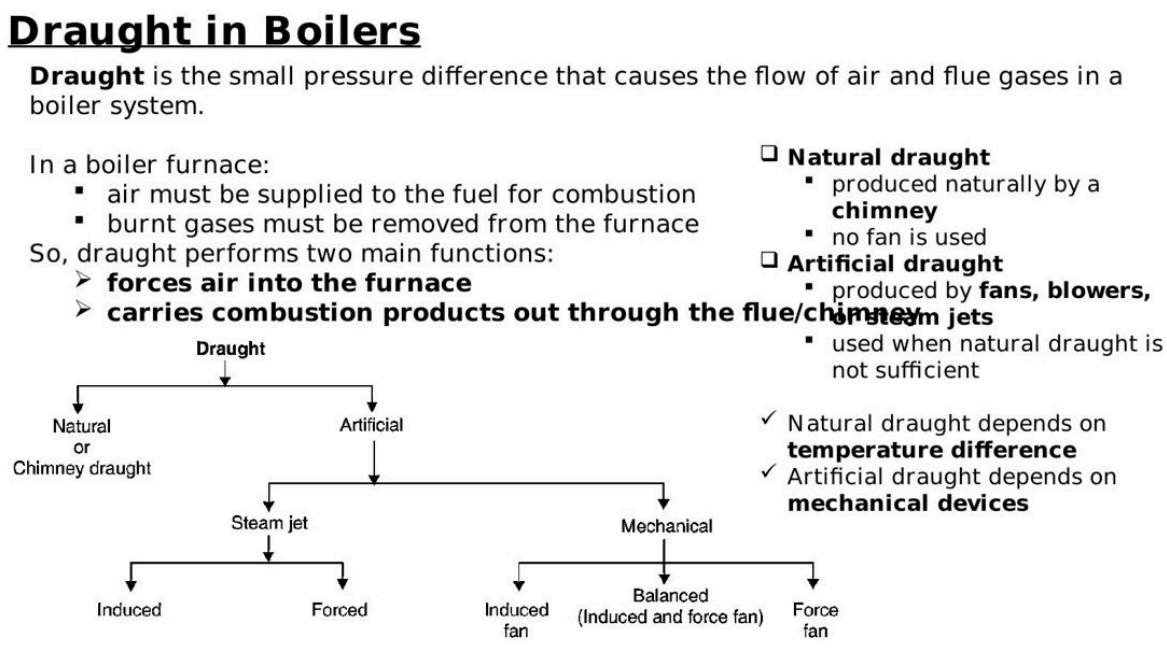
Draught is the small pressure difference that pulls air through the furnace and pushes flue gas up the chimney. Natural draught comes free from a tall column of hot (light) gas next to cold (heavy) air; mechanical draught uses fans (forced = fan at the bottom pushes, induced = fan at the top sucks, balanced = both).

Height formula: draught = $H\ g\ (\rho_{air} - \rho_{gas})$; in mm of water $h_w = 353\ H\ [1/T_a - (m+1)/(m\ T_g)]$.

Before you start: what draught means

Air must flow into the furnace to burn the coal, and hot flue gas must be pushed out through the boiler, economiser, air preheater and dust collector. That flow needs a small pressure difference, a few mm to a few hundred mm of water column. That pressure difference is **draught**.

Why draught and chimney are used (2024 8d, 2023 8b, 2020 6a)

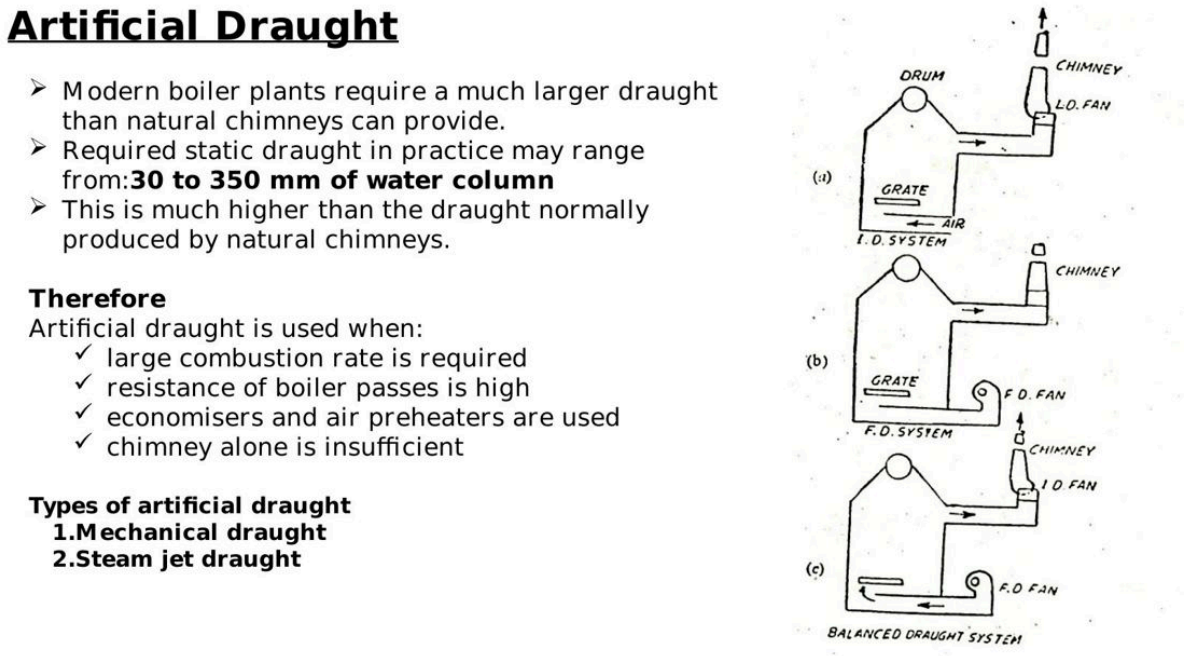


Draught: supplies air for combustion and removes burnt gases; classification into natural and artificial (mechanical: forced, induced, balanced; steam jet)

- **Draught** supplies the required quantity of **air to the furnace** for combustion, **removes the burnt gases** from the furnace through the boiler passes, and **discharges them to atmosphere** through the chimney.
- The **chimney** produces natural draught (or part of it) and discharges smoke and gases at a **great height**, so they disperse and do not pollute the area near the plant.

Natural vs mechanical draught (2023 8b, 2020 6a)

	Natural (chimney) draught	Mechanical (artificial) draught
Produced by	Density difference between hot gas inside the chimney and cold air outside	Fans: forced draught (FD), induced draught (ID) or both (balanced)
Draught available	Small, 10-20 mm of water; depends on chimney height and weather	Large, 30-350 mm of water; independent of weather
Control	Hard to control	Easily controlled to match load
Power	No power needed	Fans consume power
Cost	High chimney cost, no running cost	Lower chimney cost but running/maintenance cost
Exit gas temperature	Gases must leave hot (heat wasted)	Gases can be cooled more in economiser/air preheater → higher efficiency
Use	Small boilers	Large modern plants

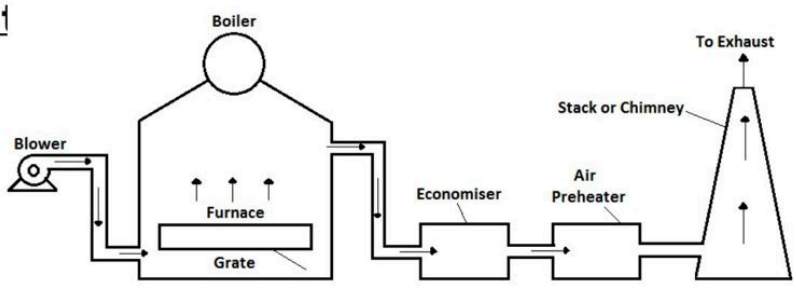


Artificial draught: (a) induced, (b) forced, (c) balanced draught systems

Forced, induced and balanced draught (2022 8b)

Artificial Draught

Types of artificial draught
1.Mechanical draught
2.Steam jet draught



Forced Draught System

(b) Forced Draught (F.D.) system

the **F.D. fan** is placed **at the air inlet side**, before the furnace.

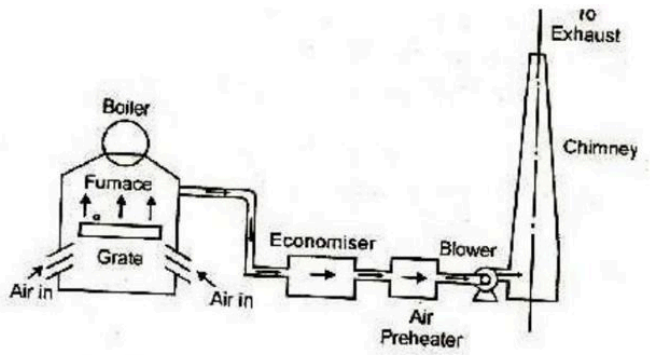
- the fan **forces air into the furnace**
- this air passes through the grate and supports combustion
- flue gases then pass through boiler and leave through chimney

Flow path

Forced draught: FD (forced draught) fan at the base pushes air into the furnace

Artificial Draught

Types of artificial draught
1.Mechanical draught
2.Steam jet draught



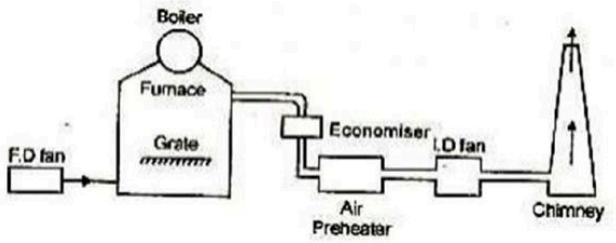
(a) Induced Draught (I.D.) system
the **I.D. fan** is placed **near the chimney / at the flue-gas exit side**.

- the fan **sucks flue gases** from the boiler
- this creates a **partial vacuum** inside the furnace
- because furnace pressure becomes lower than atmospheric pressure, **outside air enters naturally through the grate**
- gases then move through boiler → flue → I.D. fan → chimney

Induced draught: ID (induced draught) fan near the chimney sucks gases out

Artificial Draught

Types of artificial draught
1.Mechanical draught
2.Steam jet draught



(c) Balanced Draught system
In the bottom figure, **both fans are used**:
- ✓ one **F.D. fan** at the air inlet
- ✓ one **I.D. fan** near the chimney

- ➢ **F.D. fan** supplies air to the furnace
- ➢ **I.D. fan** removes flue gases from the boiler
- ➢ by proper control of both fans, furnace pressure is kept nearly neutral or slightly negative

Flow path
F.D. fan → furnace/grate → boiler → I.D. fan → chimney

Balanced draught: FD fan supplies air, ID fan removes gases

Balanced draught uses **both** an FD fan and an ID fan. The **FD fan** at the air inlet forces air through the air preheater and the fuel bed/burners; the **ID fan** near the base of the chimney sucks the flue gases through the boiler, economiser, air preheater and dust collector and discharges them into the chimney. The two fans are controlled so that the **furnace pressure is kept just below atmospheric** (about 1-2 mm of water suction). Flow path: FD fan → air preheater → furnace/grate → boiler → economiser → air preheater → dust collector → ID fan → chimney. **Advantages:** no blow-out of flame or hot gas when the furnace door is opened, no inward air leakage that would chill the furnace, good control of combustion; used in almost all large plants.

Height of chimney (2021 8b)

Step (write this)	Why / what it means
Pressure difference at base: $\Delta p = H\ g\ (\rho_a - \rho_g)$	Two columns of height H: cold heavy air outside, hot light gas inside.
$\rho_a = p/(R\ T_a) = 101{,}325/(287\ T_a) = 353/T_a$ kg/m^3	Ideal gas at 1 atm; the number 353 comes from here.
Gas: m kg air + 1 kg fuel gives (m + 1) kg gas, at nearly the same volume as m kg air at the same temperature ⇒ $\rho_g = (353/T_g) \times (m + 1)/m$	Gas density slightly higher per unit volume because of the added fuel mass.
$\Delta p = 353\ g\ H\ [1/T_a - ((m + 1)/m)(1/T_g)]$ N/m^2	
In mm of water (1 mm water = 9.81 N/m^2): **$h_w = 353\ H\ [1/T_a - ((m + 1)/m)(1/T_g)]$**	g cancels.
So **$H = h_w / (353\ [1/T_a - ((m + 1)/m)(1/T_g)])$**	Chimney height for a required draught.
Hot-gas column: $H_g = H\ [(m/(m + 1))(T_g/T_a) - 1]$	Height of hot gas that would weigh as much as the draught: $\Delta p = \rho_g\ g\ H_g$.

Memory trick: "353 over T" is the density of air at 1 atm. Draught = 353 H ($1/T_{cold}$ − $1/T_{hot} \times (m+1)/m$).

B13 Cooling towers: types, induced-draught counterflow wet tower, range, approach, efficiency + numerical

Must study · Very likely · 93% · Medium

PAST-PAPER QUESTIONS

- **2024 Q8(c)** [12] Explain the general working principle of a wet cooling tower.
- **2023 Q8(c)** [9] Write short notes on (i) approach, (ii) range, (iii) cooling efficiency of a cooling tower.
- **2022 Q8(c)** [15] What is a cooling tower? Classify it. Describe the induced draught counterflow wet cooling tower with a schematic.
- **2021 Q8(c)** [10] Water at 30 °C enters a cooling tower at 1.25 kg per kg of air. Air enters at 20 °C, 60 % RH (relative humidity) and leaves at 28 °C, 90 % RH. Make-up water is at 20 °C. Find (i) water outlet temperature, (ii) fraction of water evaporated, (iii) approach and range.

CALCULATOR **Cooling tower range, approach and efficiency**

Change any number; every step is recalculated and shown with the numbers put in.

1. Range: Range = $t_{w,in} - t_{w,out}$ = **40** - **30** = **10.0** °C
2. Approach: Approach = $t_{w,out} - t_{wb}$ = **30** - **25** = **5.0** °C
3. Cooling efficiency: η = range / (range + approach) = **10** / (**10** + **5**) × 100 = **66.7** %

The wet-bulb temperature is the limit: water can never be cooled below it. Approach of 3–5 °C is typical.

★ REMEMBER IT

- **R**ange = how much the water cooled (in − out). **A**pproach = how close it got to the wet bulb (out − wb). Efficiency = R / (R + A).

Before you start: what a cooling tower does

The condenser warms the cooling water by about 8-10 °C. Instead of throwing it away, the cooling tower cools it and sends it back. Hot water is sprinkled down over packing (fill) while air flows up. A small part (about 1-2 %) **evaporates**; evaporation takes a lot of heat (2400 kJ/kg) from the rest, so the water cools, even below the air's dry-bulb temperature.

Wet bulb temperature (WBT) = the lowest temperature water can be cooled to by evaporating into that air. It is the cooling tower's limit.

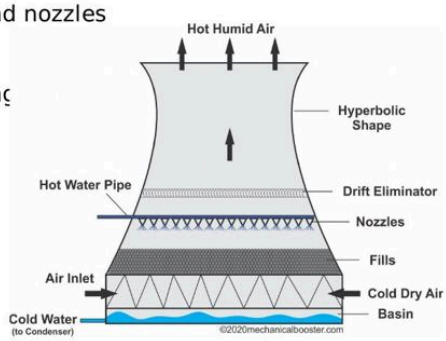
Working principle of a wet cooling tower (2024 8c)

Natural Draught Cooling Tower

In a **natural draught cooling tower**, air circulation takes place naturally without fans.

Working

- hot water from the condenser is pumped to troughs and nozzles
- it falls as droplets through the tower
- air enters through openings near the base
- air rises upward naturally and absorbs heat from falling
- cooled water collects at the bottom



Advantages

1. Low operating cost
2. Low maintenance cost
3. Nearly trouble-free operation
4. Requires relatively less ground area
5. Very large sizes are possible
6. Strong and self-supporting structure
7. Top shape helps water droplets fall out of suspension

Drawbacks

8. High initial cost
9. Performance changes with:
    - dry bulb temperature
    - relative humidity of air

Although the first cost is high, natural draught towers are often preferred for **very large power stations** because:

- no fan power is needed
- life is longer
- maintenance is less

Natural draught (hyperbolic) cooling tower

Hot water from the condenser is pumped to the top of the tower and distributed through troughs/nozzles. It falls in thin films and droplets over the **fill (packing)**, which increases the contact area and time with air. Air enters near the bottom through louvres and flows upward (counterflow) or across (crossflow), driven either by the chimney effect of the tall shell (natural draught) or by fans (mechanical draught). Heat leaves the water mainly by **evaporation** of a small fraction and partly by convection to air. Warm, humid air leaves at the top through **drift eliminators** that catch droplets. Cooled water collects in the basin at the bottom and returns to the condenser; **make-up water** replaces evaporation, drift and blow-down losses.

Classification (2022 8c)

Step (write this)	Why / what it means
`By draught: natural draught (hyperbolic shell); mechanical draught: forced draught (fan at bottom pushes air), induced draught (fan at top pulls air)`	
`By air-water flow: counterflow (air up, water down), crossflow (air horizontal)`	
`By heat transfer: wet (direct contact, evaporative), dry (water in finned tubes, no contact), hybrid wet-dry`	

Induced draught counterflow wet cooling tower (2022 8c)

Mechanical Draught Cooling Towe

B. Induced Draught Cooling Tower

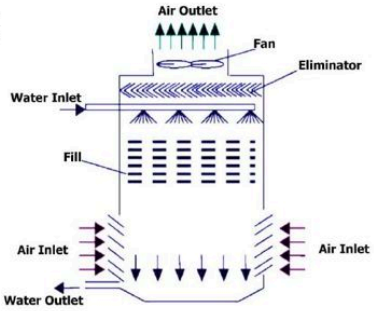
- fans are placed at the **top**
- they **draw air upward** through louvers at the base
- thus air enters from the bottom and moves upward through the packing

Advantages

1. Coldest water meets driest air
2. Warmest water meets most humid air
3. Recirculation problem is less
4. Lower first cost
5. Requires less space
6. Can cool over a wide range

Disadvantages

7. Air distribution through packing may be uneven
8. Fan motor power may be comparatively high in practice due to higher static pressure loss





Induced draught cooling tower: fan on top draws air up through the fill; water sprayed down

The **fan is at the top** of the tower and **sucks air upward**. Air enters through louvres at the base on all sides, rises through the fill, meets the hot water that is sprayed from distribution nozzles above the fill and trickles downward (**counterflow**), then passes through the drift eliminators and fan and leaves at high velocity, so it does not recirculate. Cooled water drops into the basin and is pumped back. **Advantages:** coldest water meets driest air (best heat transfer), less recirculation, less space, can cool over a wide range. **Disadvantages:** fan power is high, air distribution in the packing may be uneven.

Mechanical Draught Cooling Towers

In these towers, air flow is produced by **fans**. They are usually built in **cells or units**, and total capacity depends on the number of cells used.

A. Forced Draught Cooling Tower

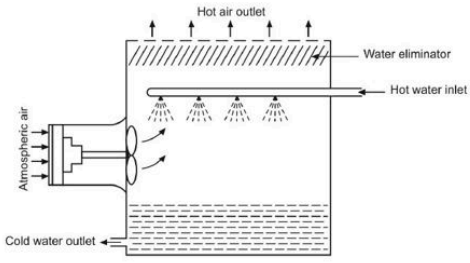
- fans are placed near the **base**
- they **push air** into the tower
- water falls downward through packings
- air moves upward and cools the water
- sides are closed and nearly air-tight
- fan handles dry incoming air

Advantages

1. Good efficiency
2. No fan blade erosion problem
3. Safer operation
4. Less vibration and noise

Disadvantages

5. Fan size is limited
6. Power requirement is high
7. In cold weather, frost and icing may occur near outlet or on nearby structures





Forced draught tower for comparison: fan at the bottom pushes air in

Range, approach, cooling efficiency (2023 8c)

Step (write this)	Why / what it means
Range = $T_{water,in}$ - $T_{water,out}$	How much the water is cooled (usually 5-10 °C).

Approach = $T_{water,out}$ - WBT (wet bulb temperature)$_{air,in}$	How close the cooled water gets to the theoretical limit (usually 3-8 °C). Smaller approach = bigger, costlier tower.
Cooling efficiency = Range / (Range + Approach) = ($T_{w,in}$ - $T_{w,out}$) / ($T_{w,in}$ - WBT)	Actual cooling / maximum possible cooling. Typically 70-75 %.

Numerical 2021 8(c): water 30 °C, 1.25 kg/kg air; air 20 °C 60 % RH in, 28 °C 90 % RH out; make-up 20 °C

Before you start: psychrometry in three formulas

Partial pressure of vapour: p_v = RH × p_{sat}(T). Specific humidity (kg water per kg dry air): ω = 0.622 p_v/(p – p_v), p = 101.325 kPa.

Enthalpy of moist air per kg dry air: h = 1.005 T + ω(2501 + 1.82 T) kJ/kg (T in °C).

p_{sat} from steam tables: 20 °C = 2.339 kPa, 28 °C = 3.782 kPa.

Step (write this)	Why / what it means
Air in: p_{v1} = 0.6 × 2.339 = 1.403 kPa; ω_1 = 0.622 × 1.403/99.92 = 0.00873 kg/kg	Inlet air is fairly dry.
h_1 = 1.005 × 20 + 0.00873(2501 + 36.4) = 42.3 kJ/kg dry air	
Air out: p_{v2} = 0.9 × 3.782 = 3.404 kPa; ω_2 = 0.622 × 3.404/97.92 = 0.02162 kg/kg	
h_2 = 1.005 × 28 + 0.02162(2501 + 51) = 83.3 kJ/kg dry air	
Water evaporated = ω_2 - ω_1 = 0.01289 kg per kg air	Mass balance on water.
(ii) Fraction evaporated = 0.01289/1.25 = **0.0103 = 1.03 %**	Of the water entering.
Energy balance: m_wc $T_{w,in}$ + h_1 = h_2 + (m_w - 0.01289) c $T_{w,out}$	Heat given by water = heat gained by air. c = 4.186 kJ/kg·K.
1.25 × 4.186 × 30 + 42.3 = 83.3 + 1.2371 × 4.186 × $T_{w,out}$ ⇒ 199.3 - 83.3 = 5.179 $T_{w,out}$	
(i) **$T_{w,out}$ = 22.4 °C**	Make-up at 20 °C is added after to replace the evaporated water.
WBT of inlet air (20 °C, 60 %) ≈ 15.2 °C (psychrometric chart)	Read where the 20 °C, 60 % point meets the wet-bulb line.
(iii) Range = 30 - 22.4 = **7.6 °C**; Approach = 22.4 - 15.2 = **7.2 °C**	Bonus: efficiency = 7.6/(7.6 + 7.2) = 51 %.

Side note: If your book reads h_1, h_2 from the chart (about 42.5 and 83.5 kJ/kg), you get the same 22.4 °C to within 0.1 °C.

Memory trick: "Range = how far it fell; Approach = how near the wet bulb".

B3 Gas turbine plant: open-cycle layout, open vs closed, applications, peak-load advantages, fuel properties Good to have Likely · 60% Easy

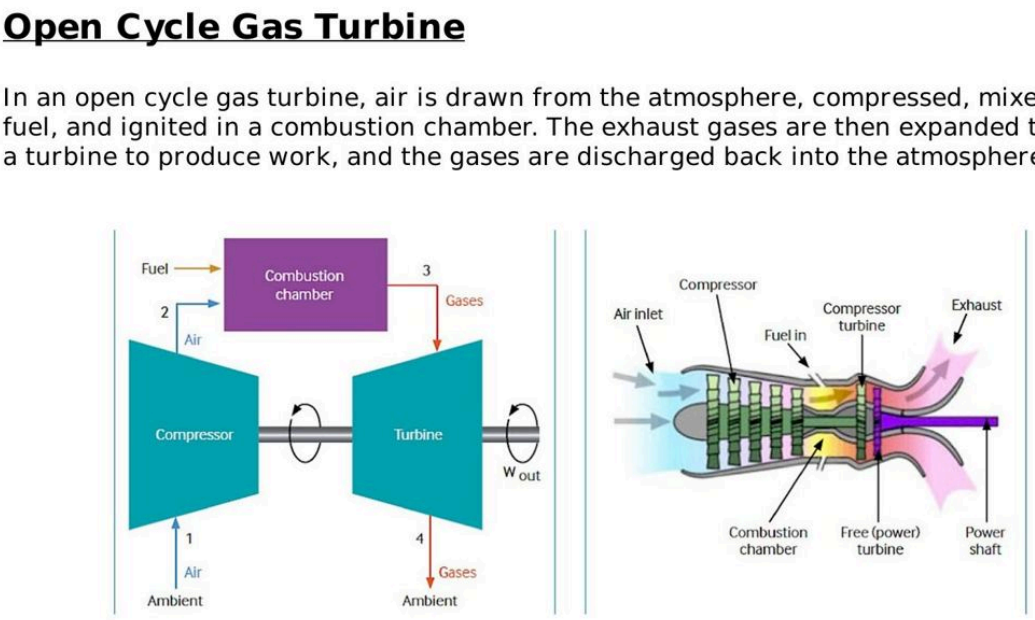
PAST-PAPER QUESTIONS

- **2023 Q6(a)** [7] What are the advantages of a gas turbine plant as a peak load plant?
- **2023 Q6(b)** [8] Draw the layout of an open cycle gas turbine power plant.
- **2022 Q6(a)** [5] Write the applications of gas turbine power plants.
- **2021 Q6(c)** [5] What properties are considered while selecting fuel for a gas turbine plant?

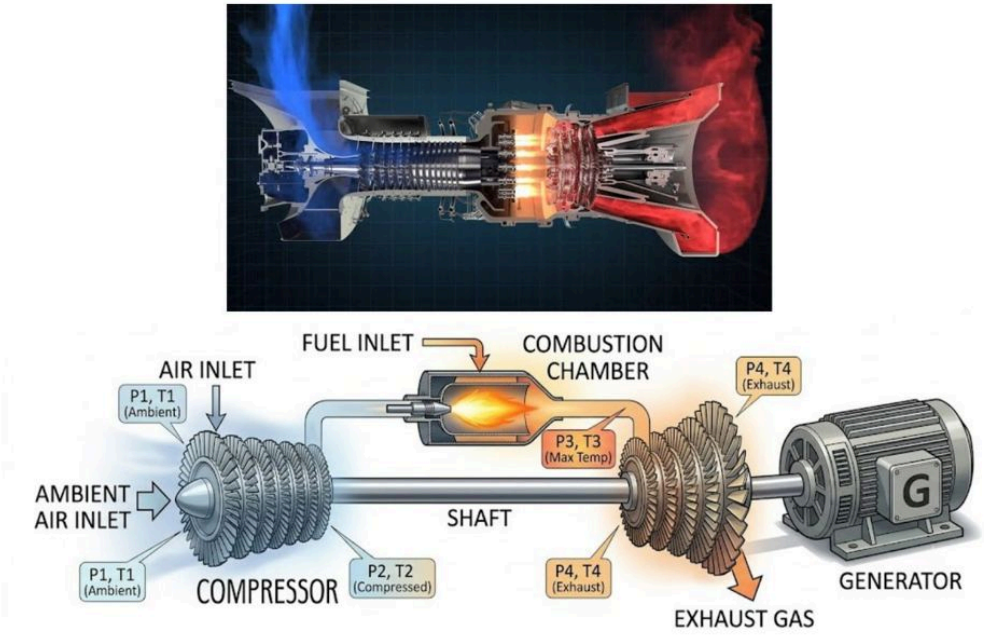
Before you start: the Brayton (gas turbine) cycle in words

1 → 2: **compressor** squeezes air (pressure rises 8-30 times). 2 → 3: fuel burns in the **combustion chamber** at constant pressure; gas gets very hot (1100-1500 °C). 3 → 4: hot gas expands in the **turbine**, which drives the compressor and the generator. 4 → 1: gas leaves (open cycle) or is cooled (closed cycle).

Open cycle gas turbine layout (2023 6b)



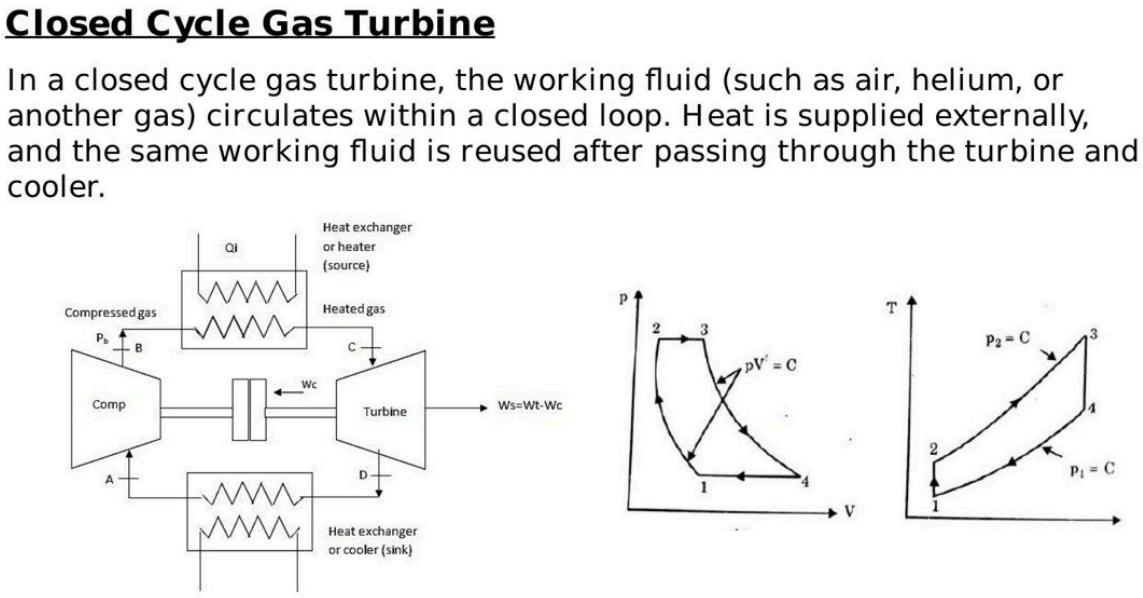
Open cycle gas turbine: air from atmosphere, compressor, combustion chamber, turbine, exhaust to atmosphere



Real machine: air inlet, compressor, combustion chamber, turbine, shaft and generator

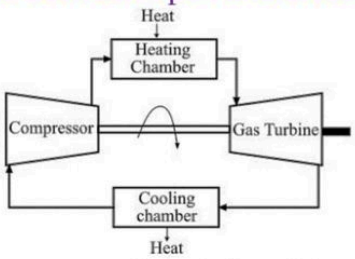
Air is drawn from the atmosphere (through a filter), compressed by an axial or centrifugal **compressor**, and delivered to the **combustion chamber** where fuel (natural gas or oil) is injected and burnt. The hot high-pressure gases expand through the **turbine**; about 60 % of turbine work drives the compressor and the rest drives the **generator**. Exhaust gas (400-600 °C) is discharged to the atmosphere through a stack. A **starting motor** brings the set up to speed before firing.

Open vs closed cycle

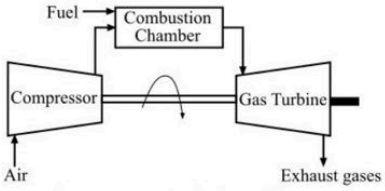


Closed cycle gas turbine with heater and cooler, and its p-v and T-s diagrams

Difference between open and closed cycle gas turbine:



1. The compressed air is heated in a heating chamber. Since the gas is heated by an external source, so the amount of gas remain the same.
2. The gas from the turbine is passed into the cooling chamber.
3. The working fluid is circulated continuously.
4. Turbine blade does not wear away earlier, as the enclosed gas does not get contaminated while flowing through the heating chamber.
5. Its maintenance cost is high.
6. The weight of installation per horse power is more.



1. The compressed air is heated in a combustion chamber. The product of combustion get mixed up in the heated air.
2. The gas from the turbine is exhausted into the atmosphere.
3. The working fluid is replaced continuously.
4. Turbine blade wears away earlier, as the air from the atmosphere get contaminated while flowing through the combustion chamber.
5. Its maintenance cost is low.
6. The weight of installation per horse power is less.

Differences between open and closed cycle gas turbines

Open cycle	Closed cycle
Fresh air taken in each cycle, products exhausted	Same working fluid (air, helium) recirculated
Fuel burnt in the working fluid (internal combustion)	Heat added through a heat exchanger (external heating), any fuel incl. nuclear/coal
No cooler needed; light, simple	Needs a heater and a cooler (pre-cooler); heavier, costlier
Turbine blades get dirty/corroded by combustion products	Clean fluid; blades last longer; can pressurise for small size
Lower weight per kW; used for peak load and aircraft	Higher part-load efficiency; any working gas

Applications (2022 6a)

- Peak load and standby plants in grids; quick-start emergency power.
- Base/intermediate load in **combined cycle** plants with a steam turbine.
- Jet aircraft engines and helicopters; ships (marine propulsion); locomotives.
- Industrial drives: gas pipeline compressors, pumps; cogeneration in industries.
- Supercharging (turbocharger) of diesel engines.

Advantages as peak load plant (2023 6a)

Application of gas turbine power plant:

1. Gas turbine plants are used as stand by plants for the hydro-electric power plants.
2. Gas turbine power plants may be used as peak load plant and stand by plants for smeller power units.
3. Gas turbine are used in jet aircraft and ships; Pulverized fuel fired plants in locomotive.

Advantages of Gras turbine power plant:

(1) It is smaller in size and weight as compared to an equivalent power plant.
(2) The initial cost and operating cast of the plant is lower than an equivalent steam power plant.
(3) The plant requires less water as compared to a condensing steam power plant
(4) The plant can be started quickly, and can be put on load in a very short time.
(5) There is no stand by losses in the gas turbine power plant.
(6) The maintenance of the plant is easier and maintenance cost is low.
(7) The lubrication of the plant is very easy.

Disadvantages:

1. Major part of the work developed in the turbine is used to drive the compressor. Therefore, net work output of the plant is low.
2. Since the temperature of the products of combustion becomes too high so service condition become complicated.

Applications, advantages and disadvantages of gas turbine plants (slide)

1. **Starts quickly** (minutes) and picks up load fast.
2. **No standby losses**; no boiler to keep hot.
3. **Low capital cost** and short construction time, so economic for few running hours.
4. Small, light, needs **little water**; can be sited near the load.
5. Simple, less maintenance, easy lubrication; can burn various fuels.

Properties of a good gas turbine fuel (2021 6c)

1. **Volatility:** should be optimum for easy starting and complete combustion.
2. **Clean combustion products:** little ash/vanadium/sodium, which deposit on and corrode blades.
3. **High energy content** (calorific value) to reduce fuel consumption.
4. **Lubricating property** for the fuel pump.
5. **Availability** in large quantity at low cost; safe handling (high flash point).

> **Memory trick:** GT (gas turbine) fuel **"VCELA"**: Volatility, Clean products, Energy content, Lubricity, Availability.

B8 Steam power plant: layout and four circuits, working, base-load advantages, site, design activities

Must study · Likely · 67% · Medium

PAST-PAPER QUESTIONS

- **2023 Q7(b)** [15] With a neat sketch describe the working principle of a steam power plant.
- **2023 Q7(c)** [8] What are the advantages of a steam power plant as a base load plant?
- **2022 Q7(a)** [15] Draw a general layout of a steam power plant and discuss its various circuits.
- **2021 Q7(b)** [15] Draw the typical layout of a steam power plant and explain its working.
- **2021 Q7(c)** [10] What activities should be carried out for designing a steam power plant?
- **2020 Q6(b)** [13] Describe the factors to be considered while selecting the site for a steam power plant.

★ REMEMBER IT

- Four circuits of a steam plant: **C-A-F-C**: Coal & ash, Air & flue gas, Feed water & steam, Cooling water. Draw the Rankine loop: boiler → turbine → condenser → pump → boiler, then hang the four circuits on it.

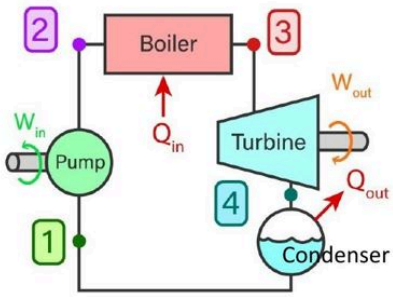
Before you start: the Rankine cycle in words

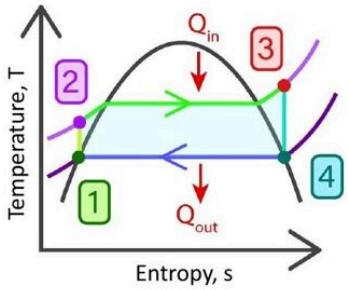
Water is **pumped** to high pressure (1→2), turned into superheated steam in the **boiler** (2→3), the steam expands in a **turbine** that drives the generator (3→4), and is condensed back to water in the **condenser** (4→1). Coal (or gas, oil) is burnt to heat the boiler.

Basic Rankine cycle

The ideal Rankine cycle has four main processes:

- **Pump (1 → 2):** Pressurizes water (condensate) to boiler pressure.
- **Boiler (2 → 3):** Adds heat to convert water into steam (often superheated).
- **Turbine (3 → 4):** Steam expands and produces work (drives generator).
- **Condenser (4 → 1):** Rejects heat and condenses exhaust steam back to water.





Basic Rankine cycle: pump, boiler, turbine, condenser and its T-s diagram

Layout and the four circuits (2022 7a, 2021 7b, 2023 7b)

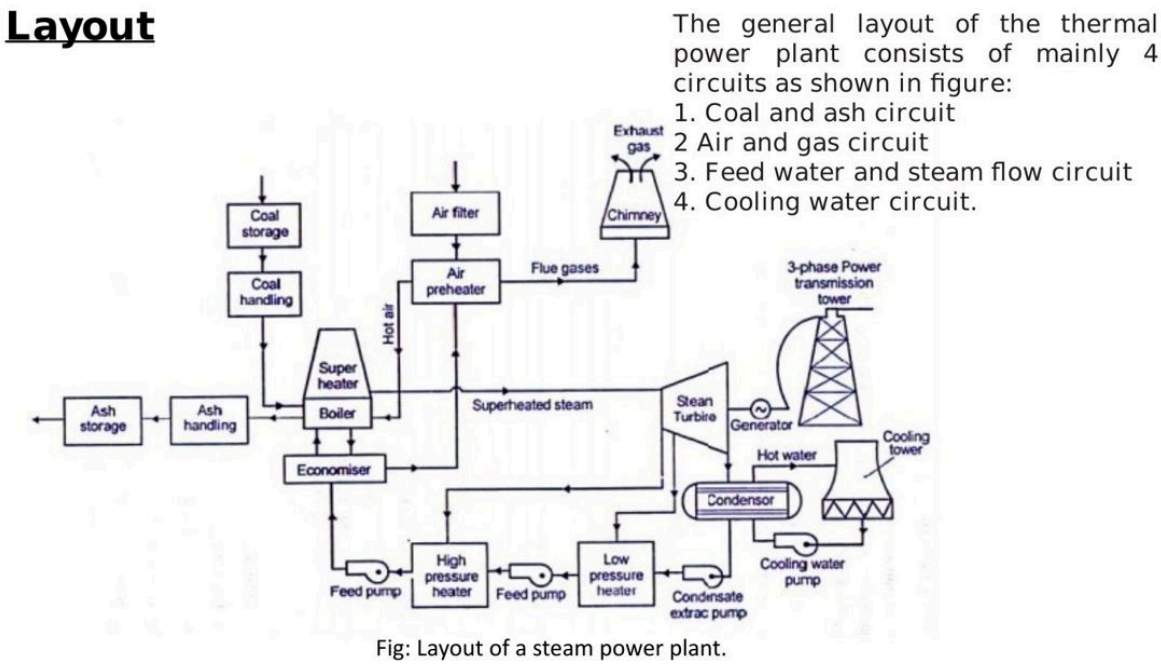


Fig: Layout of a steam power plant.

General layout of a steam power plant

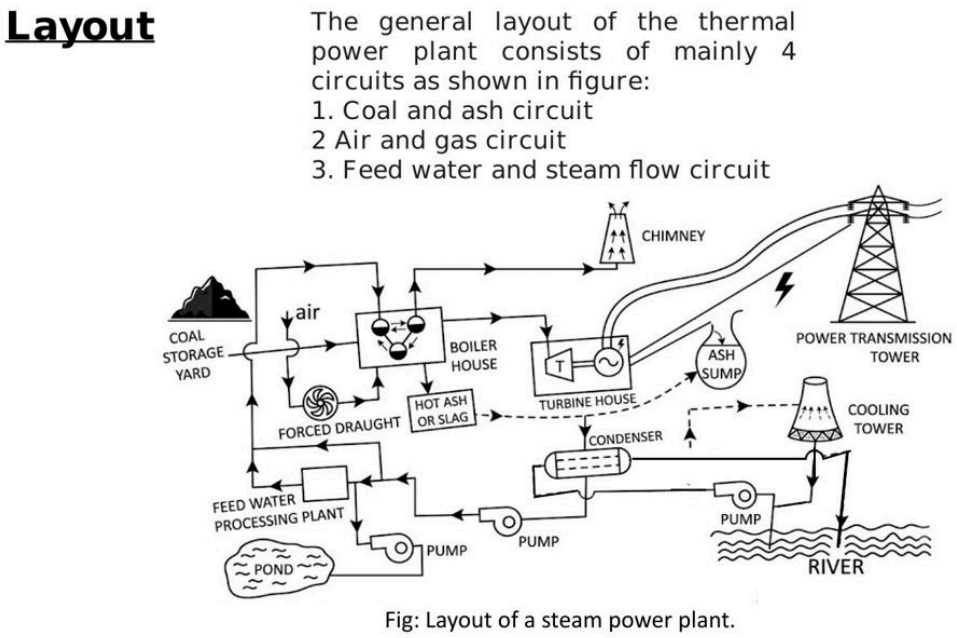


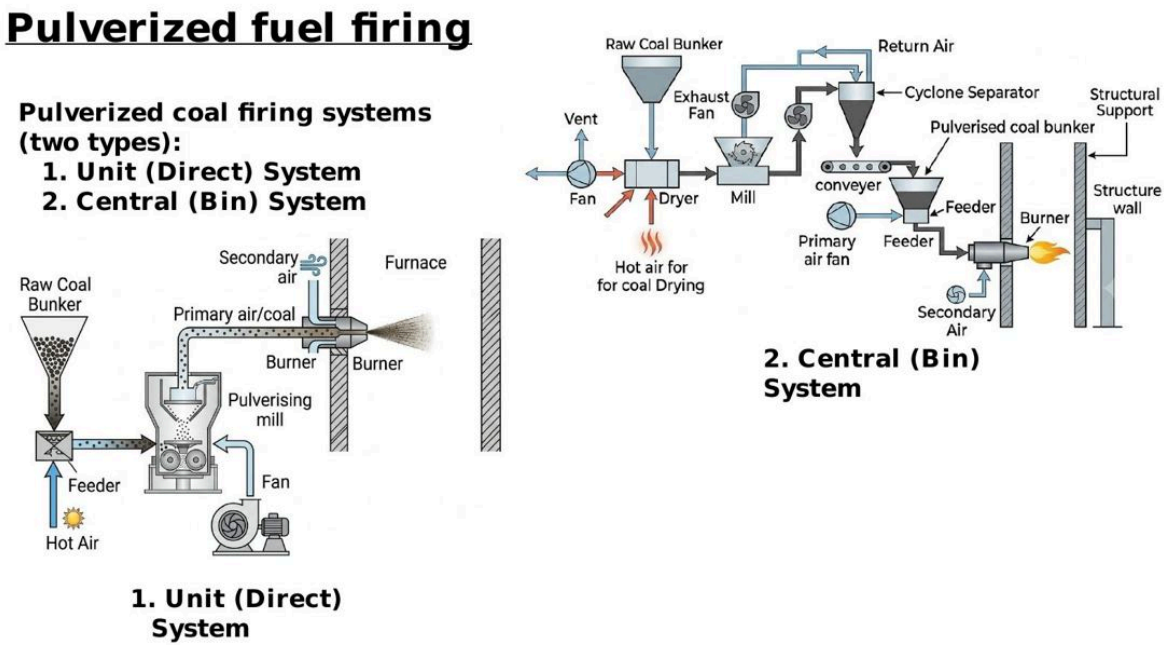
Fig: Layout of a steam power plant.

Layout showing coal storage, boiler house, chimney, turbine house, cooling tower, river and transmission

Step (write this)	Why / what it means
`1. Coal and ash circuit`	Coal from storage → coal handling (crushing, conveying) → boiler furnace; ash from furnace → ash handling → ash storage/disposal.
`2. Air and flue gas circuit`	Air by forced draught fan → air preheater → furnace; flue gas → superheater, economiser, air preheater → dust collector → induced draught fan → chimney.
`3. Feed water and steam circuit`	Condensate → feed pump → feed water heaters → economiser → boiler drum → superheater → turbine → condenser. Make-up water from treatment plant.
`4. Cooling water circuit`	Water from river/cooling tower → condenser (absorbs heat of exhaust steam) → back to river or

	cooling tower.

Working: coal burnt in the furnace releases heat that converts feed water into high-pressure steam in the boiler; the superheater raises its temperature. The steam expands through the turbine, producing shaft work that drives the alternator. Exhaust steam is condensed in the condenser under vacuum (which raises the work obtained), and the condensate is pumped back through feed heaters and the economiser to the boiler. Flue gases give up heat in the economiser and air preheater, are cleaned in the dust collector (ESP) and leave via the chimney.



Pulverized fuel firing (unit and central system), part of the coal circuit

Advantages as base load plant (2023 7c)

1. **Large unit sizes** (500-1000 MW) with high efficiency at steady full load.
2. **Low fuel cost** (coal) gives low running cost per kWh (kilowatt-hour = 1 unit of electrical energy), suited to many running hours.
3. Slow start-up and load change are not a problem for steady base load.
4. **Long life** (25-40 years) and reliable.
5. Can be built near coal fields or ports; independent of rainfall (unlike hydro).
6. High capital cost is spread over many kWh at a high load factor.

Site selection for a steam plant (2020 6b)

1. **Supply of fuel:** near coal mines or a port; cheap transport.
2. **Plenty of water** for condenser cooling and boiler feed (river, sea).
3. **Transport facilities** (rail, river) for coal and heavy machinery.
4. **Cheap land** with good bearing capacity; space for coal yard, ash pond and extension.
5. **Near load centre** to reduce transmission cost.
6. **Ash disposal** facilities.
7. **Away from populated areas** (smoke, noise, dust); consider wind direction.
8. Availability of **labour**.

Activities for designing a steam power plant (2021 7c)

1. **Load survey and forecasting**: maximum demand, load curve, growth.
2. **Selection of site** (factors above) and of **fuel** (type, source, transport).

3. **Choice of plant capacity**, number and size of units, steam conditions (pressure, temperature, reheat, regeneration).
4. **Selection of main equipment**: boiler, turbine, generator, condenser, cooling system.
5. **Design of auxiliaries**: coal and ash handling, draught system, feed water treatment, dust collection, cooling towers.
6. **Layout** of buildings and equipment; electrical system (transformers, switchyard).
7. **Economic analysis** (capital, cost per kWh) and **environmental impact assessment**; instrumentation and control; safety.

> **Memory trick:** Four circuits **"CAFe Cool"**: Coal-ash, Air-flue gas, Feed water-steam, Cooling water.

B5 Brayton cycle numericals and the maximum-work condition

Must study

Likely · 67% Hard

PAST-PAPER QUESTIONS

- **2023 Q6(d)** [10] Show that the net work output of a gas turbine plant is maximum when the exit temperatures of compressor and turbine are the same.
- **2022 Q7(c)** [12] Ideal gas turbine with two stages of compression and two of expansion, pressure ratio 2.83 per stage, air enters each compressor stage at 300 K and each turbine stage at 1300 K. Find back work ratio and thermal efficiency (i) without regenerator, (ii) with an ideal regenerator.
- **2021 Q6(b)** [15] Brayton cycle: pressure ratio 8, compressor inlet 300 K, turbine inlet 1300 K, ηc = 80 %, ηt = 85 %. Find (i) exit temperatures, (ii) back work ratio, (iii) thermal efficiency, (iv) efficiency with an 80 % effective regenerator.
- **2020 Q5(b)** [14] Ideal Brayton cycle: pressure ratio 9, compressor inlet 310 K, turbine inlet 1100 K. Using air-standard assumptions find the thermal efficiency.

UNDERSTAND **Read this to understand; you do not need to write it**

Write one number first: $\mathbf{x = r_p^{0.2857}}$ (because $(\gamma-1)/\gamma = 0.4/1.4 = 0.2857$). Then $T_{2s} = T_1 x$ and $T_{4s} = T_3/x$. Everything else is $c_p \times$ temperature difference.

Real compressor: $T_2 = T_1 + (T_{2s} - T_1)/\eta_c$ (divide: it needs more). Real turbine: $T_4 = T_3 - \eta_t(T_3 - T_{4s})$ (multiply: it gives less).

Max work proof (2023 6d): $w_{net} = c_p[T_3 - T_3/x - T_1 x + T_1]$; $dw/dx = 0$ gives $x = \sqrt{(T_3/T_1)}$, so $T_2 = T_4 = \sqrt{(T_1 T_3)}$.

Symbols used here

Symbol	Name and meaning	Unit
r_p	Pressure ratio p_2/p_1	–
T_1, T_3	Compressor inlet, turbine inlet temperature	K
T_{2s}, T_{4s}	Ideal (isentropic) exit temperatures	K
η_c, η_t	Compressor, turbine isentropic efficiency	–
c_p	Specific heat of air	1.005 kJ/kg K
w_c, w_t	Compressor and turbine work	kJ/kg
r_{bw}	Back work ratio = w_c/w_t	–
ε	Regenerator effectiveness	–

CALCULATOR **Brayton cycle calculator (any year's data)**

Change any number; every step is recalculated and shown with the numbers put in.

1. Temperature ratio: x = $r_p^{0.2857}$ = pow(**8**,0.2857) = **1.8114**
2. Ideal compressor exit: **300** × **1.811** = **543.4** K
3. Actual compressor exit: **300** + (**543.4** - **300**) / (**80** / 100) = **604.3** K
4. Ideal turbine exit: **1300** / **1.811** = **717.7** K
5. Actual turbine exit: **1300** - **85** / 100 × (**1300** - **717.7**) = **805.0** K
6. Compressor work: 1.005 × (**604.3** - **300**) = **305.8** kJ/kg
7. Turbine work: 1.005 × (**1300** - **805**) = **497.4** kJ/kg
8. Back work ratio: **305.8** / **497.4** = **0.615**
9. Net work: **497.4** - **305.8** = **191.7** kJ/kg
10. Heat supplied, no regenerator: 1.005 × (**1300** - **604.3**) = **699.2** kJ/kg
11. Thermal efficiency, no regenerator: **191.7** / **699.2** × 100 = **27.4** %
12. Air after regenerator: **604.3** + **80** / 100 × (**805** - **604.3**) = **764.9** K
13. Efficiency with regenerator: **191.7** / (1.005 × (**1300** - **764.9**)) × 100 = **35.6** %

Set both efficiencies to 100 % and regenerator to 0 for the ideal cycle (2020 5b: r_p 9, 310 K, 1100 K → 46.6 %).

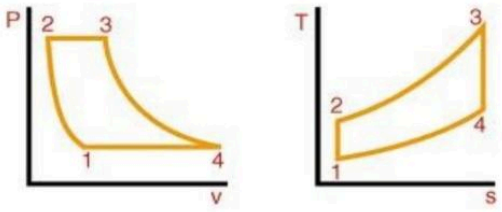
Before you start: the two formulas for every Brayton problem

Isentropic means ideal compression/expansion with no friction and no heat loss. For air (γ = 1.4): T2s/T1 = T3/T4s = $r_p^{(\gamma-1)/\gamma}$ = $r_p^{0.2857}$.

Work or heat for air = c_p × temperature change, c_p = 1.005 kJ/kg·K.

Real machines: compressor needs more work, so T2 = T1 + (T2s − T1)/η_c; turbine gives less, so T4 = T3 − η_t(T3 − T4s).

Back work ratio = w_c/w_t (fraction of turbine work eaten by the compressor).



Net work done by the turbine = $C_P[(T_3 - T_4) - (T_2 - T_1)]$

Heat supplied = $C_P(T_3 - T_2)$

Thermal Efficiency $\eta_{th} = \frac{C_P[(T_3-T_4)-(T_2-T_1)]}{C_P(T_3-T_2)} = \frac{[(T_3-T_4)-(T_2-T_1)]}{(T_3-T_2)}$

Pressure ratio, $r = \frac{P_2}{P_1} = \frac{P_3}{P_4}$

For Isentropic compression (1-2)

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = (r)^{\frac{\gamma-1}{\gamma}}$$

$$T_2 = T_1(r)^{\frac{\gamma-1}{\gamma}}$$

Brayton cycle P-v and T-s; net work, heat supplied and efficiency

For Isentropic compression (1-2)

$$\frac{T_3}{T_4} = \left(\frac{P_3}{P_4}\right)^{\frac{\gamma-1}{\gamma}} = (r)^{\frac{\gamma-1}{\gamma}}$$

$$T_3 = T_4(r)^{\frac{\gamma-1}{\gamma}}$$

Putting the value of T_2 and T_3 in the above equation

$$\eta_{th} = \frac{\left[\left(T_4(r)^{\frac{\gamma-1}{\gamma}} - T_4\right) - \left(T_1(r)^{\frac{\gamma-1}{\gamma}} - T_1\right)\right]}{\left(T_4(r)^{\frac{\gamma-1}{\gamma}} - T_1(r)^{\frac{\gamma-1}{\gamma}}\right)}$$

$$\eta_{th} = \frac{\left[T_4\left((r)^{\frac{\gamma-1}{\gamma}} - 1\right) - T_1\left((r)^{\frac{\gamma-1}{\gamma}} - 1\right)\right]}{(T_4 - T_1)(r)^{\frac{\gamma-1}{\gamma}}}$$

$$\eta_{th} = \frac{(T_4 - T_1)\left((r)^{\frac{\gamma-1}{\gamma}} - 1\right)}{(T_4 - T_1)(r)^{\frac{\gamma-1}{\gamma}}} = \frac{\left((r)^{\frac{\gamma-1}{\gamma}} - 1\right)}{(r)^{\frac{\gamma-1}{\gamma}}}$$

$$= 1 - \left(\frac{1}{r}\right)^{\frac{\gamma-1}{\gamma}}$$

Derivation of ideal efficiency $\eta = 1 - 1/r_p^{(\gamma-1)/\gamma}$

2020 5(b): ideal, r_p = 9, T1 = 310 K, T3 = 1100 K

Step (write this)	Why / what it means
$x = r_p^{0.2857} = 9^{0.2857} = 1.873$	Do this once; use it for both compressor and turbine.
T2 = 310 × 1.873 = 580.8 K; T4 = 1100 / 1.873 = 587.2 K	
n = 1 - 1/x = 1 - 1/1.873 = **0.466 = 46.6 %**	Ideal Brayton efficiency depends only on pressure ratio.
Check: w_{net} = 1.005[(1100 - 587.2) - (580.8 - 310)] = 243.3 kJ/kg; q_{in} = 1.005(1100 - 580.8) = 521.8 kJ/kg; 243.3/521.8 = 46.6 %	Same answer by the long route.

2021 6(b): r_p = 8, 300 K, 1300 K, η_c 80 %, η_t 85 %, regenerator 80 %

Step (write this)	Why / what it means
$x = 8^{0.2857} = 1.811$	
T2s = 300 × 1.811 = 543.4 K; T2 = 300 + 243.4/0.80 = **604.3 K**	(i) Compressor exit. Real compressor needs more work, so exit is hotter.
T4s = 1300/1.811 = 717.7 K; T4 = 1300 - 0.85(1300 - 717.7) = **805.0 K**	(i) Turbine exit.
w_c = 1.005(604.3 - 300) = 305.8 kJ/kg; w_t = 1.005(1300 - 805) = 497.5 kJ/kg	
(ii) Back work ratio = 305.8/497.5 = **0.615**	61.5 % of turbine work drives the compressor.
q_{in} = 1.005(1300 - 604.3) = 699.2 kJ/kg; w_{net} = 191.7 kJ/kg	
(iii) n = 191.7/699.2 = **27.4 %**	
With regenerator: T5 = T2 + ε(T4 - T2) = 604.3 + 0.8(805 - 604.3) = 764.9 K	Air is preheated by exhaust from 604.3 to 764.9 K.

q_{in} = 1.005(1300 - 764.9) = 537.8 kJ/kg; (iv) n = 191.7/537.8 = **35.6 %**	Work unchanged, heat less.

Side note: Cengel (air tables) gives T2 = 605 K, T4 = 804 K, r_{bw} = 0.617, η = 26.6 % and 36.9 % with regenerator. Either method gets full marks; state c_p = 1.005 kJ/kg·K constant.

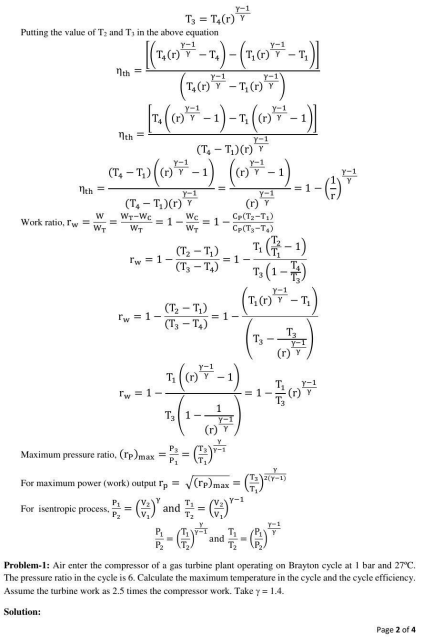
2022 7(c): two-stage compression and expansion, 2.83 per stage, 300 K and 1300 K, ideal

Step (write this)	Why / what it means
$x = 2.83^{0.2857} = 1.346$	Each stage has pressure ratio 2.83 (overall 8).
Each compressor stage: 300 → 300 × 1.346 = 403.8 K	Intercooled back to 300 K before stage 2.
Each turbine stage: 1300 → 1300/1.346 = 965.7 K	Reheated back to 1300 K before stage 2.
w_c = 2 × 1.005(403.8 - 300) = 208.7 kJ/kg; w_t = 2 × 1.005(1300 - 965.7) = 671.9 kJ/kg	Two equal stages each.
Back work ratio = 208.7/671.9 = **0.311**	
(i) No regenerator: q_{in} = 1.005[(1300 - 403.8) + (1300 - 965.7)] = 1236.7 kJ/kg; n = 463.2/1236.7 = **37.5 %**	Heat in combustor + reheater.
(ii) Ideal regenerator: air heated to 965.7 K by exhaust; q_{in} = 1.005[(1300 - 965.7) × 2] = 671.9 kJ/kg; n = 463.2/671.9 = **68.9 %**	Now heat added = turbine work, so a huge rise.

Side note: Cengel Example 9-8 with air tables: r_{bw} = 0.304, η = 35.7 % and 69.5 %.

Maximum net work when T2 = T4 (2023 6d)

Step (write this)	Why / what it means
$w_{net} = c_p[(T3 - T4) - (T2 - T1)]$	Turbine work minus compressor work.
Put $x = r_p^{(\gamma-1)/\gamma}$: $T2 = T1\ x$, $T4 = T3 / x$	Isentropic relations.
$w_{net} = c_p[T3 - T3/x - T1\ x + T1]$	Only x can change (we choose the pressure ratio).
$dw_{net}/dx = c_p[T3/x^2 - T1] = 0$	For a maximum the slope is zero.
$x^2 = T3/T1 \Rightarrow x = \sqrt{(T3/T1)}$	Optimum pressure ratio: $r_{p,opt} = (T3/T1)^{\gamma/[2(\gamma-1)]}$.
$T2 = T1\ x = \sqrt{(T1T3)}$ and $T4 = T3/x = \sqrt{(T1T3)} \Rightarrow$ **T2 = T4**	Proved: compressor exit = turbine exit temperature.
$w_{max} = c_p(\sqrt{T3} - \sqrt{T1})^2$	Substitute back. Second derivative $-2T3/x^3 < 0$, so it is a maximum.



$$T_3 = T_4(r)^{\frac{\gamma-1}{\gamma}}$$

Putting the value of T_2 and T_3 in the above equation

$$\eta_{th} = \frac{\left[\left(T_4(r)^{\frac{\gamma-1}{\gamma}} - T_4\right) - \left(T_1(r)^{\frac{\gamma-1}{\gamma}} - T_1\right)\right]}{\left(T_4(r)^{\frac{\gamma-1}{\gamma}} - T_1(r)^{\frac{\gamma-1}{\gamma}}\right)}$$

$$\eta_{th} = \frac{\left[T_4\left((r)^{\frac{\gamma-1}{\gamma}} - 1\right) - T_1\left((r)^{\frac{\gamma-1}{\gamma}} - 1\right)\right]}{(T_4 - T_1)(r)^{\frac{\gamma-1}{\gamma}}}$$

$$\eta_{th} = \frac{(T_4 - T_1)\left((r)^{\frac{\gamma-1}{\gamma}} - 1\right)}{(T_4 - T_1)(r)^{\frac{\gamma-1}{\gamma}}} = \frac{\left((r)^{\frac{\gamma-1}{\gamma}} - 1\right)}{(r)^{\frac{\gamma-1}{\gamma}}} = 1 - \left(\frac{1}{r}\right)^{\frac{\gamma-1}{\gamma}}$$

Work ratio, $r_w = \frac{W}{W_T} = \frac{W_T - W_C}{W_T} = 1 - \frac{W_C}{W_T} = 1 - \frac{c_p(T_2 - T_1)}{c_p(T_3 - T_4)}$

$$r_w = 1 - \frac{(T_2 - T_1)}{(T_3 - T_4)} = 1 - \frac{T_1\left(\frac{T_2}{T_1} - 1\right)}{T_3\left(1 - \frac{T_4}{T_3}\right)}$$

$$r_w = 1 - \frac{(T_2 - T_1)}{(T_3 - T_4)} = 1 - \frac{\left(T_1(r)^{\frac{\gamma-1}{\gamma}} - T_1\right)}{\left(T_3 - \frac{T_3}{(r)^{\frac{\gamma-1}{\gamma}}}\right)}$$

$$r_w = 1 - \frac{T_1\left((r)^{\frac{\gamma-1}{\gamma}} - 1\right)}{T_3\left(1 - \frac{1}{(r)^{\frac{\gamma-1}{\gamma}}}\right)} = 1 - \frac{T_1}{T_3}(r)^{\frac{\gamma-1}{\gamma}}$$

Maximum pressure ratio, $(r_P)_{max} = \frac{P_3}{P_1} = \left(\frac{T_3}{T_1}\right)^{\frac{\gamma}{\gamma-1}}$

For maximum power (work) output $r_p = \sqrt{(r_P)_{max}} = \left(\frac{T_3}{T_1}\right)^{\frac{\gamma}{2(\gamma-1)}}$

For isentropic process, $\frac{P_1}{P_2} = \left(\frac{V_2}{V_1}\right)^{\gamma}$ and $\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1}$

$$\frac{P_1}{P_2} = \left(\frac{T_1}{T_2}\right)^{\frac{\gamma}{\gamma-1}} \text{ and } \frac{T_1}{T_2} = \left(\frac{P_1}{P_2}\right)^{\frac{\gamma-1}{\gamma}}$$

Problem-1: Air enter the compressor of a gas turbine plant operating on Brayton cycle at 1 bar and 27°C. The pressure ratio in the cycle is 6. Calculate the maximum temperature in the cycle and the cycle efficiency. Assume the turbine work as 2.5 times the compressor work. Take $\gamma = 1.4$.

Solution:

Page 2 of 4

Problem sheet: same derivation and optimum pressure ratio

Memory trick: "x = 1.4 rule": power 0.2857 = 2/7. Compressor multiplies by x, turbine divides by x.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2021 6(b) Given: $r_p = 8$, $T_1 = 300$ K, $T_3 = 1300$ K, $\eta_c = 0.80$, $\eta_t = 0.85$, regenerator $\varepsilon = 0.80$, $c_p = 1.005$ kJ/kg K

$x = 8^{0.2857} = 1.811$

Compressor exit: $T_2 = T_1 + T_1(x - 1)/\eta_c$

$= 300 + 243.4/0.80$

$=$ **604.3 K**

Turbine exit: $T_4 = T_3 - \eta_t(T_3 - T_3/x)$

$= 1300 - 0.85(1300 - 717.7)$

$=$ **805.0 K**

Back work ratio: $w_c/w_t = (T_2 - T_1)/(T_3 - T_4)$

$= 304.3 / 495.0$

$=$ **0.615**

Efficiency: $w_{net}/q_{in} = 1.005(495.0 - 304.3) / 1.005(1300 - 604.3)$

$= 191.7 / 699.2$

$=$ **27.4 %**

With regenerator: $T_5 = T_2 + \varepsilon(T_4 - T_2) = 764.9$ K; $\eta = 191.7 / 1.005(1300 - 764.9)$

$= 191.7 / 537.8$

$=$ **35.6 %**

B2 Diesel plant cooling (necessity, thermostat cooling) and starting methods

Must study · Possible · 53% · Easy

PAST-PAPER QUESTIONS

- **2023 Q5(c)** [10] What are the methods used to start a diesel power plant? Briefly explain.
- **2023 Q5(d)** [8] What are the necessities of a cooling system in a diesel plant? Describe thermostat cooling with a figure.
- **2022 Q5(c)** [15] What is the necessity of a cooling system in a diesel plant? Describe the thermostat cooling system with a figure.
- **2020 Q4(b)** [10] Write the disadvantages of a diesel power plant. Why is cooling necessary?
- **2020 Q4(c)** [10] Describe the methods used to start a diesel power plant.

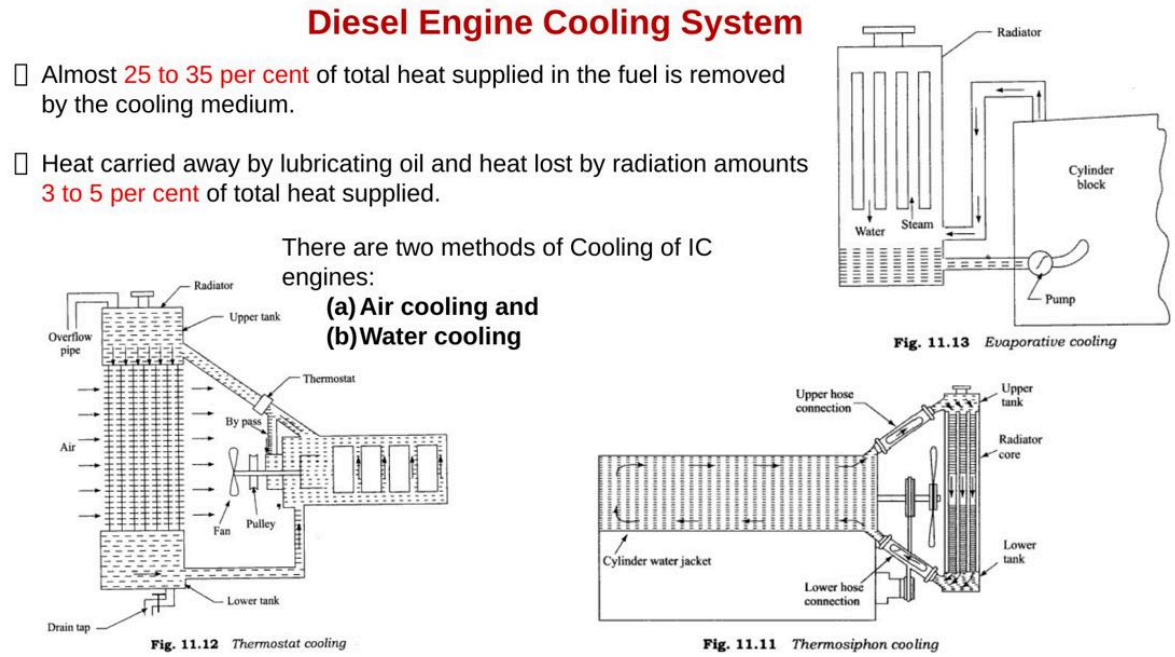
★ REMEMBER IT

- Starting methods: compressed air (most common for big engines), electric motor (battery, small engines), auxiliary petrol engine. Thermostat cooling: a wax valve keeps water in a short loop until the engine reaches about 80 °C, then opens to the radiator.

Before you start: why an engine needs cooling

Burning gas in the cylinder reaches about 2750 °C. Only about one-third of the fuel heat becomes work; 25-35 % flows into the cylinder walls and must be removed, otherwise metal and oil overheat.

Necessity of cooling (2023 5d, 2022 5c, 2020 4b)

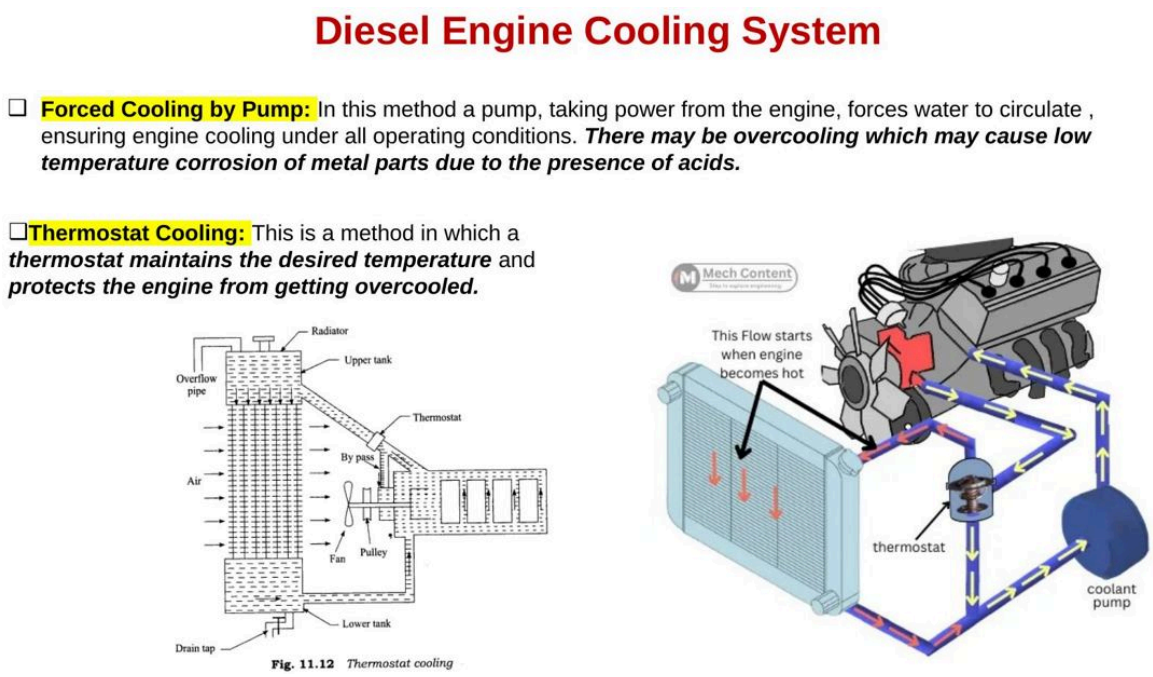


Cooling removes about 25-35 % of the heat supplied; two methods: air cooling and water cooling

1. Heat flux into the chamber walls can reach about 10 MW/m^2; without cooling, **thermal stresses** crack the cylinder and head. Walls must be kept below about 400 °C (cast iron) or 300 °C (aluminium).

2. The **lubricating oil film** on the cylinder wall burns or loses its properties above about 160-180 °C, causing piston seizure.
3. Valves (especially exhaust) overheat and warp; hot spots cause pre-ignition and knock.
4. **Volumetric efficiency** and power fall if the incoming air is heated by hot walls.
5. Too much cooling is also bad: it wastes heat, raises fuel use and causes corrosion from condensed acids, so the temperature must be **controlled**, not just lowered.

Thermostat cooling system (2023 5d, 2022 5c)



Thermostat cooling: a thermostat valve between the engine water jacket and the radiator

It is a **pump-forced water cooling** system with a **thermostat** (temperature-sensitive valve) placed in the outlet line from the cylinder head to the radiator.

Step (write this)	Why / what it means
`1. Engine cold: thermostat closed`	Water only circulates around the jacket (through a by-pass) and warms up quickly.
`2. Water reaches set temperature (about 70-85 °C): thermostat opens`	A wax or bellows element expands and lifts the valve.
`3. Hot water flows to the top tank of the radiator`	Pump keeps the flow going.
`4. Water flows down radiator tubes; fan air removes heat`	Cooled water collects in the bottom tank.
`5. Pump returns cooled water to the jacket`	If the engine cools, thermostat closes partly and reduces the flow.

Advantage: keeps the engine at its best working temperature, gives quick warm-up, prevents over-cooling and saves fuel. Other water cooling methods: thermosiphon (natural circulation by density difference, slow), forced (pump) cooling, pressurised cooling (1.5-2 bar to raise boiling point), evaporative cooling.

Diesel Engine Cooling System

The cylinder and its head are enclosed in a water jacket which is connected to a radiator. Water flowing in the jacket carries away the heat from the engine and becomes heated. ***The hot water then flows into the radiator and gets cooled by rejecting heat to air from the radiator walls***. Cooled water is again circulated in the water jacket.

Various methods used for circulating the water around the cylinder are the following.

- ❑ **Thermosiphon Cooling:** In this method water flow is caused by density difference. The rate of circulation is however ***slow and insufficient***.

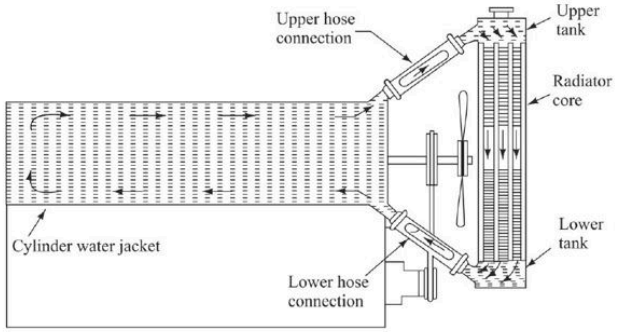


Fig. 11.11 *Thermosiphon cooling*

Thermosiphon cooling for comparison

Starting methods (2023 5c, 2020 4c)

Step (write this)	Why / what it means
`1. Compressed air starting`	Used for large diesel plants. Air at about 17 bar from storage bottles (filled by a compressor) is admitted to some cylinders through a distributor during the power stroke; it turns the crankshaft until fuel injection starts and the engine fires.
`2. Electric starting`	A small electric motor powered by a battery cranks the engine through a pinion on the flywheel; for small engines (up to about 100 hp).
`3. Starting by an auxiliary engine`	A small petrol engine is clutched to the main engine to crank it; for medium engines.

Starting procedure: check fuel, lubricating oil and cooling water supply; check air pressure (or battery); remove load; crank at slow speed for a few minutes; then let it run up and check all systems before loading.

> **Memory trick:** Starting: **"AEA"** = Air, Electric, Auxiliary engine. Big engine → Air.

B9 Reheat Rankine cycle and binary vapour cycle

Good to have · Possible · 53% · Medium

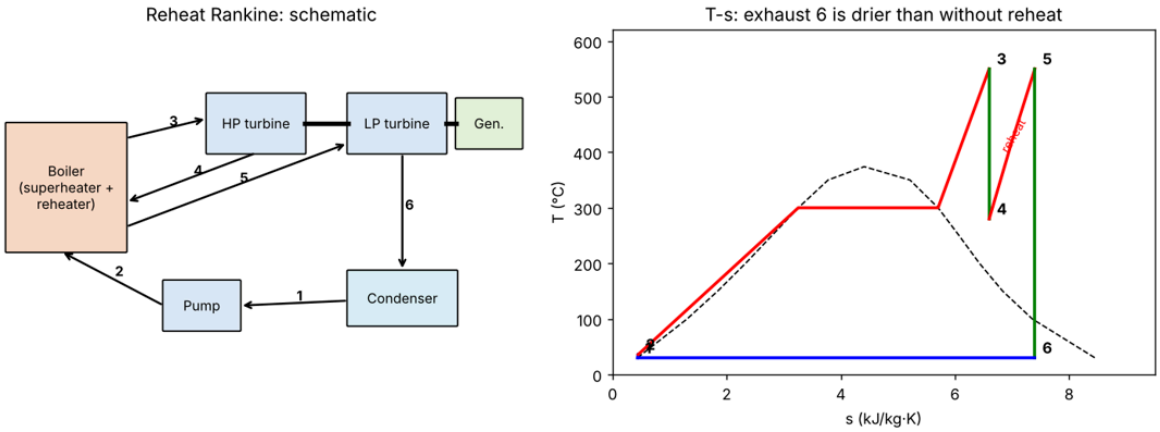
PAST-PAPER QUESTIONS

- **2024 Q7(a)** [15] Explain the working principle of a steam power plant on the reheat Rankine cycle with a schematic.
- **2024 Q7(b)** [10] Why is a binary vapour cycle used in a power plant? Why is it not widely used?
- **2022 Q7(b)** [11] With a schematic describe the working of a binary vapour cycle power plant.

Before you start: why reheat

In a simple Rankine cycle, if you raise boiler pressure for more efficiency, the steam leaving the turbine becomes very wet (more than 10-12 % water droplets), and droplets erode the last turbine blades. Reheat fixes this.

Reheat Rankine cycle (2024 7a)



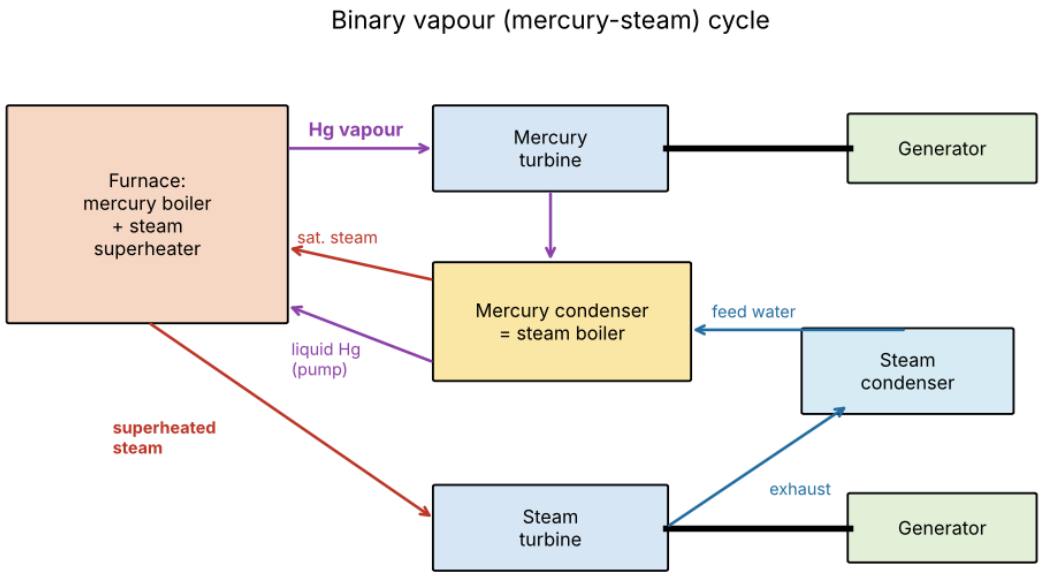
Reheat Rankine cycle: schematic and T-s diagram (1-2 pump, 2-3 boiler, 3-4 HP (high pressure) turbine, 4-5 reheater, 5-6 LP (low pressure) turbine, 6-1 condenser)

Step (write this)	Why / what it means
`1→2 Pump: condensate pumped to boiler pressure`	$w_p = v(P2 - P1)$, very small.
`2→3 Boiler + superheater: water to superheated steam`	$q = h3 - h2$.
`3→4 HP turbine: steam expands to an intermediate pressure`	$w = h3 - h4$.
`4→5 Reheater: steam returns to the boiler and is reheated (usually to the original temperature) at constant pressure`	Extra heat $q = h5 - h4$.
`5→6 LP turbine: expands to condenser pressure`	$w = h5 - h6$; exit steam is much drier.
`6→1 Condenser: steam condensed to water`	Heat rejected to cooling water.
`n = [(h3 - h4) + (h5 - h6) - wp] / [(h3 - h2) + (h5 - h4)]`	Total turbine work minus pump work, over total heat.

Advantages: drier steam at turbine exit (less blade erosion), more work per kg of steam (smaller steam flow and plant size), efficiency rises about 4-5 % and allows higher boiler pressures.

Drawbacks: extra piping, reheater and controls, higher cost; used in large plants above about 100 MW.

Binary vapour cycle (2024 7b, 2022 7b)



Binary vapour cycle: mercury cycle on top, its condenser is the steam boiler of the water cycle below

Why it is used: water is a poor working fluid at high temperature: to get steam at about 550 °C its pressure must be very high, and its critical temperature (374 °C) is low, so most heat is added at fairly low temperature. A fluid like **mercury** has a high critical temperature (about 1490 °C) and a low saturation pressure at high temperature (about 10 bar at 515 °C). So two cycles are combined:

Step (write this)	Why / what it means
`Mercury boiler: mercury vaporised by the furnace at high temperature, low pressure`	High-temperature heat addition.
`Mercury turbine: mercury vapour expands, drives a generator`	
`Mercury condenser = steam boiler: condensing mercury (about 230 °C) boils water`	Heat rejected by the top cycle is the heat input of the bottom cycle.
`Steam superheated in the furnace, expands in steam turbine, condensed, pumped back`	Normal Rankine cycle.

Heat is thus added at a much higher average temperature, so the **overall efficiency is higher** (closer to Carnot): about 50 % vs 35-40 %.

Why not widely used: mercury is **very costly and scarce**; its vapour is **highly poisonous** (leaks are dangerous); it **does not wet** metal surfaces, so heat transfer is poor (needs additives like magnesium, titanium); low latent heat means a large mass flow; complex, costly plant and maintenance. Today supercritical steam plants and gas-steam combined cycles give similar efficiency more cheaply.

> **Memory trick:** Binary = **"Mercury on top, water below"**. Drawback words: **"Costly, Toxic, Doesn't wet"**.

B12 Numericals: chimney draught and chimney height Good to have

Possible · 33% Medium

PAST-PAPER QUESTIONS

- **2023 Q8(d)** [8] Chimney 28 m high, hot gas 320 °C, outside air 27 °C, 15 kg air per kg coal. Find (i) draught in mm of water, (ii) draught as height of hot gas column.
- **2020 Q6(c)** [9] Find chimney height to produce a static draught of 22 mm of water. Mean flue gas temperature 290 °C, ambient 20 °C, R_air = 29.26 and R_gas = 26.2 kgf·m/kg·K, barometer 760 mm Hg.

CALCULATOR **Chimney draught calculator**

Change any number; every step is recalculated and shown with the numbers put in.

1. Gas in kelvin: **320** + 273 = **593** K
2. Air in kelvin: **27** + 273 = **300** K
3. Draught: h_w = 353 H $[1/T_a - (m+1)/(m\ T_g)]$ = 353 × **28** × (1 / **300** - (**15**+1) / **15** / **593**) = **15.17** mm of water
4. Draught as a hot-gas column: $H_g = H[(m/(m+1))(T_g/T_a) - 1]$ = **28** × (**15** / (**15**+1) × **593** / **300** - 1) = **23.89** m of hot gas
5. Draught in pascal: 1 mm water = 9.81 Pa = **15.17** × 9.81 = **148.8** Pa

Before you start: tips before the numbers

Convert temperatures to kelvin: T(K) = T(°C) + 273. Use the formulas from B11. m = kg of air supplied per kg of fuel.

2023 8(d): H = 28 m, gas 320 °C, air 27 °C, 15 kg air/kg coal

Step (write this)	Why / what it means
T_g = 593 K, T_a = 300 K, m = 15, (m + 1)/m = 16/15 = 1.0667	
h_w = 353 × 28 [1/300 - 1.0667/593]	Draught formula.
= 9884 [0.003333 - 0.001799] = 9884 × 0.0015345 = **15.17 mm of water**	(i)
H_g = 28 [(15/16)(593/300) - 1] = 28 [1.8531 - 1] = **23.9 m of hot gas**	(ii) Draught expressed as a hot-gas column.

2020 6(c): draught 22 mm water, T_g = 290 °C, T_a = 20 °C, R_{air} = 29.26, R_{gas} = 26.2 kgf·m/kg·K, 760 mm Hg

Side note: Here R is in kgf·m/kg·K (gravitational units), so pressure must be in kgf/m^2. 760 mm Hg = 1.0332 kgf/cm^2 = 10,332 kgf/m^2. Also, 1 mm of water = 1 kgf/m^2, which makes this very easy.

Step (write this)	Why / what it means
$\rho_a = p/(R_a T_a)$ = 10,332/(29.26 × 293) = 1.2052 kg/m^3	T_a = 293 K.

$\rho_g = 10{,}332/(26.2 \times 563) = 0.7004\ kg/m^3$	$T_g = 563$ K.
Draught $(kgf/m^2) = H\ (\rho_a - \rho_g) \Rightarrow 22 = H \times 0.5048$	Weight difference of the two columns per m^2.
H = 43.6 m	

Memory trick: 1 mm of water = 1 kgf/m² = 9.81 Pa.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2023 8(d) Given: H = 28 m, $T_g = 320 + 273 = 593$ K, $T_a = 27 + 273 = 300$ K, m = 15 kg air/kg coal

(i) Draught: $h_w = 353\ H\ [1/T_a - (m+1)/(m\ T_g)]$

$= 353 \times 28\ [1/300 - 16/(15 \times 593)] = 9884 \times 0.0015345$

= **15.17 mm of water**

(ii) Hot gas column: $H_g = H[(m/(m+1))(T_g/T_a) - 1]$

$= 28[(15/16)(593/300) - 1] = 28 \times 0.8531$

= **23.9 m**

B7 Exergy analysis numericals (2024): Diesel cycle and closed Brayton cycle

Skip if short | Possible · 33% | Hard

PAST-PAPER QUESTIONS

- **2024 Q5(c)** [25] A diesel plant operates on a simple Diesel cycle with air, compression ratio 16. Air is at 323 K and 100 kPa at the start of compression and 1723 K at the end of heat addition. Compression efficiency 85 %, expansion efficiency 90 %, T_source = 2000 K, T0 = 300 K. (i) Draw schematic and T-s; (ii) write mass, energy, entropy and exergy balances for each process; (iii) find all state properties (P, T, v, u, h, s, e); (iv) total entropy generation, exergy destruction, net work, energy and exergy efficiency.
- **2024 Q6(a)** [25] A closed gas turbine plant on the simple Brayton cycle has pressure ratio 16. Air enters the compressor at atmospheric pressure and 300 K at 10 kg/s; it leaves the turbine at 973 K; the gas exits the combustion chamber at 1800 K. Compressor and turbine isentropic efficiency 85 %, variable specific heats, T_s = 2500 K, T0 = 293 K, P0 = 100 kPa. (i) Schematic and T-s; (ii) balances for all components; (iii) state properties; (iv) total entropy generation, exergy destruction, net work, energy and exergy efficiency.

LEARN IT FROM ZERO

Exergy, from zero

Energy is never destroyed, so the first law cannot tell you where a plant *wastes* its chance to make work. **Exergy** is the part of energy that can still be turned into work, measured against the surroundings (the **dead state**, T_0 and P_0). Heat Q at temperature T_s carries exergy $Q(1 - T_0/T_s)$: very hot heat is valuable, heat near room temperature is almost worthless.

Every real process (friction, heat flowing across a big temperature gap, mixing) **generates entropy** s_{gen} and destroys exergy: $\mathbf{X_{destroyed} = T_0\, s_{gen}}$. The exam recipe for each process: (1) find the states, (2) Δs from $c_p \ln(T_2/T_1) - R \ln(P_2/P_1)$, (3) $s_{gen} = \Delta s - q/T_{boundary}$, (4) $X_{destroyed} = T_0 s_{gen}$, (5) second-law efficiency = w_{net} / X_{in}.

Check that always works: $\mathbf{X_{in} = w_{net} + \Sigma X_{destroyed}}$.

CALCULATOR **Entropy generation and exergy destroyed in one process**

Change any number; every step is recalculated and shown with the numbers put in.

1. Entropy change of air: $\Delta s = c \ln(T_b/T_a) - R \ln(\text{ratio})$ = **1.005** × ln(**1723** / **1095**) - **0.287** × ln(**1**) = **0.4557** kJ/kg K
2. Entropy generated: $s_{gen} = \Delta s - q/T_b$ = **0.4557** - **631.2** / **2000** = **0.1401** kJ/kg K
3. Exergy destroyed: $X_d = T_0 s_{gen}$ = **300** × **0.1401** = **42.0** kJ/kg
4. Exergy of the heat: $X = q(1 - T_0/T_s)$ = **631.2** × (1 - **300** / **2000**) = **536.5** kJ/kg

Default = 2024 5(c) heat addition 2→3 (constant pressure): s_{gen} = 0.140 kJ/kg K, X_d = 42.0 kJ/kg.

For a constant-volume process put c_v = 0.718 and ratio 1.

Before you start: exergy explained for a first-timer

Energy is never lost (first law). But not all energy is equally useful: 1 kJ of heat at 2000 K can make much more work than 1 kJ at 310 K. **Exergy** is the maximum useful work a system or heat can give before it reaches the **dead state** (the surroundings, T0 and P0).

Every real process (friction, heat transfer across a big temperature difference, mixing) **generates entropy** $s_{gen} \geq 0$ and **destroys exergy**: $\mathbf{X_{destroyed} = T0 \times s_{gen}}$ (Gouy-Stodola).

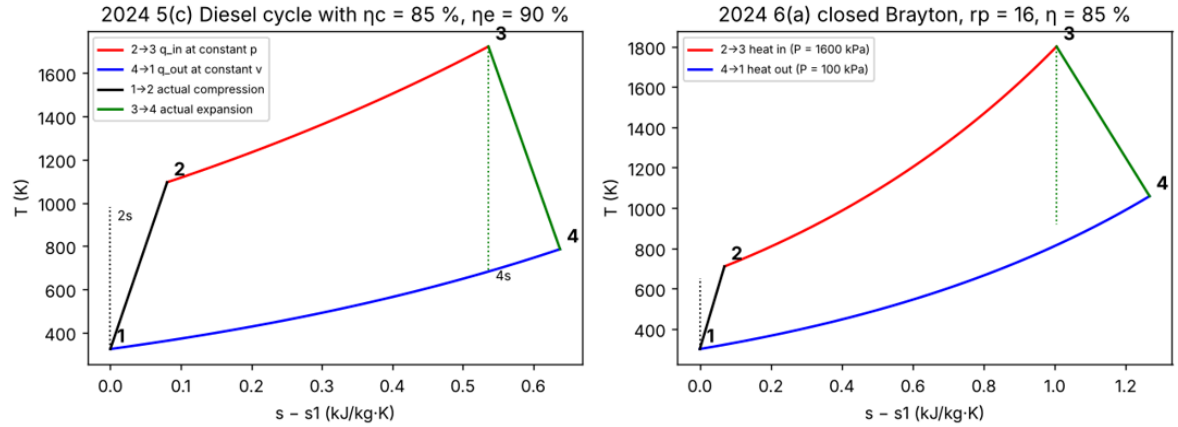
Exergy of heat Q taken from a source at T_s: $X = Q(1 - T0/T_s)$. Energy efficiency = w_{net}/q_{in}; exergy (second-law) efficiency = w_{net}/X_{in}.

Air as ideal gas: $\Delta s = c_p \ln(T2/T1) - R \ln(P2/P1) = c_v \ln(T2/T1) + R \ln(v2/v1)$. Data: $c_p = 1.005$, $c_v = 0.718$, $R = 0.287$ kJ/kg·K, $\gamma = 1.4$.

Balance equations to write for every process (part ii of both questions)

Step (write this)	Why / what it means
Mass: $m_{in} = m_{out}$ (closed system: m constant)	Same air goes all round the cycle.
Energy (closed, Diesel): $q - w = u2 - u1$. Energy (steady flow, Brayton): $q - w = h2 - h1$	First law per kg.
Entropy: $s2 - s1 = \Sigma\, q/T_{boundary} + s_{gen}$	Adiabatic: $s_{gen} = s2 - s1$. Heat in from source T_s: $s_{gen} = (s3 - s2) - q_{in}/T_s$. Heat out to T0: $s_{gen} = (s1 - s4) + q_{out}/T0$.
Exergy: $X_{heat\ in} - X_{heat\ out} - w = \Delta(\text{exergy}) + X_{destroyed}$, $X_{destroyed} = T0\ s_{gen}$	For the whole cycle $\Delta(\text{exergy}) = 0$, so $X_{in} = w_{net} + \Sigma X_{destroyed}$.

2024 5(c): Diesel cycle, r = 16, 323 K, 100 kPa, T3 = 1723 K, η_c 85 %, η_e 90 %, T_s = 2000 K, T0 = 300 K



Left: T-s of this Diesel cycle (dotted = isentropic 2s and 4s). Right: the closed Brayton cycle of 6(a)

Schematic: piston-cylinder: 1→2 compression (adiabatic, η_c 85 %), 2→3 heat added at constant pressure from a 2000 K source, 3→4 expansion (adiabatic, η_e 90 %), 4→1 heat rejected at constant volume to surroundings at 300 K. Cold-air standard, constant specific heats.

Step (write this)	Why / what it means
$v1 = RT1/P1 = 0.287 \times 323/100 = 0.927\ m^3/kg$; $v2 = v1/16 = 0.0579$	Volumes fixed by compression ratio.
$T2s = T1\ r^{\gamma-1} = 323 \times 16^{0.4} = 979.2$ K	Isentropic compression.

$T2 = T1 + (T2s - T1)/n_c = 323 + 656.2/0.85 =$ **1094.9 K**	Real compression gets hotter.
$P2 = RT2/v2 = 0.287 \times 1094.9/0.0579 =$ **5424 kPa**	Ideal gas law at state 2.
$P3 = P2$; $v3 = RT3/P3 = 0.0912$ m^3/kg; cut-off ratio $r_c = v3/v2 = T3/T2 = 1.574$	Constant pressure heating.
$T4s = T3\ (v3/v4)^{0.4} = 1723 \times (1/10.17)^{0.4} = 681.4$ K	Expansion ratio v4/v3 = 16/1.574 = 10.17.
$T4 = T3 - n_e(T3 - T4s) = 1723 - 0.9 \times 1041.6 =$ **785.5 K**; $P4 = RT4/v1 = 243.2$ kPa	

State table (u = c_vT, h = c_pT, s measured from dead state 300 K, 100 kPa; e = closed-system exergy = (u – u0) + P0(v – v0) − T0(s – s0)):

State	P (kPa)	T (K)	v (m^3/kg)	u (kJ/kg)	h (kJ/kg)	s – s0 (kJ/kg·K)	e (kJ/kg)
1	100	323	0.927	231.9	324.6	0.0742	0.8
2	5424	1094.9	0.0579	786.2	1100.4	0.1550	444.0
3	5424	1723	0.0912	1237.1	1731.6	0.6107	761.5
4	243.2	785.5	0.927	564.0	789.5	0.7123	141.5

Step (write this)	Why / what it means
$w_{12,in} = c_v(T2 - T1) = 0.718 \times 771.9 = 554.3$ kJ/kg	Adiabatic: work in = rise in u.
$q_{in} = c_p(T3 - T2) = 1.005 \times 628.1 =$ **631.2 kJ/kg**	Constant pressure: q = Δh. (Boundary work PΔv = 180.3 kJ/kg.)
$w_{34,out} = c_v(T3 - T4) = 0.718 \times 937.5 = 673.1$ kJ/kg	
$q_{out} = c_v(T4 - T1) = 0.718 \times 462.5 = 332.1$ kJ/kg	Constant volume rejection.
$w_{net} = q_{in} - q_{out} =$ **299.1 kJ/kg**; $n_{energy} = 299.1/631.2 =$ **47.4 %**	Check: 673.1 + 180.3 − 554.3 = 299.1.
$s_{gen,12} = c_v \ln(T2/T1) + R \ln(v2/v1) = 0.8765 - 0.7957 = 0.0808$ kJ/kg·K	Friction in compression.
$s_{gen,23} = c_p \ln(T3/T2) - q_{in}/T_s = 0.4557 - 0.3156 = 0.1400$	Heat crossing from 2000 K source to cooler gas.
$s_{gen,34} = c_v \ln(T4/T3) + R \ln(v4/v3) = -0.5639 + 0.6656 = 0.1017$	Friction in expansion.
$s_{gen,41} = c_v \ln(T1/T4) + q_{out}/T0 = -0.6382 + 1.1070 = 0.4689$	Hot gas dumping heat to 300 K surroundings.
Total $s_{gen} = 0.7914$ kJ/kg·K	
$X_{destroyed} = T0\ s_{gen}$: 24.2 + 42.0 + 30.5 + 140.7 = **237.4 kJ/kg**	Biggest loss: heat rejection.
$X_{in} = q_{in}(1 - T0/T_s) = 631.2 \times 0.85 = 536.5$ kJ/kg	Exergy supplied with the heat.
$n_{exergy} = w_{net}/X_{in} = 299.1/536.5 =$ **55.7 %**	Check: 536.5 − 237.4 = 299.1 = w_{net}. Balanced.

2024 6(a): closed Brayton, r_p = 16, 300 K, 100 kPa, 10 kg/s, T3 = 1800 K, η = 85 %, variable c_p, T_s = 2500 K, T0 = 293 K

> **Side note:** Variable specific heats means read h and s° from the air table (Cengel A-17) instead of using $c_p\Delta T$. Isentropic step: $s°_{2s} = s°_1 + R \ln(P2/P1)$ (or use relative pressure $P_{r2} = P_{r1} \times r_p$). Values below are from the table polynomial, within 0.5 % of the printed table.

Schematic: compressor (1→2) → heater fed from a 2500 K source (2→3) → turbine (3→4) driving compressor and generator → cooler rejecting heat to surroundings at 293 K (4→1) → back to compressor.

Step (write this)	Why / what it means
State 1: 300 K, 100 kPa: h1 = 300.19 kJ/kg, s°1 = 1.7020 kJ/kg·K	Table.
2s: s°2s = 1.7020 + 0.287 ln16 = 2.4978 ⇒ T2s = 650 K, h2s = 661.7	Find T with this s° in the table.
h2 = h1 + (h2s - h1)/n_c = 300.19 + 361.5/0.85 = **725.5 kJ/kg** ⇒ T2 = 709 K	Real compressor.
State 3: 1800 K, 1600 kPa: h3 = 2005.4, s°3 = 3.6722	Table.
4s: s°4s = 3.6722 - 0.287 ln16 = 2.8764 ⇒ T4s = 919 K, h4s = 956.4	
h4 = h3 - n_t(h3 - h4s) = 2005.4 - 0.85 × 1049 = **1113.7 kJ/kg** ⇒ T4 = 1058 K	Real turbine.

State	P (kPa)	T (K)	v (m^3/kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)	ψ flow exergy (kJ/kg)
1	100	300	0.861	214.1	300.2	1.7020	0.1
2	1600	709.4	0.127	521.9	725.5	1.7960	397.9
3	1600	1800	0.323	1488.8	2005.4	2.8765	1361.2
4	100	1058.2	3.037	810.0	1113.7	3.0358	422.8

Step (write this)	Why / what it means
w_c = h2 - h1 = 425.3 kJ/kg; w_t = h3 - h4 = 891.7 kJ/kg	Steady-flow energy balance, adiabatic.
w_{net} = 466.3 kJ/kg ⇒ W_{net} = 10 × 466.3 = **4.66 MW**	
q_{in} = h3 - h2 = 1279.8 kJ/kg (12.8 MW); q_{out} = h4 - h1 = 813.5 kJ/kg	
n_{energy} = 466.3/1279.8 = **36.4 %**	
s_{gen}: compressor s2 - s1 = 0.0939; heater (s3 - s2) - q_{in}/2500 = 0.5686; turbine s4 - s3 = 0.1593; cooler (s1 - s4) + q_{out}/293 = 1.4428	kJ/kg·K.
Total s_{gen} = 2.2646 kJ/kg·K = 22.65 kW/K	× 10 kg/s.
$X_{destroyed}$ = 293 × s_{gen}: 27.5 + 166.6 + 46.7 + 422.7 = **663.5 kJ/kg = 6.64 MW**	Cooler (heat dumped) is the largest loss, then the heater.
X_{in} = q_{in}(1 - 293/2500) = 1129.8 kJ/kg; n_{exergy} = 466.3/1129.8 = **41.3 %**	Check: 1129.8 − 663.5 = 466.3.

Side note: The paper also says the gas leaves the turbine at 973 K. That is not possible with η_t = 85 % (it would need $\eta_t \approx$ 94 %). If you use 973 K: h4 = 1016.6, w_t = 988.7, w_{net} = 563.4 kJ/kg (5.63 MW), η = 44.0 %, s_{gen} = 1.933 kJ/kg·K, $X_{destroyed}$ = 566.4 kJ/kg, η_{exergy} = 49.9 %. Write one line saying which data you used.

Memory trick: Exergy problems in four moves: **"States, Work, Entropy, Exergy"**. And always finish with the check $X_{in} - X_{destroyed} = w_{net}$.